THE WEIGHTED GOREINOV–TYRTYSHNIKOV–ZAMARASHKIN SELECTION PROBLEM

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ABSTRACT. A weighted design of size m and rank k over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ consists of vectors $g_1, \dots, g_m \in \mathbb{F}^k$ and positive weights t_c summing to one with $\sum_c t_c\, g_c g_c^{\mathsf{H}} = I_k$. The selection problem asks for k of the vectors that dominate the identity on their own, $\sum_{c \in C} g_c g_c^{\mathsf{H}} \succeq I_k$. We solve the problem over $\mathbb{C}$: for $k \geq 2$ such a subset exists in every design precisely when $m \leq k+1$, and at $m = k+1$ the bound 1 is attained at every weight vector. Over $\mathbb{R}$ we settle rank at most two and corank at most two, and reduce each rank to finitely many cells. We then prove a variational principle. Grant the statement at the two cells one size below; if it fails at a cell, it fails at a global minimizer of the selection objective over a compactification of the design space, and that minimizer has strictly positive weights. Rank three over $\mathbb{R}$ thereby reduces to two cells, $(6, 3)$ and $(7, 3)$, and at each of them to ruling out interior critical points below the threshold. The threshold is attained at every size, so no argument for the conjecture can carry a uniform margin. Statements whose head carries the tag *Lean* are verified in the Lean 4 proof assistant.

CONTENTS

1. Introduction

Let $\mathbb{F}$ denote $\mathbb{R}$ or $\mathbb{C}$. A *weighted design* of size m and rank k consists of vectors $g_1, \ldots, g_m \in \mathbb{F}^k$ and weights $t_1, \ldots, t_m > 0$ with $\sum_c t_c = 1$ subject to the Parseval condition

$$\sum_{c=1}^{m} t_c\, g_c g_c^{\mathsf{H}} \;=\; I_k. \tag{1}$$

The condition says that the vectors resolve the identity on average. Do k of them resolve it on their own? The selection problem asks for a $C \subseteq \{1, \ldots, m\}$ with $|C| = k$ and

$$S_C \;:=\; \sum_{c\in C} g_c g_c^{\mathsf{H}} \;\succeq\; I_k. \tag{2}$$

Such a C *dominates*. We write $\mathrm{GTZ}(m,k)$ for the assertion that every design of size m and rank k over $\mathbb{R}$ admits a dominating k-subset, and $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ for its complex analogue.

At uniform weights the problem is classical. Let $U \in \mathbb{F}^{m\times k}$ satisfy $U^{\mathsf{H}}U = I_k$, let u_c^{H} be its c-th row, and put $g_c := \sqrt{m}\, u_c$, $t_c := 1/m$; then (1) holds, and for U_C the $k \times k$ submatrix on the rows C the requirement (2) reads

$$U_C^{\mathsf{H}} U_C \;\succeq\; \tfrac{1}{m} I_k, \qquad \text{that is} \qquad \sigma_{\min}(U_C) \;\geq\; m^{-1/2}, \qquad \text{that is} \qquad \|U_C^{-1}\|_2 \;\leq\; \sqrt{m}.$$

Some $k \times k$ submatrix of a matrix with orthonormal columns has inverse of spectral norm at most $\sqrt{m}$: this is the conjecture of Goreinov, Tyrtyshnikov and Zamarashkin [11], who raised it while bounding the error of pseudoskeleton approximation. It is the analytic content of the maximal-volume principle, and it remains open.

The principle does prove a weaker bound [12, 10]. Among the finitely many k-subsets choose one of maximal volume, that is one maximizing $|\det U_C|$. It is nonsingular: by the Cauchy–Binet formula $\sum_{|C|=k} |\det U_C|^2 = \det(U^{\mathsf{H}}U) = 1$, so the volumes do not all vanish. Put $B := U U_C^{-1}$. Replacing the j-th row of U_C by the r-th row of U replaces that row by its expansion $\sum_i B_{ri}(U_C)_i$ in the selected rows, and multilinearity leaves one surviving term, so the new determinant is $B_{rj} \det U_C$. An entry of B of modulus greater than 1 would therefore name a k-subset of strictly larger volume, and maximality gives $|B_{rj}| \leq 1$ throughout. The rows of B indexed by C form I_k, whence

$$\operatorname{tr}\big((U_C^{\mathsf{H}} U_C)^{-1}\big) \;=\; \|U_C^{-1}\|_F^2 \;=\; \|B\|_F^2 \;\leq\; k + k(m-k),$$

using $\|UX\|_F = \|X\|_F$ in the middle step. Moreover $U_C^{\mathsf{H}} U_C \preceq U^{\mathsf{H}}U = I_k$, so $(U_C^{\mathsf{H}} U_C)^{-1} \succeq I_k$; a Hermitian $k \times k$ matrix dominating the identity has all its eigenvalues at least 1 and so has largest eigenvalue at most its trace diminished by $k-1$. Therefore $\|U_C^{-1}\|_2^2 \leq k(m-k)+1$ and $S_C \succeq \frac{m}{k(m-k)+1} I_k$. The bound and the conjecture differ by a single term:

$$k(m-k)+1 \;=\; m \;+\; (k-1)(m-k-1), \tag{3}$$

a ring identity in the substitution $k = a+1$, $m-k = b+1$. The deficit $(k-1)(m-k-1)$ vanishes precisely when $k = 1$ or $m = k+1$, and is positive otherwise; at $(m,k) = (7,3)$ it equals 6, and the principle delivers $\sigma_{\min}^2 \geq 1/13$ where $1/7$ is asked. The classical argument therefore suffices exactly on the range

$$k \leq 1 \quad \text{or} \quad m \leq k+1. \tag{4}$$

Beyond (4) the known cases are few. Nesterenko [20] settled $(m,k) = (4,2)$ over $\mathbb{R}$ by an argument in Plücker coordinates, and reformulated the rank-two problem as an isoperimetric question for closed polygons [21]; Sengupta and Pautov [27] then settled rank two over $\mathbb{R}$ for every m. Nesterenko also described the equality locus at rank two over $\mathbb{R}$ [24], and built configurations attaining the bound 1 at every pair (m,k) from the star spaces of series-parallel graphs [22]. Over $\mathbb{C}$ the conjecture is false at rank two: at uniform weights some pair always satisfies $S_C \succeq \alpha I_2$ with

$\alpha = 2 - 2/\sqrt{3} < 1$, and α cannot be raised, being attained when $4 \mid m$ and approached otherwise [23]. The restricted-invertibility theorems [3, 28, 19] select subsets of size strictly below the rank, with a least eigenvalue controlled by a stable-rank ratio; that control degenerates as the subset size approaches the rank, which is where the present question sits. Maximizing the volume is itself NP-hard, and NP-hard to approximate [6], so the selection principle asserts existence without exhibiting a subset, and the algorithmic literature settles for the polynomial factors of [13].

We study the problem with general weights. The weights then become a coordinate of the problem rather than a fixed normalization: vectors and weights vary independently subject to (1), and the extremal configurations appear as sections over the weight simplex instead of isolated matrices. Under the substitution $v_c := \sqrt{t_c}\, g_c$ the condition (1) says that the matrix V with rows v_c^{H} is an isometry, and (2) compares a principal block of the projection $P = VV^{\mathsf{H}}$ with the corresponding block of $\operatorname{diag}(t)$. These coordinates put the problem on a compact set and open it to variational methods. At a fixed pair (m, k) the weighted statement is formally stronger than the classical one; over all sizes the two agree, since a design with rational weights is a uniform design of larger size with repeated vectors, and arbitrary weights follow by a limit.

Over $\mathbb{C}$ the problem is settled. For $k \geq 2$ selection succeeds in every complex design when $m \leq k + 1$ and fails whenever $m \geq k + 2$ (Theorem 4.9); for $k \leq 1$ it always succeeds. Against (4) this reads: over $\mathbb{C}$ the conjecture holds just where the maximal-volume count already proves it, and fails everywhere else. Nothing remains open over $\mathbb{C}$, and no field-independent argument can settle the real conjecture.

Over $\mathbb{R}$, rank two holds at every size (Theorem 5.10), corank one and corank two hold at every rank (Theorems 3.19 and 5.11), and at each fixed rank the problem is finite: if $\mathrm{GTZ}(m, k)$ holds for all $m \leq k(k + 1)/2 + 1$ then it holds for all m (Theorem 5.15). That bound is one more than the dimension of the space of real symmetric $k \times k$ matrices, the size at which a support-reduction argument applied to (1) starts to bite. Rank three leaves the cells $(6, 3)$ and $(7, 3)$, and no symmetry reduces the pair further: $(6, 3)$ is self-dual, and duality carries $(7, 3)$ to $(7, 4)$, again of corank three (Corollary 5.19).

The threshold in (2) is attained at every size $m \geq k + 1$: there are designs in which some k-subset dominates and none dominates strictly (Theorem 6.4). At corank one these designs are classified, and they form a section over the whole open weight simplex, a set of dimension $m - 1$ (Theorem 3.19). The regular tetrahedron of Example 2.11 is one of them; another, at $(5, 3)$, has pairwise non-parallel vectors (Example 6.9), so splitting a vector of a smaller design does not produce the whole locus.

The space of designs is not compact: its weights range over an open simplex, and by (1) a vector diverges as its weight tends to zero. In the coordinates (P, t) that space becomes the interior of a compact one, on which the objective $\max_{|C|=k} \lambda_{\min}(W[C])$, $W = P - \operatorname{diag}(t)$, extends continuously. The smaller cells discharge the added boundary points. Grant the same statement at $(m - 1, k)$ and at $(m - 1, k - 1)$: then every point with a vanishing weight carries a dominating k-subset (Theorem 7.14), and if $\mathrm{GTZ}(m, k)$ fails the objective attains its minimum at an interior point, that is at a genuine weighted design (Theorem 7.18). At $(6, 3)$ both hypotheses are theorems of §5; at $(7, 3)$ one of them is the cell $(6, 3)$.

Rank three then reduces to excluding critical points below the threshold at the two surviving cells. §8 derives the system such a point satisfies: a multiplier Λ obeying a quadric law $g_c^{\mathsf{H}} \Lambda g_c = \nu$ at every c, a coverage law, and three exclusions that cut down the possible configurations of active subsets. A congruence relates the objective minimized in §7 to the one differentiated in §8, and the two need not share their minimizers. Definition 8.2 therefore makes the step from one to the other a hypothesis, and Theorem 8.23 is the only statement that carries it.

Because the threshold is attained, and attained on a set of full dimension in the weights, no proof of GTZ can carry a uniform margin: for every $\varepsilon > 0$ there are designs admitting no k-subset with $S_C \succeq (1+\varepsilon)I_k$ (Corollary 6.8). That costs several familiar strategies. An induction on the size or on the rank cannot transport a quantitative slack across a step, there being none to transport at the source. A certificate for (2), whether a sum-of-squares identity, a dual multiplier, or a determinantal or interlacing bound, must be exact on the extremal locus, so any strategy whose output is a strict inequality fails at the outset. A minimizing sequence cannot be handled term by term, since the identity available at a degenerate limit fails at every positive weight, however small (Theorem 7.21); hence the variational principle runs through a compactification rather than through a density argument. Appendix A collects these strategies with the exact rational witnesses that refute them.

The cells $(6,3)$ and $(7,3)$ remain open (Conjectures 8.21 and 8.22), and with them the real problem at rank three and above; over $\mathbb{C}$ the sharp constant at rank three is bracketed but not determined (Remark 4.14). §10 lists these questions together with what the constraints above require of any solution.

Tags. A statement whose head carries the tag *Lean* is machine-verified; §9 gives the correspondence. One carrying *conditional* rests on a conjecture, stated at that point and not proved here; a hypothesis discharged inside the paper is carried in the statement and does not earn the tag. Untagged statements are proved in the ordinary way. Every numerical assertion below rests on exact rational or exact algebraic arithmetic.

Notation. For $x, y \in \mathbb{F}^k$ we write $\langle x, y\rangle := x^{\mathsf{H}} y$, conjugate-linear in the first argument and linear in the second, and $\|x\|^2 = \langle x, x\rangle$. We write $A \succeq 0$ for a positive semidefinite Hermitian A and $A \succ 0$ for a positive definite one; $A[C]$ is the principal submatrix on the index set C, and spec A the spectrum. A *k-subset* always means a subset of $\{1,\dots,m\}$ of cardinality k. The *corank* of a cell (m,k) is $m-k$. A pair (m,k) with $m<k$ carries no design (Lemma 2.2), so every statement quantifying over such a pair is vacuous.

Each symbol in the following table has one meaning throughout, and is never rebound, not even inside a proof. Symbols outside the table, such as x, y, u, A, B, Y and Z, are proof-local bound variables; they are reused across disjoint proofs, and none of them shadows a symbol of the table.

m, k	size and rank	P	the chart projection VV^{H}
c, d	indices of atoms	D	the diagonal $\operatorname{diag}(t)$
C	a k-subset	W	the gap $P-D$
$\mathcal{D}$	a design	Gram_C	the Gram matrix of C
g_c	the c-th atom	M_C	the matrix of rows g_c^{H}, $c \in C$
t_c	its weight	S_C	the subset sum $\sum_{c\in C} g_c g_c^{\mathsf{H}}$
ℓ_c	its leverage $\|g_c\|^2$	F, G	the design and chart objectives
s_c	its leverage score $t_c\ell_c$	ν	the critical value of F
v_c	the chart atom $\sqrt{t_c}\, g_c$	Λ	the stationarity multiplier
V	the isometry of rows v_c^{H}	λ_i	its scalar multipliers
$S^{\perp}$	the dual frame operator	π_C	volume sampling, $\det P[C]$
$N(k)$	the window $k(k+1)/2+1$	α	the extremal constants
$\mathcal{T}$	the tangent space	$\mathcal{G}$	a graph

Transpose and adjoint are written A^{T} and A^{H}, as in numerical linear algebra, and are set upright so that neither can be mistaken for a variable. Over $\mathbb{R}$ the two agree; the paper writes T in the sections that are real throughout and H where the field is $\mathbb{F}$, with no other decoration for either operation.

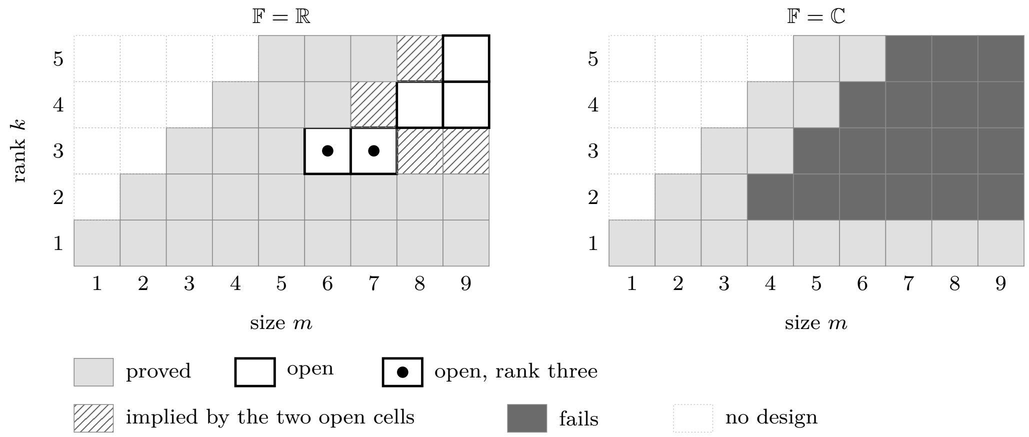


FIGURE 1. The decision board. Rows are the rank k, columns the size m; cells with $m < k$ carry no design. Over $\mathbb{R}$ the two open cells at rank three are $(6, 3)$ and $(7, 3)$; the hatched region reduces to them by Theorem 5.15. Over $\mathbb{C}$ the answer is $m \leq k + 1$ for every $k \geq 2$.

2. Elementary cases

The condition (1) yields three elementary consequences on its own: a trace identity, an upper bound on each leverage score, and a comparison of the unweighted sum of the $g_c g_c^{\mathsf{H}}$ with the identity. Between them they settle the cells $m = k$ and $k = 1$, and no others. §2.4 says why they stop there, and Example 2.11 exhibits what the rest of the paper works against: a design at which every k-subset sits on the threshold.

2.1. The objects.

Definition 2.1. A *weighted design* of size m and rank k over $\mathbb{F}$ is a pair $((g_c)_{c\leq m}, (t_c)_{c\leq m})$ with $g_c \in \mathbb{F}^k$, $t_c > 0$ and $\sum_c t_c = 1$, satisfying (1). For $C \subseteq \{1, \ldots, m\}$ put $S_C := \sum_{c\in C} g_c g_c^{\mathsf{H}}$; the subset C *dominates* if $S_C - I_k \succeq 0$. The assertion $\mathrm{GTZ}(m, k)$ is that every design of size m and rank k over $\mathbb{R}$ admits a dominating k-subset, and $\mathrm{GTZ}_{\mathbb{C}}(m, k)$ is the same assertion over $\mathbb{C}$.

Repeated vectors and the zero vector are admitted, and excluding them would simplify nothing: a k-subset containing two parallel vectors is rank-deficient and cannot dominate, which is the mechanism of Example 2.10. The weights are real even over $\mathbb{C}$, and they enter nowhere but (1); the matrix S_C does not involve them, so any operation that keeps the vectors and moves the weights preserves domination.

Since $g_c g_c^{\mathsf{H}} \succeq 0$ for every c, enlarging C can only increase S_C in the Loewner order, so any superset of a dominating set dominates. Fewer than k vectors span a proper subspace and leave S_C singular, while the full index set dominates in every design (Lemma 2.5). Cardinality k is therefore the smallest at which domination is possible, and by that monotonicity the hardest.

Lemma 2.2 (Lean)**.** *The vectors of a design span $\mathbb{F}^k$. In particular $k \leq m$, and there is no design with $m < k$.*

Proof. Let $u \in \mathbb{F}^k$ satisfy $\langle u, g_c\rangle = 0$ for every c. Pairing (1) with u gives

$$\|u\|^2 \;=\; u^{\mathsf{H}} I_k u \;=\; \sum_c t_c\, u^{\mathsf{H}} g_c g_c^{\mathsf{H}} u \;=\; \sum_c t_c\, |\langle u, g_c\rangle|^2 \;=\; 0,$$

so $u = 0$. The orthogonal complement of the span is trivial, and a spanning family in $\mathbb{F}^k$ has at least k members. $\square$

2.2. Three counting identities.

The first identity is the trace of (1).

Lemma 2.3 (Lean). $\sum_c s_c = \sum_c t_c \ell_c = k$. *Hence* $\max_c \ell_c \geq k$.

Proof. $\operatorname{tr}(g_c g_c^{\mathsf{H}}) = \|g_c\|^2 = \ell_c$, and $\operatorname{tr} I_k = k$; take traces in (1) term by term. The weights are positive and sum to 1, so $k = \sum_c t_c \ell_c$ is a weighted mean of the ℓ_c and does not exceed their maximum. $\square$

Each leverage score is also bounded above.

Lemma 2.4 (Lean). $0 \leq s_c \leq 1$ *for every* c. *With Lemma 2.3 the leverage scores therefore lie in* $[0, 1]$ *and sum to* k, *which gives* $m \geq k$ *a second time, with equality only if* $s_c = 1$ *throughout.*

Proof. Fix e and separate the e-th term of (1):

$$I_k - t_e\, g_e g_e^{\mathsf{H}} \;=\; \sum_{c \neq e} t_c\, g_c g_c^{\mathsf{H}} \;\succeq\; 0. \tag{5}$$

Evaluating the quadratic form of (5) at g_e gives $0 \leq \ell_e - t_e \ell_e^2$. If $\ell_e = 0$ then $s_e = 0 \leq 1$; otherwise divide by $\ell_e > 0$. $\square$

The third identity compares the unweighted sum over all indices with the weighted one.

Lemma 2.5 (Lean). *For every design,*

$$S_{\{1,\dots,m\}} - I_k \;=\; \sum_c (1 - t_c)\, g_c g_c^{\mathsf{H}}. \tag{6}$$

The right-hand side is positive semidefinite, and positive definite as soon as $m \geq 2$.

Proof. Subtract (1) from $\sum_c g_c g_c^{\mathsf{H}}$. Each t_c is one summand of a sum of positive numbers equal to 1, so $t_c \leq 1$ and the coefficients $1 - t_c$ are nonnegative; a nonnegative combination of the positive semidefinite matrices $g_c g_c^{\mathsf{H}}$ is positive semidefinite. If $m \geq 2$ then $t_c < 1$ for every c, since $t_c + t_{c'} \leq 1$ with $t_{c'} > 0$ for any second index c'. Put $\sigma := \min_c (1 - t_c)/t_c > 0$. Every coefficient of

$$\sum_c \big((1 - t_c) - \sigma t_c\big)\, g_c g_c^{\mathsf{H}} \;=\; \sum_c (1 - t_c)\, g_c g_c^{\mathsf{H}} \;-\; \sigma I_k$$

is nonnegative by the choice of σ, the displayed equality being (1) scaled by σ. Hence $\sum_c (1 - t_c) g_c g_c^{\mathsf{H}} \succeq \sigma I_k \succ 0$. $\square$

The operator on the right of (6) returns in §3, where its inverse defines the pivot of a vector and its positive definiteness supplies the whitener behind Theorem 3.14.

2.3. The cases that close by counting.

Proposition 2.6 (Lean). $\mathrm{GTZ}(k, k)$ *and* $\mathrm{GTZ}_{\mathbb{C}}(k, k)$ *hold.*

Proof. By Lemma 2.2 a design of rank k has $m \geq k$ vectors, so $m = k$ leaves exactly one k-subset, namely $\{1, \dots, m\}$, and it dominates by Lemma 2.5. $\square$

Proposition 2.7 (Lean). $\mathrm{GTZ}(m, 1)$ *and* $\mathrm{GTZ}_{\mathbb{C}}(m, 1)$ *hold for every* m.

Proof. At $k = 1$ the matrices in (1) are scalars and the condition reads $\sum_c t_c |g_c|^2 = 1$; this is Lemma 2.3 at $k = 1$. A weighted mean does not exceed its maximum, so $|g_{c_0}|^2 \geq 1$ for some c_0, and the singleton $\{c_0\}$, which has cardinality $k = 1$, dominates. $\square$

That exhausts the counting identities. One further column lies inside the maximal-volume range (4), namely $m = k + 1$; it too reduces to the scalar mechanism of Proposition 2.7, but only after the vectors are traded for the kernel of the associated projection. That change of variables is Theorem 3.19, where the elementary methods stop.

Uniform weights recover the classical statement.

Proposition 2.8 (Lean)**.** *Let $n \geq 1$ and let $A \in \mathbb{F}^{n\times k}$ satisfy $A^{\mathsf{H}}A = I_k$. If* $\mathrm{GTZ}(n,k)$ *holds, respectively* $\mathrm{GTZ}_{\mathbb{C}}(n,k)$*, then there is an injection $\iota : \{1,\dots,k\} \to \{1,\dots,n\}$ with*

$$(A_\iota)^{\mathsf{H}} A_\iota \;\succeq\; \tfrac{1}{n} I_k, \qquad A_\iota := \big(A_{\iota(1)},\dots,A_{\iota(k)}\big)^{\mathsf{H}},$$

where A_r^{H} denotes the r-th row of A.

Proof. Put $g_r := \sqrt{n}\,A_r$ and $t_r := 1/n$. Then $\sum_r t_r g_r g_r^{\mathsf{H}} = \sum_r A_r A_r^{\mathsf{H}} = A^{\mathsf{H}}A = I_k$, so this is a design of size n and rank k. Let C dominate and let ι enumerate C in increasing order. Then $(A_\iota)^{\mathsf{H}} A_\iota = \sum_{r\in C} A_r A_r^{\mathsf{H}} = \frac{1}{n} S_C$, so

$$(A_\iota)^{\mathsf{H}} A_\iota - \tfrac{1}{n} I_k \;=\; \tfrac{1}{n}(S_C - I_k) \;\succeq\; 0. \qquad \square$$

At a fixed (m,k) the implication runs only this way, so the weighted statement is the stronger of the two. Recovering it from the classical one costs a change of size, since a design with rational weights is a uniform design of larger size with repeated vectors (§1).

2.4. **Where counting stops.** Domination can be stated without matrices. A k-subset C dominates if and only if

$$\sum_{c\in C} |\langle x, g_c\rangle|^2 \;\geq\; \|x\|^2 \qquad \text{for every } x \in \mathbb{F}^k, \tag{7}$$

since the left-hand side is the quadratic form of S_C at x. In the same language (1) says that $\sum_c t_c |\langle x, g_c\rangle|^2 = \|x\|^2$ for every x: the squared projections of a fixed direction on the vectors of a design average to the squared length of that direction. The proof of Proposition 2.7 now applies one direction at a time.

Lemma 2.9 (Lean)**.** *For every design and every $x \in \mathbb{F}^k$ there is an index c with $|\langle x, g_c\rangle|^2 \geq \|x\|^2$.*

Proof. The numbers $|\langle x, g_c\rangle|^2$ have weighted mean $\|x\|^2$ by (1), and a weighted mean with positive weights summing to 1 does not exceed the maximum of the numbers averaged. $\square$

At $k = 1$ this is the theorem: there is one direction up to scalars, and the witnessing index may be chosen once and for all. At $k \geq 2$ the witness of Lemma 2.9 depends on x, while (7) demands k indices that work simultaneously for every x on a sphere of dimension $k-1$. Domination is a matrix inequality, and (1) determines only the linear functionals of the family; the least eigenvalue is not one of them.

The trace is one of them. If C dominates then all k eigenvalues of S_C are at least 1, so

$$\operatorname{tr} S_C \;=\; \sum_{c\in C} \ell_c \;\geq\; k. \tag{8}$$

A subset satisfying (8) costs nothing to find: by Lemma 2.3 some $\ell_{c_0} \geq k$, and any k-subset containing c_0 will do. The trace test therefore holds somewhere in every design and distinguishes nothing. In the design below it passes at all twenty triples, and twelve of them do not dominate.

Example 2.10. Let $k = 3$, let the six vectors be $\pm\sqrt{3}\,e_1$, $\pm\sqrt{3}\,e_2$, $\pm\sqrt{3}\,e_3$ and let every weight be $\frac{1}{6}$. Then

$$\sum_c t_c\, g_c g_c^{\mathsf{T}} \;=\; \tfrac{1}{6}\cdot 3\cdot 2 \sum_{i=1}^{3} e_i e_i^{\mathsf{T}} \;=\; I_3,$$

so this is a design at $(6,3)$, with $\ell_c = 3$ and $s_c = \frac{1}{2}$ for every c, in accordance with Lemmas 2.3 and 2.4. Every one of the twenty triples satisfies $\operatorname{tr} S_C = 9 \geq 3$. Eight of them take one vector from each axis pair and have $S_C = 3I_3 \succ I_3$; the remaining twelve repeat an axis, as $\{\sqrt{3}e_1, -\sqrt{3}e_1, \sqrt{3}e_2\}$ does with $S_C = 6\,e_1 e_1^{\mathsf{T}} + 3\,e_2 e_2^{\mathsf{T}}$, and are singular in the direction they omit. The trace is blind to which direction that is.

Selection is therefore a combinatorial problem about the directions a k-subset leaves uncovered, constrained by the metric identity (1). The remaining sections develop two instruments for it: a change of coordinates turning (2) into a statement about principal blocks of a projection (§3), and, where that does not decide the cell, a variational analysis of the worst design (§7 and §8).

2.5. **The tetrahedron.** In Example 2.10 eight triples dominate strictly. In the next design every triple dominates and none dominates strictly.

Example 2.11 (Lean)**.** Let $k = 3$ and take the four vectors

$$g_0 = (1,1,1), \quad g_1 = (1,-1,-1), \quad g_2 = (-1,1,-1), \quad g_3 = (-1,-1,1)$$

with equal weights $t_c = \frac{1}{4}$. Each diagonal entry of $\sum_c g_c g_c^{\mathsf{T}}$ is 4, and each off-diagonal entry is $1 - 1 - 1 + 1 = 0$, so

$$\sum_{c=0}^{3} g_c g_c^{\mathsf{T}} \;=\; 4I_3, \qquad \text{hence} \qquad \sum_{c=0}^{3} \tfrac{1}{4}\, g_c g_c^{\mathsf{T}} \;=\; I_3: \tag{9}$$

a design at $(4,3)$. Here $\ell_c = 3$ and $s_c = \frac{3}{4}$ for every c, so Lemma 2.3 holds with equal terms, $4 \cdot \frac{3}{4} = 3 = k$, and $\langle g_i, g_j \rangle = -1$ for $i \neq j$. The four vectors point to the vertices of a regular tetrahedron and each has length $\sqrt{3}$.

Let $C = \{0,1,2,3\} \setminus \{c_0\}$ be one of the four triples. Subtracting the omitted term from (9),

$$S_C \;=\; 4I_3 - g_{c_0} g_{c_0}^{\mathsf{T}}.$$

The matrix $g_{c_0} g_{c_0}^{\mathsf{T}}$ has eigenvalue $\ell_{c_0} = 3$ on the line $\mathbb{R} g_{c_0}$ and 0 on its orthogonal complement, so

$$\operatorname{spec} S_C = \{1,4,4\}, \qquad \lambda_{\min}(S_C) = 1. \tag{10}$$

For $c_0 = 0$ the certificate is a sum of squares: writing $x = (x_1, x_2, x_3)$,

$$x^{\mathsf{T}}(S_C - I_3)x \;=\; 3\|x\|^2 - \langle x, g_0\rangle^2 \;=\; 3\sum_i x_i^2 - \Big(\sum_i x_i\Big)^2 \;=\; \sum_{i<j} (x_i - x_j)^2,$$

which vanishes exactly on $\mathbb{R} g_0$, the direction omitted from C.

The tetrahedron is the smallest design at rank three with $\lambda_{\min}(S_C) = 1$ at every triple, and it recurs throughout what follows. It is the equality case of the corank-one classification: the tie law (21) of §3 reads $3 \cdot \frac{1}{4} \cdot 3 = \frac{9}{4} = 3 - 1 + \frac{1}{4}$ here. Splitting its vectors produces extremal designs at every size $m \geq 4$ and rank 3, and that is the route by which the no-margin principle of §6 reaches the open cells. Its volume-sampling mixture is $(x-1)(x-4)^2$ (§6), whose least root sits at the threshold. It also calibrates the first-order system of §8, where $\Lambda = \frac{1}{3} I_3$ and $\nu = 1$ with each of the four triples carrying multiplier $\frac{1}{4}$.

3. The projection chart

Absorbing the weights into the vectors replaces a design by a projection on the index space, and replaces the domination condition (2) by a comparison between a principal block of that projection and a diagonal matrix. Every argument in the sequel is carried out in these coordinates.

Throughout, $k \geq 1$ and $m \geq k$; the second inequality is Lemma 2.2. For an index set C of size r, listed in increasing order, M_C denotes the $r \times k$ matrix whose rows are g_c^{H}, $c \in C$, and

$$\mathrm{Gram}_C \;:=\; M_C M_C^{\mathsf{H}} \;=\; \big(\langle g_c, g_d\rangle\big)_{c,d \in C}$$

their Gram matrix. The two products of M_C carry different information:

$$M_C^{\mathsf{H}} M_C \;=\; \sum_{c \in C} g_c g_c^{\mathsf{H}} \;=\; S_C \quad (k \times k), \qquad M_C M_C^{\mathsf{H}} \;=\; \mathrm{Gram}_C \quad (r \times r). \tag{11}$$

Both identities are the definition of matrix multiplication read in the two possible orders.

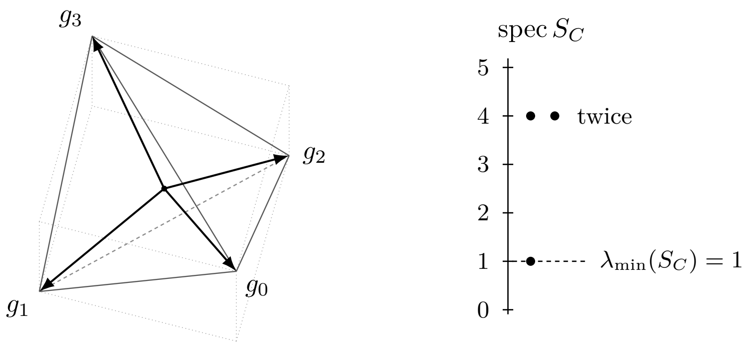


FIGURE 2. The design of Example 2.11: four vectors to the vertices of a regular tetrahedron, drawn inside the cube whose even-sign vertices they are. Every triple omits one vector and has spectrum $\{1, 4, 4\}$ by (10), so it dominates with nothing to spare.

3.1. **Construction.** Put

$$v_c \;:=\; \sqrt{t_c}\, g_c \in \mathbb{F}^k,$$

and let V be the $m \times k$ matrix whose c-th row is v_c^{H}; thus $V = \operatorname{diag}(\sqrt{t_1}, \ldots, \sqrt{t_m})\, M_{\{1,\ldots,m\}}$. The Parseval condition (1) is exactly the orthonormality of the columns of V:

$$V^{\mathsf{H}} V \;=\; \sum_c v_c v_c^{\mathsf{H}} \;=\; \sum_c t_c\, g_c g_c^{\mathsf{H}} \;=\; I_k. \tag{12}$$

Accordingly

$$P \;:=\; V V^{\mathsf{H}} \tag{13}$$

is a matrix on the *index* space $\mathbb{F}^m$. Write $D := \operatorname{diag}(t_1, \ldots, t_m)$ and

$$W \;:=\; P - D. \tag{14}$$

The pair (P, t) is the *chart* of the design and W its *gap*.

Proposition 3.1 (Lean)**.** *P is the orthogonal projection onto the column space of V, of rank k, and*

$$P_{cd} = \sqrt{t_c t_d}\, \langle g_c, g_d \rangle, \qquad P_{cc} = s_c, \qquad \operatorname{tr} P = k.$$

Consequently $W_{cc} = t_c(\ell_c - 1)$ and $\sum_c W_{cc} = k - 1$.

Proof. $P^{\mathsf{H}} = (VV^{\mathsf{H}})^{\mathsf{H}} = VV^{\mathsf{H}} = P$, and, using (12) once, $P^2 = V(V^{\mathsf{H}}V)V^{\mathsf{H}} = VIV^{\mathsf{H}} = P$. A Hermitian idempotent is the orthogonal projection onto its range, which is the column space of V because $V^{\mathsf{H}}V = I_k$ forces V to have rank k. The entries are $P_{cd} = \langle v_c, v_d \rangle = \sqrt{t_c t_d} \langle g_c, g_d \rangle$; at $c = d$ this is $t_c \ell_c = s_c$. Finally $\operatorname{tr} P = \operatorname{tr}(VV^{\mathsf{H}}) = \operatorname{tr}(V^{\mathsf{H}}V) = \operatorname{tr} I_k = k$. The diagonal of W is $s_c - t_c = t_c(\ell_c - 1)$, and $\sum_c (s_c - t_c) = k - 1$ by Lemma 2.3. ☐

Lemma 3.2. *Fix m, k and a weight vector t with $t_c > 0$ and $\sum_c t_c = 1$. The map sending a design with weight vector t to its chart P is a bijection from the set of such designs modulo the action $g_c \mapsto U g_c$ of the unitary group of $\mathbb{F}^k$ onto the set of orthogonal projections of rank k on $\mathbb{F}^m$.*

Proof. Let P be an orthogonal projection of rank k and let V be the $m \times k$ matrix whose columns are an orthonormal basis of the range of P; then $V^{\mathsf{H}}V = I_k$ and $VV^{\mathsf{H}} = P$. Writing v_c^{H} for the c-th row of V and setting $g_c := t_c^{-1/2} v_c$ produces a family with $\sum_c t_c g_c g_c^{\mathsf{H}} = \sum_c v_c v_c^{\mathsf{H}} = V^{\mathsf{H}}V = I_k$, a design with weight vector t and chart P. If two designs with weight vector t have the same chart, their matrices V and V' have the same column space and orthonormal columns, so $V' = VU$ for a unitary U of $\mathbb{F}^k$, whence $g_c' = U^{\mathsf{H}} g_c$; conversely replacing g_c by $U g_c$ replaces V by VU^{H} and leaves $P = VV^{\mathsf{H}}$ unchanged. ☐

Remark 3.3. In frame language (12) says that $(v_c)_{c\le m}$ is a Parseval frame for $\mathbb{F}^k$ and P is the associated projection of the ambient $\mathbb{F}^m$ onto it [5]. The classical problem is the uniform slice: for $t_c \equiv 1/n$ and $g_c := \sqrt{n}\, a_c$, where a_c^{H} is the c-th row of a matrix A with $A^{\mathsf{H}}A = I_k$, the factor $\sqrt{n}$ in the vectors cancels the factor $1/n$ in the weights and $P = AA^{\mathsf{H}}$; domination of C reads $\sum_{c\in C} a_c a_c^{\mathsf{H}} \succeq n^{-1}I_k$, which is the row-selection problem of [11]. In that slice $D = n^{-1}I$ and the chart criterion below is the condition $\lambda_{\min}(P[C]) \ge 1/n$ used in [20, 22]. The general weight vector is new: D is then a genuine diagonal, and the comparison is Loewner rather than scalar.

3.2. **The criterion.** The passage from S_C to the block $W[C]$ is a congruence by a positive diagonal followed by an exchange of the two products in (11). We record the two steps separately: the first holds at every selection size, the second only at $|C| = k$.

Lemma 3.4 (Lean)**.** *For every index set C of size r,*

$$W[C] \;=\; \operatorname{diag}(\sqrt{t_c})_{c\in C}\, \left(\mathrm{Gram}_C - I_r\right)\, \operatorname{diag}(\sqrt{t_c})_{c\in C}, \tag{15}$$

and therefore $W[C] \succeq 0$ if and only if $\mathrm{Gram}_C \succeq I_r$.

Proof. Entrywise, for $c, d \in C$,

$$W[C]_{cd} \;=\; P_{cd} - t_c\delta_{cd} \;=\; \sqrt{t_c t_d}\,\langle g_c, g_d\rangle - \sqrt{t_c t_d}\,\delta_{cd} \;=\; \sqrt{t_c t_d}\,\left(\mathrm{Gram}_C - I_r\right)_{cd},$$

where the middle step uses $\sqrt{t_c t_d}\,\delta_{cd} = t_c\delta_{cd}$. This is (15). The matrix $B := \operatorname{diag}(\sqrt{t_c})_{c\in C}$ is invertible because every $t_c > 0$, and for invertible B one has $X \succeq 0$ if and only if $B^{\mathsf{H}}XB \succeq 0$: the substitutions $x \mapsto Bx$ and $x \mapsto B^{-1}x$ carry each quadratic form to the other. □

Lemma 3.5 (Lean)**.** *Let X be a matrix over $\mathbb{F}$, of any shape. Then*

$$I - X^{\mathsf{H}}X \succeq 0 \quad\Longleftrightarrow\quad I - XX^{\mathsf{H}} \succeq 0.$$

If M is square, then $M^{\mathsf{H}}M \succeq I$ if and only if $MM^{\mathsf{H}} \succeq I$.

Proof. The first statement is symmetric in X and X^{H}, so one implication suffices. Assume $\|Xu\| \le \|u\|$ for every u, which is $I - X^{\mathsf{H}}X \succeq 0$. For any w, by Cauchy–Schwarz and then the hypothesis at $u = X^{\mathsf{H}}w$,

$$\|X^{\mathsf{H}}w\|^4 \;=\; \langle w, XX^{\mathsf{H}}w\rangle^2 \;\le\; \|w\|^2\,\|X(X^{\mathsf{H}}w)\|^2 \;\le\; \|w\|^2\,\|X^{\mathsf{H}}w\|^2.$$

If $X^{\mathsf{H}}w \ne 0$ divide by $\|X^{\mathsf{H}}w\|^2$; in either case $\|X^{\mathsf{H}}w\| \le \|w\|$, which is $I - XX^{\mathsf{H}} \succeq 0$.

For the second statement let M be $n \times n$ with $M^{\mathsf{H}}M \succeq I$, that is $\|Mu\| \ge \|u\|$ for every u. Then $Mu = 0$ forces $u = 0$, so M is invertible. Put $X := M^{-1}$ and substitute $u = Xw$: $\|w\| \ge \|Xw\|$, so $I - X^{\mathsf{H}}X \succeq 0$. By the first statement $I - XX^{\mathsf{H}} \succeq 0$, that is $\|(M^{\mathsf{H}})^{-1}w\| \le \|w\|$ for every w; substituting $w = M^{\mathsf{H}}u$ gives $\|u\| \le \|M^{\mathsf{H}}u\|$, which is $MM^{\mathsf{H}} \succeq I$. The converse is the same argument with M and M^{H} exchanged. □

Lemma 3.6 (Lean)**.** *For every k-subset C,*

$$S_C \succeq I_k \quad\Longleftrightarrow\quad W[C] \succeq 0.$$

Proof. For $|C| = k$ the matrix M_C is square, so Lemma 3.5 applies to (11) and gives $S_C = M_C^{\mathsf{H}}M_C \succeq I_k$ if and only if $\mathrm{Gram}_C = M_C M_C^{\mathsf{H}} \succeq I_k$. By Lemma 3.4 the latter is $W[C] \succeq 0$. □

The problem therefore reads: *some $k \times k$ principal block of a rank-k orthogonal projection dominates the corresponding block of a positive diagonal matrix of trace one.* The weights enter only through D, so the projection and the weights may be varied independently, and $W[C] \succeq 0$ is a closed condition in both arguments. The compactification of §7 rests on both facts.

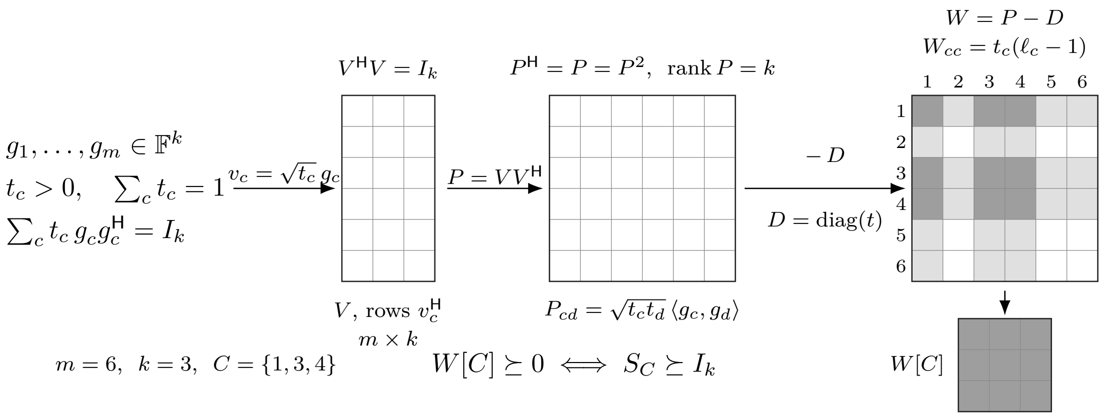


FIGURE 3. The chart, at $m = 6$, $k = 3$ and $C = \{1, 3, 4\}$. The design data pass through the isometry V of (12) to the projection $P = VV^{\mathsf{H}}$ of (13) and the gap $W = P - D$ of (14). Selecting C is taking a principal block, and Lemma 3.6 identifies its semidefiniteness with domination.

Remark 3.7. The equivalence of Lemma 3.6 depends on the hypothesis $|C| = k$. Domination is the $k \times k$ statement $M_C^{\mathsf{H}} M_C \succeq I_k$, while the block condition is the $r \times r$ statement $M_C M_C^{\mathsf{H}} \succeq I_r$ with $r = |C|$. For $r < k$ the left-hand matrix has rank at most $r < k$ and can never dominate I_k, while the block condition is unconstrained; for $r > k$ the block matrix has rank at most $k < r$ and never dominates I_r, while S_C may. Both failures are rank effects, and $r = k$ is the only size at which the equivalence has content in both directions.

Taking determinants in (15) converts the criterion into an exact scalar test, one principal minor at a time.

Corollary 3.8 (Lean)**.** *For every index set C,*

$$\det W[C] \;=\; \Big(\prod_{c \in C} t_c\Big)\, \det(\operatorname{Gram}_C - I).$$

The factor is strictly positive, so $W[C]$ and $\operatorname{Gram}_C - I$ have the same principal minors up to positive factors and are semidefinite together, minor by minor. Over the rationals the minors decide the question by exact arithmetic, with no eigenvalue computation and no genericity assumption.

3.3. **Inertia, and what it does not decide.** The gap W has a fixed signature, which by itself decides nothing.

Lemma 3.9 (Lean)**.** *Let $m \geq 2$. If $Px = x$ then $\langle x, Wx \rangle = \sum_c (1 - t_c)|x_c|^2$, which is positive for $x \neq 0$. If $Px = 0$ then $\langle x, Wx \rangle = -\sum_c t_c |x_c|^2$, which is negative for $x \neq 0$.*

Proof. $\langle x, Dx \rangle = \sum_c t_c |x_c|^2$ in both cases. If $Px = x$ then $\langle x, Px \rangle = \|x\|^2 = \sum_c |x_c|^2$, and subtracting gives the first identity; each coefficient $1 - t_c$ is strictly positive because $m \geq 2$ forces $t_c < 1$. If $Px = 0$ then $\langle x, Px \rangle = 0$ and the second identity follows; each t_c is strictly positive. □

Lemma 3.10. *For $m \geq 2$ the matrix W has inertia $(k,\, m - k,\, 0)$.*

Proof. The subspaces $\operatorname{ran} P$ and $\ker P$ are complementary of dimensions k and $m - k$, and by Lemma 3.9 the form $\langle x, Wx \rangle$ is positive definite on the first and negative definite on the second. Sylvester's law of inertia [16, §4.5] gives the count. □

The hypothesis $m \geq 2$ is necessary: at $m = k = 1$ the only design has $t_1 = 1$ and $\ell_1 = 1$, so $P = D = 1$ and $W = 0$, of inertia $(0, 0, 1)$.

Remark 3.11 (Lean). A positive semidefinite matrix has a nonnegative diagonal. Hence a Hermitian matrix whose diagonal entries are all negative has no positive semidefinite principal submatrix of any nonempty size. Such matrices realize the inertia of Lemma 3.10 at the two smallest cells. At $(m,k)=(2,1)$ take

$$B_2 = \begin{pmatrix} -1 & 3 \\ 3 & -1 \end{pmatrix}, \qquad \det B_2 = 1-9=-8.$$

Its form is $\langle x, B_2x\rangle = -x_1^2 - x_2^2 + 6x_1x_2$, which takes the value 4 at $(1,1)$ and -8 at $(1,-1)$: the inertia is $(1,1,0)$, which is the inertia of a chart gap at $(2,1)$. At $(m,k)=(3,2)$ take

$$B_3 = \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & -2 \\ 2 & -2 & -1 \end{pmatrix}, \qquad \det B_3 = -5.$$

On the plane spanned by $(1,1,0)$ and $(1,0,1)$, parameterized as $x=(a+b,a,b)$,

$$\langle x, B_3x\rangle \;=\; -(a+b)^2 - a^2 - b^2 + 4a(a+b) + 4b(a+b) - 4ab \;=\; 2a^2+2ab+2b^2,$$

positive definite there; so B_3 has at least two positive eigenvalues, and $\det B_3 < 0$ excludes three, leaving inertia $(2,1,0)$, the inertia of a chart gap at $(3,2)$. Every diagonal entry of B_2 and of B_3 is -1, so neither matrix has a positive semidefinite principal block of the required size. No statement of the form "a Hermitian matrix of inertia $(k, m-k, 0)$ possesses a positive semidefinite $k\times k$ principal block" is therefore available, and no proof of GTZ can run on the signature of W alone.

A chart gap differs from B_2 and B_3 in its diagonal: by Proposition 3.1 it has $W_{cc} = t_c(\ell_c - 1) \geq -t_c > -1$ at every index and $\sum_c W_{cc} = k-1 \geq 0$, so at least one diagonal entry is nonnegative.

3.4. **Duality.** Since $I-P$ is the orthogonal projection onto $\ker P$, selecting a block of P and deleting the complementary block of $I-P$ are the same operation. With the correct normalization that operation turns a design of rank k into a design of rank $m-k$, exchanging domination for domination at the complement.

Fix a design at (m,k) with $k\geq 1$ and $m \geq k+1$, and put $r := m-k \geq 1$; then $m\geq 2$ and every $t_c < 1$. Choose an $m\times r$ matrix Z whose columns are an orthonormal basis of $\ker P$, so that

$$Z^{\mathsf{H}}Z = I_r, \qquad ZZ^{\mathsf{H}} = I_m - P,$$

and write z_c^{H} for the c-th row of Z. As in (12) and Proposition 3.1,

$$\sum_c z_c z_c^{\mathsf{H}} = Z^{\mathsf{H}}Z = I_r, \qquad \|z_c\|^2 = (ZZ^{\mathsf{H}})_{cc} = 1 - P_{cc} = 1 - s_c.$$

Rescale by the co-weights: put

$$w_c \;:=\; \frac{z_c}{\sqrt{1-t_c}} \in \mathbb{F}^r,$$

which is legitimate because $t_c < 1$. The first display becomes

$$\sum_c (1-t_c)\, w_c w_c^{\mathsf{H}} \;=\; I_r. \tag{16}$$

This is not a Parseval condition, since the coefficients $1-t_c$ sum to $m-1$ rather than to 1. The correct rescaling is by the operator

$$S^{\perp} \;:=\; \sum_c t_c\, w_c w_c^{\mathsf{H}}. \tag{17}$$

Lemma 3.12. *$S^{\perp} \succ 0$.*

Proof. $S^{\perp} \succeq 0$ as a nonnegative combination of rank-one positive semidefinite matrices. If $S^{\perp}u = 0$ then $\langle u, S^{\perp}u\rangle = \sum_c t_c |\langle w_c, u\rangle|^2 = 0$, so $\langle w_c, u\rangle = 0$ for every c because every $t_c > 0$; then (16) applied to u gives $u = \sum_c (1-t_c)\langle w_c, u\rangle\, w_c = 0$. ☐

Lemma 3.13 (Lean)**.** *Every $A \succ 0$ admits an invertible R with $R^{\mathsf{H}}AR = I$.*

Proof. Write $A = LL^{\mathsf{H}}$ with L invertible, by Cholesky factorization, and put $R := (L^{\mathsf{H}})^{-1}$. Then $R^{\mathsf{H}}AR = L^{-1}LL^{\mathsf{H}}(L^{\mathsf{H}})^{-1} = I$. □

Theorem 3.14 (Lean)**.** *Let $k \geq 1$ and $m \geq k+1$, and let $\mathcal{D}$ be a design at (m,k) over $\mathbb{F}$ with weight vector t. With w_c, $S^{\perp}$ and R as above, the vectors*

$$h_c \;:=\; R^{\mathsf{H}}w_c \in \mathbb{F}^{m-k}$$

with the same *weights t_c form a design $\mathcal{D}^{\perp}$ at $(m, m-k)$, and for every k-subset C,*

$$C \text{ dominates in } \mathcal{D} \quad\Longleftrightarrow\quad C^{\complement} \text{ dominates in } \mathcal{D}^{\perp}.$$

Proof. The weights are unchanged, so they are positive and sum to one, and

$$\sum_c t_c\, h_c h_c^{\mathsf{H}} \;=\; R^{\mathsf{H}}\Big(\sum_c t_c\, w_c w_c^{\mathsf{H}}\Big)R \;=\; R^{\mathsf{H}}S^{\perp}R \;=\; I_{m-k}$$

by (17) and Lemma 3.13; here the conjugation identity $R^{\mathsf{H}}(ww^{\mathsf{H}})R = (R^{\mathsf{H}}w)(R^{\mathsf{H}}w)^{\mathsf{H}}$ is used termwise. So $\mathcal{D}^{\perp}$ is a design at $(m, m-k)$.

Fix C with $|C| = k$ and put $E := C^{\complement}$, of size $r = m-k$. Let N be the $k \times r$ matrix whose rows are w_c^{H}, $c \in C$; equivalently $N = \operatorname{diag}\big((1-t_c)^{-1/2}\big)_{c\in C}\, Z_C$, where Z_C is the submatrix of Z on the rows C. Then, as in (11),

$$NN^{\mathsf{H}} = \big(\langle w_c, w_d\rangle\big)_{c,d\in C}, \qquad N^{\mathsf{H}}N = \sum_{c\in C} w_c w_c^{\mathsf{H}}.$$

Now run the equivalences. Since $P = I - ZZ^{\mathsf{H}}$,

$$W[C] \;=\; \operatorname{diag}(1-t_c)_{c\in C} - (ZZ^{\mathsf{H}})[C],$$

and $(ZZ^{\mathsf{H}})[C] = Z_C Z_C^{\mathsf{H}}$. Conjugating by the invertible diagonal $\operatorname{diag}\big((1-t_c)^{-1/2}\big)_{c\in C}$ as in the proof of Lemma 3.4,

$$W[C] \succeq 0 \quad\Longleftrightarrow\quad I_k - NN^{\mathsf{H}} \succeq 0 \quad\Longleftrightarrow\quad I_r - N^{\mathsf{H}}N \succeq 0,$$

the second step by Lemma 3.5 applied to $X := N^{\mathsf{H}}$. By (16) the right-hand condition reads

$$\sum_{c\in C} w_c w_c^{\mathsf{H}} \;\preceq\; \sum_c (1-t_c)\, w_c w_c^{\mathsf{H}}.$$

Subtracting both sides from $\sum_c w_c w_c^{\mathsf{H}}$ turns this into

$$\sum_{c\in E} w_c w_c^{\mathsf{H}} \;\succeq\; \sum_c t_c\, w_c w_c^{\mathsf{H}} \;=\; S^{\perp},$$

and conjugating by the invertible R turns it into $\sum_{c\in E} h_c h_c^{\mathsf{H}} \succeq R^{\mathsf{H}}S^{\perp}R = I_{m-k}$, which is domination of E in $\mathcal{D}^{\perp}$. Every step is an equivalence, and $|E| = m-k$ is the rank of $\mathcal{D}^{\perp}$, so Lemma 3.6 applies at both ends. □

Corollary 3.15 (Lean)**.** *For $k \geq 1$ and $m \geq k+1$,* $\mathrm{GTZ}(m,k) \Leftrightarrow \mathrm{GTZ}(m, m-k)$*, and likewise over $\mathbb{C}$.*

Proof. If every design at $(m, m-k)$ has a dominating $(m-k)$-subset, apply this to $\mathcal{D}^{\perp}$ and complement the resulting subset. The construction is symmetric in the two cells. □

The dual design is not canonical: Z is determined up to a unitary of $\mathbb{F}^r$ and R up to a unitary factor, and both ambiguities act on the h_c by a common unitary, which changes no domination. The normalization, by contrast, is forced.

Remark 3.16. Rescaling the kernel rows by the weights rather than the co-weights, so that the dual vectors are $u_c := z_c/\sqrt{t_c}$, also produces a design at $(m, m-k)$, since $\sum_c t_c u_c u_c^{\mathsf{H}} = \sum_c z_c z_c^{\mathsf{H}} = I_{m-k}$, and it is the construction one writes down first. It does not satisfy the conclusion of Theorem 3.14. At uniform weights the two agree: for $t_c \equiv 1/m$ one has $S^\perp = \frac{1}{m-1} I$ by (16), hence $R = \sqrt{m-1}\, I$ and $h_c = \sqrt{m-1}\, w_c = \sqrt{m}\, z_c = u_c$. At non-uniform weights they do not.

Example 3.17. Take $m = 2$, $k = 1$, $t = (\frac{1}{5}, \frac{4}{5})$ and $g = (2, \frac{1}{2})$; the Parseval condition holds, $\frac{1}{5} \cdot 4 + \frac{4}{5} \cdot \frac{1}{4} = 1$. Here $v = (2/\sqrt{5},\ 1/\sqrt{5})$, of norm one, so $\ker P$ is spanned by the unit vector $z = (-1/\sqrt{5},\ 2/\sqrt{5})$, giving $|z_1|^2 = \frac{1}{5}$ and $|z_2|^2 = \frac{4}{5}$. The singleton $\{1\}$ dominates, $g_1^2 = 4 \geq 1$, and $\{2\}$ does not, $g_2^2 = \frac{1}{4} < 1$.

The correct dual: $w_1^2 = \frac{1/5}{4/5} = \frac{1}{4}$, $w_2^2 = \frac{4/5}{1/5} = 4$, so $S^\perp = \frac{1}{5} \cdot \frac{1}{4} + \frac{4}{5} \cdot 4 = \frac{13}{4}$ and $h_c^2 = w_c^2/S^\perp$, that is $h_1^2 = \frac{1}{13}$ and $h_2^2 = \frac{16}{13}$. Parseval holds, $\frac{1}{5} \cdot \frac{1}{13} + \frac{4}{5} \cdot \frac{16}{13} = 1$, and the correspondence is exact: $\{2\}$ dominates in $\mathcal{D}^\perp$ because $\frac{16}{13} \geq 1$, matching $\{1\}$ upstairs, and $\{1\}$ fails downstairs because $\frac{1}{13} < 1$, matching $\{2\}$ upstairs.

The naive dual of Remark 3.16: $u_1^2 = \frac{1/5}{1/5} = 1$ and $u_2^2 = \frac{4/5}{4/5} = 1$. Both singletons dominate, so the correspondence fails at $C = \{2\}$, which does not dominate upstairs while its complement does downstairs.

3.5. **Corank one.** Let $m = k + 1$ with $k \geq 1$. Then $\ker P$ is a line, spanned by a unit vector $z \in \mathbb{F}^m$, and $P = I - zz^{\mathsf{H}}$. Set

$$p_c \;:=\; \frac{|z_c|^2}{1 - t_c} \;=\; \frac{1 - t_c \ell_c}{1 - t_c}, \tag{18}$$

the second expression by $|z_c|^2 = 1 - P_{cc} = 1 - s_c$ (Proposition 3.1). Since z is a unit vector,

$$\sum_c p_c\,(1 - t_c) \;=\; \sum_c |z_c|^2 \;=\; 1, \qquad \text{while} \qquad \sum_c (1 - t_c) \;=\; m - 1 \;=\; k. \tag{19}$$

The p_c therefore have co-weighted mean $1/k$. Domination compares a positive diagonal with a rank-one perturbation of it, and one scalar decides such a comparison.

Lemma 3.18 (Lean)**.** *Let $A \succ 0$ and let b be a vector, and put $q := b^{\mathsf{H}} A^{-1} b$. Then*

$$A - bb^{\mathsf{H}} \succeq 0 \iff q \leq 1, \qquad A - bb^{\mathsf{H}} \succ 0 \iff q < 1.$$

Proof. Put $y := A^{-1} b$, so that $Ay = b$ and $q = \langle y, Ay \rangle \geq 0$. The form of the perturbed matrix is $\langle x, (A - bb^{\mathsf{H}})x \rangle = \langle x, Ax \rangle - |\langle b, x \rangle|^2$.

Suppose $q \leq 1$. The form $\langle \cdot, A \cdot \rangle$ is an inner product, and $\langle b, x \rangle = \langle Ay, x \rangle = \langle y, Ax \rangle$ is the A-inner product of y and x; Cauchy–Schwarz in it gives $|\langle b, x \rangle|^2 \leq q\, \langle x, Ax \rangle$. Hence $\langle x, (A - bb^{\mathsf{H}})x \rangle \geq (1 - q)\langle x, Ax \rangle$, which is nonnegative, and strictly positive for $x \neq 0$ when $q < 1$.

Conversely, evaluate at $x = y$: the form equals $q - q^2 = q(1 - q)$. If $A - bb^{\mathsf{H}} \succeq 0$ this is nonnegative, which forces $q \leq 1$ since $q \geq 0$. If moreover $q = 1$ then $y \neq 0$, because $\langle y, Ay \rangle = 1$, and the form vanishes at y, so $A - bb^{\mathsf{H}}$ is not positive definite. □

Theorem 3.19 (Lean)**.** *Let $m = k + 1$, $k \geq 1$, over either field.*

(i) *The complement of $\{c_0\}$ dominates if and only if*

$$p_{c_0} \;\geq\; \sum_c t_c\, p_c, \tag{20}$$

and it dominates strictly if and only if the inequality is strict.

(ii) $\mathrm{GTZ}(k+1, k)$ *and* $\mathrm{GTZ}_{\mathbb{C}}(k+1, k)$ *hold.*

(iii) *Equality holds in* (20) *for every* c_0 *if and only if* $p_c \equiv 1/k$*, if and only if*

$$k\, t_c\, \ell_c \;=\; k - 1 + t_c \qquad (1 \le c \le m), \tag{21}$$

and in that case every k*-subset dominates and none dominates strictly.*

(iv) *For every weight vector* t *on* $k+1$ *atoms there is a design with weights exactly* t *satisfying* (21)*.*

Proof. (i) Put $C := \{1, \dots, m\} \setminus \{c_0\}$, of size k. By Lemma 3.6 the subset C dominates precisely when $W[C] = (I - zz^{\mathsf{H}} - D)[C] \succeq 0$, that is when $A - bb^{\mathsf{H}} \succeq 0$ with $A := \operatorname{diag}(1 - t_c)_{c \in C}$ and $b := (z_c)_{c \in C}$. Every $t_c < 1$, so $A \succ 0$ and $A^{-1} = \operatorname{diag}\big((1 - t_c)^{-1}\big)_{c \in C}$, whence

$$q \;=\; b^{\mathsf{H}} A^{-1} b \;=\; \sum_{c \ne c_0} \frac{|z_c|^2}{1 - t_c} \;=\; \sum_{c \ne c_0} p_c \;=\; \sum_c p_c - p_{c_0}.$$

Lemma 3.18 turns domination into $\sum_c p_c - p_{c_0} \le 1$, and strict domination into the strict inequality. Expanding (19) as $\sum_c p_c - \sum_c t_c p_c = 1$ and substituting yields (20).

(ii) The numbers t_c are nonnegative and sum to one, so the weighted mean $\sum_c t_c p_c$ does not exceed $\max_c p_c$. Any index c_0 attaining the maximum satisfies (20), and its complement is a dominating k-subset. No step used the field.

(iii) Equality at every c_0 says that every p_{c_0} equals the fixed number $\sum_c t_c p_c$, so p is constant, say $p_c \equiv \bar p$; then (19) gives $\bar p k = 1$. Conversely a constant p makes both sides of (20) equal. For the second equivalence, by (18),

$$p_c = \frac{1}{k} \iff k\,(1 - t_c \ell_c) = 1 - t_c \iff k\, t_c \ell_c = k - 1 + t_c,$$

which is (21); equivalently $\ell_c = (k - 1 + t_c)/(k\, t_c)$. At corank one every k-subset is the complement of a singleton, so by (i) equality everywhere means that every k-subset dominates and none does so strictly.

(iv) Given t, the system (21) prescribes $|z_c|^2 = (1 - t_c)/k$, and these numbers sum to $k/k = 1$ by (19); so put

$$z_c \;:=\; \sqrt{\frac{1 - t_c}{k}} \;>\; 0.$$

It remains to realize $P = I - zz^{\mathsf{H}}$ by a design with weight vector t, which is Lemma 3.2; explicitly, let e_1 be the first standard basis vector and $u := z - e_1$, nonzero because $z_1^2 = (1 - t_1)/k < 1$, and put

$$Q \;:=\; I - \frac{2\, uu^{\mathsf{H}}}{\|u\|^2}, \qquad \|u\|^2 = 2(1 - z_1),$$

the Householder reflection. It is Hermitian and involutive, hence unitary, and $He_1 = e_1 + u = z$ because $u^{\mathsf{H}} e_1 = z_1 - 1$. Its remaining k columns are therefore an orthonormal basis of $z^{\perp}$; let V be the $m \times k$ matrix formed by them and $g_c := t_c^{-1/2} v_c$ as in Lemma 3.2. Then $V^{\mathsf{H}} V = I_k$ and $VV^{\mathsf{H}} = HH^{\mathsf{H}} - zz^{\mathsf{H}} = I - zz^{\mathsf{H}}$, so the design so obtained has weight vector t and chart $P = I - zz^{\mathsf{H}}$, whose diagonal gives $t_c \ell_c = 1 - (1 - t_c)/k$, that is (21). □

No step of the argument sees the field, so the complex collapse of §4 must wait one size longer. Part (iv) produces a family of designs attaining the threshold with no slack, one over each point of the open weight simplex; §6 draws the consequence.

Example 3.20. The tetrahedron of Example 2.11 is the uniform member at $k = 3$. There $t_c = \frac14$ and (21) reads $3 \cdot \frac14 \cdot \ell_c = 3 - 1 + \frac14 = \frac94$, so $\ell_c = 3$, as computed there. The kernel vector is $z = (\frac12, \frac12, \frac12, \frac12)$, since $|z_c|^2 = (1 - \frac14)/3 = \frac14$, and the chart is

$$P \;=\; I_4 - \tfrac14 J_4, \qquad P_{cc} = \tfrac34 = s_c, \qquad P_{cd} = -\tfrac14 = \tfrac14 \langle g_c, g_d \rangle \ (c \ne d),$$

where J_4 is the all-ones matrix; $P^2 = I - \frac{1}{2}J + \frac{1}{16}J^2 = I - \frac{1}{2}J + \frac{1}{4}J = P$ and $\operatorname{tr} P = 3$. Every p_c equals $\frac{1/4}{3/4} = \frac{1}{3} = 1/k$, so (20) holds with equality at every c_0; by part (iii) every triple dominates and none dominates strictly, which is the spectrum $\{1, 4, 4\}$ recorded in Example 2.11.

Remark 3.21. At $k = 2$ the law (21) reads $\ell_c = \frac{1}{2} + \frac{1}{2t_c}$. Its uniform-weight shadow is classical: for $t_c = p/n$ the right-hand side is $\frac{1}{2} + \frac{n}{2p}$, and these are the magnitudes appearing in the equality description of [24, Proposition 1], whose three clusters of sizes $p + q + r = n$ correspond to the three vectors of a corank-one design with weights p/n, q/n, r/n.

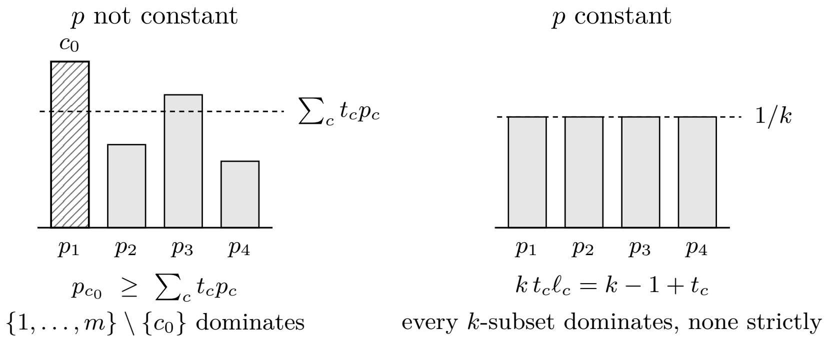


FIGURE 4. Theorem 3.19 at $(4, 3)$. The pivots p_c of (18) have t-weighted mean $1/k$ by (19). Left: when p is not constant the largest pivot exceeds that mean, and by (20) its complement dominates. Right: when p is constant the tie law (21) holds, every k-subset dominates and none dominates strictly.

Corank two is one further step: Corollary 3.15 converts it into rank two, which is Theorem 5.10, and §5.3 carries out the conversion once that input is proved.

4. The complex problem

Over $\mathbb{C}$ the problem is solved, and the answer stops exactly where the elementary methods of §2–§3 stop: $\mathrm{GTZ}_{\mathbb{C}}(m, k)$ holds for $k \geq 2$ if and only if $m \leq k + 1$. Half of that is already proved. Nothing in §2 or §3 consulted the field, so their positive results stand over $\mathbb{C}$ unchanged, and they cover $m \leq k + 1$ (§4.2).

The other half needs one counterexample and a way to move it. A single design of size four and rank two fails, and two operations propagate the failure across the board: one raises the size and the rank together, the other raises the size alone, and each keeps a bound on the least eigenvalue that any k-subset can achieve. That bound is the quantity to track, so §4.1 defines it before §4.3 exhibits the seed, §4.4 the operations and §4.5 the assembly.

4.1. **The value of a design.** Once the Boolean question is answered *no*, what remains to study is a number, and the operations of §4.4 propagate it. Fix a design $\mathcal{D}$ at (m, k) over $\mathbb{F}$. For every k-subset C and every $x \in \mathbb{F}^k$,

$$x^{\mathsf{H}} S_C x \;=\; \sum_{c \in C} |\langle g_c, x\rangle|^2, \qquad \text{so} \qquad S_C \succeq s\, I_k \iff s\|x\|^2 \leq \sum_{c \in C} |\langle g_c, x\rangle|^2 \;\text{ for all } x. \tag{22}$$

Both readings are used below: the left one to recognize a design, the right one to transport a bound along a map of index sets. Put

$$\operatorname{val}(\mathcal{D}) \;:=\; \max_{|C| = k} \lambda_{\min}(S_C), \tag{23}$$

a maximum over a finite set. Since $S_C \succeq sI_k$ is equivalent to $\lambda_{\min}(S_C) \geq s$, the design $\mathcal{D}$ admits a dominating k-subset if and only if $\mathrm{val}(\mathcal{D}) \geq 1$. We say $\mathcal{D}$ *has value at most* b when $S_C - sI_k$ fails to be positive semidefinite for every k-subset C and every $s > b$; this is $\mathrm{val}(\mathcal{D}) \leq b$. Finally set

$$\alpha_k(\mathbb{F}) \ := \ \inf_m \ \inf_{\mathcal{D}} \ \mathrm{val}(\mathcal{D}),$$

the infimum over all sizes and all designs of size m and rank k over $\mathbb{F}$. Thus $\mathrm{GTZ}(m,k)$ holds for every m precisely when $\alpha_k(\mathbb{R}) \geq 1$, and likewise over $\mathbb{C}$; and since Theorem 3.19(iii)–(iv) produces designs of value exactly 1 at every rank, the conjecture over $\mathbb{R}$ is the assertion $\alpha_k(\mathbb{R}) = 1$ for every k.

At every rank and over either field the value has a floor, which limits how far the complex counterexamples can push it.

Proposition 4.1 (Lean)**.** *Every weighted design of size m and rank k over $\mathbb{F}$ admits a k-subset C with $S_C \succeq \frac{1}{k} I_k$. Hence $\alpha_k(\mathbb{F}) \geq 1/k$.*

Proof. Let A be the $m \times k$ matrix whose c-th row is g_c^{H}, so that $(Ax)_c = \langle g_c, x\rangle$, and let $D = \mathrm{diag}(t_1, \dots, t_m)$. Condition (1) reads $A^{\mathsf{H}} D A = I_k$; in particular A has rank k, so some k-subset indexes an invertible $k \times k$ submatrix. Choose C maximizing $|\det A_C|$ among all k-subsets, put $M := A_C$ and $B := AM^{-1}$. Then $A = BM$, so the c-th row of B holds the coefficients expressing g_c^{H} in the rows of M, and the rows of B indexed by C form I_k. By Cramer's rule, $B_{ci} \det M$ is the determinant of the matrix obtained from M by substituting g_c^{H} for its i-th row; that matrix is, up to a permutation of rows, $A_{C'}$ with $C' := (C \setminus \{c_i\}) \cup \{c\}$, where c_i is the i-th element of C, and it is singular if $c \in C \setminus \{c_i\}$. Hence

$$|B_{ci}| \ = \ \frac{|\det A_{C'}|}{|\det A_C|} \ \leq \ 1$$

for every c and i, by maximality of the volume [10]. By Cauchy–Schwarz, for every $u \in \mathbb{F}^k$,

$$|(Bu)_c|^2 \ \leq \ \Big(\sum_{i \leq k} |B_{ci}|^2\Big) \|u\|^2 \ \leq \ k\|u\|^2.$$

Fix $x \in \mathbb{F}^k$ and take $u := Mx$, so that $Ax = Bu$ and $\|u\|^2 = \sum_{c \in C} |\langle g_c, x\rangle|^2$. Then

$$\|x\|^2 \ = \ x^{\mathsf{H}} A^{\mathsf{H}} D A x \ = \ \sum_c t_c |(Bu)_c|^2 \ \leq \ k\|u\|^2 \sum_c t_c \ = \ k \sum_{c \in C} |\langle g_c, x\rangle|^2,$$

which is (22) at $s = 1/k$. □

4.2. **The positive half.**

Proposition 4.2 (Lean)**.** *Let $k \geq 1$ and $m \leq k+1$. Then $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ holds, as does $\mathrm{GTZ}(m,k)$.*

Proof. The vectors of a design span, so $m \geq k$ and there is nothing to prove when $m < k$: no design exists. If $m = k$ the assertion is Proposition 2.6, and if $m = k+1$ it is Theorem 3.19. Neither argument uses the field: the first is the inequality $t_c \leq 1$ applied to (1), the second is the rank-one Schur criterion together with the fact that a weighted mean does not exceed its maximum. □

The bound 1 is attained at the last size at which it holds, and attained on a set of full dimension in the weights: by Theorem 3.19(iii)–(iv), for every $k \geq 1$, every weight vector t on $k+1$ atoms and over either field there is a design at $(k+1, k)$ with weights exactly t in which every k-subset dominates and none dominates strictly. Its chart is $P = I_{k+1} - zz^{\mathsf{H}}$ with $|z_c|^2 = (1 - t_c)/k$, and its leverages obey the tie law (21).

One size lower the behaviour differs, and the two cells are easily confused: at $m = k$ every design has value bounded away from 1, and 1 is approached only in a degenerate limit.

Proposition 4.3. *Let $\mathcal{D}$ be a design at (k,k) over $\mathbb{F}$. Then $\ell_c = 1/t_c$ for every c, the unique k-subset satisfies*

$$\lambda_{\min}(S_{\{1,\dots,k\}}) \;=\; \frac{1}{\max_c t_c},$$

and this quantity lies in $(1,k]$ for $k \geq 2$. Every weight vector is realized, so for $k \geq 2$ the infimum of val *over designs at (k,k) equals 1 and is not attained.*

Proof. In the chart of §3 the matrix V is square with $V^{\mathsf{H}}V = I_k$, hence unitary, so its rows are orthonormal: $\langle v_c, v_d\rangle = \delta_{cd}$ with $v_c = \sqrt{t_c}\, g_c$. Thus $\ell_c = \|v_c\|^2/t_c = 1/t_c$ and

$$S_{\{1,\dots,k\}} \;=\; \sum_c g_c g_c^{\mathsf{H}} \;=\; \sum_c \frac{1}{t_c}\, v_c v_c^{\mathsf{H}},$$

which is diagonal in the orthonormal basis (v_c) with eigenvalues $1/t_c$. Its least eigenvalue is $1/\max_c t_c$. Since $\sum_c t_c = 1$ over $k \geq 2$ terms, $1/k \leq \max_c t_c < 1$, whence the stated range. Conversely $g_c := t_c^{-1/2} e_c$ is a design at (k,k) with weights t, and letting $\max_c t_c \to 1$ drives the value to 1 from above. □

4.3. **The seed.** Let $\omega := e^{2\pi i/3}$, so that $1 + \omega + \omega^2 = 0$ and $\omega + \bar\omega = -1$.

Example 4.4 (Lean)**.** Put

$$g_0 := (\sqrt{2},\, 0), \qquad g_j := \Big(\sqrt{\tfrac{2}{3}},\ \sqrt{\tfrac{4}{3}}\,\omega^{\,j-1}\Big) \quad (j = 1,2,3), \qquad t_c := \tfrac{1}{4}. \tag{24}$$

Each vector has $\ell_c = 2$: for $j \geq 1$, $\frac{2}{3} + \frac{4}{3} = 2$. The atom sum is

$$\sum_{j=1}^{3} g_j g_j^{\mathsf{H}} \;=\; \begin{pmatrix} 2 & \sqrt{8/9}\,\sum_j \overline{\omega^{\,j-1}} \\ \sqrt{8/9}\,\sum_j \omega^{\,j-1} & 4 \end{pmatrix} \;=\; \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix},$$

by $1+\omega+\omega^2 = 0$, so $\sum_c g_c g_c^{\mathsf{H}} = 4I_2$ and (1) holds at the uniform weight $\frac{1}{4}$. This is the $\mathbb{C}^2$ symmetric informationally complete configuration: four equiangular lines, and indeed $|\langle g_a, g_b\rangle|^2 = \frac{4}{3}$ for every $a \neq b$. For $b = 0$ this is $(\sqrt{2}\cdot\sqrt{2/3})^2 = 4/3$; for $1 \leq a < b \leq 3$ it is

$$\big(\tfrac{2}{3} + \tfrac{4}{3}\omega^{\,r}\big)\big(\tfrac{2}{3} + \tfrac{4}{3}\overline{\omega^{\,r}}\big) \;=\; \tfrac{4}{9} - \tfrac{8}{9} + \tfrac{16}{9} \;=\; \tfrac{4}{3}, \qquad r = b - a \not\equiv 0,$$

using $\omega^r + \overline{\omega^r} = -1$.

Lemma 4.5 (Lean)**.** *The design* (24) *has value exactly*

$$\alpha_{\mathrm{SIC}} \;:=\; 2 - \frac{2}{\sqrt{3}} \;=\; 0.8452994616\ldots,$$

the least root of $z^2 - 4z + \frac{8}{3}$. In particular $\mathrm{GTZ}_{\mathbb{C}}(4,2)$ *is false, and $\alpha_2(\mathbb{C}) \leq \alpha_{\mathrm{SIC}} < \frac{9}{10}$.*

Proof. Let $C = \{a,b\}$ be a pair and let A be the 2×2 matrix with rows $g_a^{\mathsf{H}}, g_b^{\mathsf{H}}$, so that $S_C = A^{\mathsf{H}}A$ and

$$\det S_C = |\det A|^2 = \det AA^{\mathsf{H}} = \ell_a\ell_b - |\langle g_a, g_b\rangle|^2 = 4 - \tfrac{4}{3} = \tfrac{8}{3}, \qquad \operatorname{tr} S_C = \ell_a + \ell_b = 4.$$

Hence $\det(S_C - zI_2) = z^2 - 4z + \frac{8}{3}$ for every one of the six pairs, with roots $2 \pm 2/\sqrt{3}$, and $\lambda_{\min}(S_C) = \alpha_{\mathrm{SIC}}$. The value is the maximum of these six equal numbers. Since $\alpha_{\mathrm{SIC}} < 1$ no pair dominates. The inequality $\alpha_{\mathrm{SIC}} < \frac{9}{10}$ is $\frac{11}{10} < \frac{2}{\sqrt{3}}$, that is $11\sqrt{3} < 20$, that is $363 < 400$. □

Here the two fields part: over $\mathbb{R}$ rank two holds at every size (Theorem 5.10), so $\alpha_2(\mathbb{R}) = 1$, while over $\mathbb{C}$ the same cell has value below 1 already at size four. No proof of GTZ can therefore be field-agnostic. The constant α_{SIC} is moreover sharp: by [23] no uniform-weight design of rank two has value below $2 - 2/\sqrt{3}$, the bound being attained exactly when $4 \mid m$, at a repeated copy of (24). The bound passes to arbitrary weights in two steps. A design with rational weights is obtained from a uniform design of larger size by merging equal vectors, so by Lemma 4.7, read backwards, it has the same value and the bound holds there. For arbitrary weights, hold the chart P fixed: by Lemma 3.2 every weight vector t' with positive entries carries a design with that chart, namely $g'_c = \sqrt{t_c/t'_c}\, g_c$, and val is a continuous function of t' there, being a maximum of finitely many least eigenvalues of matrices depending continuously on t'. Rational weight vectors are dense in the open simplex, so the bound persists in the limit. Hence $\alpha_2(\mathbb{C}) = 2 - 2/\sqrt{3}$.

4.4. **Two padding operations.** The first operation adjoins one coordinate together with one atom along it, raising the size and the rank by one. The second lists an existing atom twice, raising the size alone. Each preserves (1) by a one-line computation, and each preserves a value bound; the inequality preserved is displayed in the statement.

Lemma 4.6 (Lean)**.** *Let $\mathcal{D}$ be a complex weighted design at (m, k) and let $0 < w < 1$. Define $\widehat{\mathcal{D}}$ at $(m+1, k+1)$ by*

$$\widehat{g}_c := \Big(\frac{g_c}{\sqrt{1-w}}, 0\Big) \quad (c \le m), \qquad \widehat{g}_{m+1} := \frac{1}{\sqrt{w}}\, e_{k+1}, \qquad \widehat{t}_c := (1-w)t_c, \quad \widehat{t}_{m+1} := w. \tag{25}$$

Then $\widehat{\mathcal{D}}$ is a design, and if $\mathcal{D}$ has value at most $b \ge 0$ then $\widehat{\mathcal{D}}$ has value at most $b/(1-w)$.

Proof. The weights are positive and sum to $(1-w) + w = 1$. For the Parseval condition, the first m terms contribute

$$\sum_{c \le m} (1-w)t_c\, \frac{1}{1-w} \begin{pmatrix} g_c g_c^{\mathsf{H}} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

and the last contributes $w \cdot w^{-1} e_{k+1} e_{k+1}^{\mathsf{H}}$; the sum is I_{k+1}.

Now let C be a $(k+1)$-subset and $s > b/(1-w)$; we show $S_C - sI_{k+1}$ is not positive semidefinite. Suppose first $m+1 \notin C$. Every $\widehat{g}_c$ with $c \le m$ vanishes in the last coordinate, so the $(k+1, k+1)$ entry of $S_C - sI_{k+1}$ is $-s < 0$, and a positive semidefinite matrix has nonnegative diagonal. Suppose instead $m+1 \in C$ and put $C' := C \setminus \{m+1\}$, a k-subset of $\{1, \dots, m\}$. Test on the flat directions $\widehat{x} := (x, 0)$ with $x \in \mathbb{C}^k$. There

$$\langle \widehat{g}_{m+1}, \widehat{x} \rangle = 0, \qquad \langle \widehat{g}_c, \widehat{x} \rangle = \frac{\langle g_c, x \rangle}{\sqrt{1-w}} \quad (c \le m), \qquad \|\widehat{x}\| = \|x\|.$$

If $S_C \succeq sI_{k+1}$ then, by the right-hand form of (22) applied to $\widehat{x}$,

$$s\|x\|^2 \;\le\; \sum_{c \in C} |\langle \widehat{g}_c, \widehat{x} \rangle|^2 \;=\; \frac{1}{1-w} \sum_{c \in C'} |\langle g_c, x \rangle|^2 \qquad \text{for every } x \in \mathbb{C}^k,$$

that is $S_{C'} \succeq (1-w)s\, I_k$ in the original design. But $(1-w)s > b$, contradicting the hypothesis on $\mathcal{D}$. ∎

The spike is invisible in the flat directions and is the only atom visible in the new one. The factor $1/(1-w)$ costs something real: for $b > 0$ every admissible w has $b < b/(1-w)$, so no member of the family attains the level b, though the levels descend to it as $w \to 0$.

Lemma 4.7 (Lean)**.** *Let $\mathcal{D}$ be a complex weighted design at (m, k), let $c_0 \le m$ and let $0 < f < 1$. Define $\widetilde{\mathcal{D}}$ at $(m+1, k)$ by listing the vector g_{c_0} twice,*

$$\widetilde{g}_c := g_c \ (c \le m), \quad \widetilde{g}_{m+1} := g_{c_0}, \qquad \widetilde{t}_{c_0} := f\, t_{c_0}, \quad \widetilde{t}_{m+1} := (1-f)t_{c_0}, \quad \widetilde{t}_c := t_c \text{ otherwise.} \tag{26}$$

Then $\widetilde{\mathcal{D}}$ is a design, and if $\mathcal{D}$ has value at most $b \ge 0$ then so does $\widetilde{\mathcal{D}}$.

Proof. The weights are positive and still sum to one, and in (1) the term $t_{c_0} g_{c_0} g_{c_0}^{\mathsf{H}}$ is replaced by $f t_{c_0} g_{c_0} g_{c_0}^{\mathsf{H}} + (1-f) t_{c_0} g_{c_0} g_{c_0}^{\mathsf{H}}$, which is the same matrix.

Let C be a k-subset of $\{1, \dots, m+1\}$ and $s > b \ge 0$. If C fails to contain both labels c_0 and $m+1$, then replacing $m+1$ by c_0 where necessary carries C to a k-subset C' of $\{1, \dots, m\}$ with the same family of vectors, so $S_C = S_{C'}$ and $S_C - sI_k$ is not positive semidefinite by the hypothesis on $\mathcal{D}$. If C contains both labels, its k vectors include two equal ones, hence span a subspace of dimension at most $k-1$; choosing $x \neq 0$ orthogonal to that subspace gives $x^{\mathsf{H}} S_C x = 0$ by (22), while $x^{\mathsf{H}}(sI_k)x = s\|x\|^2 > 0$. $\square$

4.5. **The ladder.**

Theorem 4.8 (Lean)**.** *Let $k \geq 2$ and $m \geq k+2$. There is a complex weighted design of size m and rank k whose value is at most*

$$\tfrac{10}{9}\,\alpha_{\mathrm{SIC}} \;=\; \frac{20}{9} - \frac{20\sqrt{3}}{27} \;=\; 0.9392216240\ldots \;<\; 1.$$

In particular $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ *is false.*

Proof. Set $e := k-2 \geq 0$. If $e = 0$ start with the design (24) itself, of size $4 = k+2$ and value α_{SIC} by Lemma 4.5. If $e \geq 1$, apply Lemma 4.6 e times to (24) with the common spike weight

$$w \;:=\; 1 - \left(\tfrac{9}{10}\right)^{1/e}, \qquad \text{so that} \qquad (1-w)^e = \tfrac{9}{10}.$$

Each application raises the size and the rank by one, so after e steps the size is $4 + e = k+2$ and the rank is $2 + e = k$; and each application divides the value bound by $1-w$, so the resulting bound is

$$\frac{\alpha_{\mathrm{SIC}}}{(1-w)^e} \;=\; \frac{10}{9}\,\alpha_{\mathrm{SIC}}.$$

This is the asserted bound, and it is below 1 because $\alpha_{\mathrm{SIC}} < \frac{9}{10}$ (Lemma 4.5). Finally apply Lemma 4.7 $m-k-2$ times, which raises the size to m without raising the rank and without raising the value bound. A design of value below 1 has no dominating k-subset. $\square$

At $k = 3$ the theorem is the one-spike padded configuration of size five, whose value is exactly $\frac{10}{9}\alpha_{\mathrm{SIC}}$; that cell, $(5,3)$, is the smallest at which rank three fails over $\mathbb{C}$, since Proposition 4.2 covers $m \leq 4$. The bound is not the best available at larger sizes: §4.6 exhibits designs of rank three with value $0.7639\ldots$ and $0.7018\ldots$.

Theorem 4.9 (Lean)**.** *Let $k \geq 2$. Then*

$$\mathrm{GTZ}_{\mathbb{C}}(m,k) \quad\Longleftrightarrow\quad m \leq k+1.$$

For $k \leq 1$, $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ *holds for every m.*

Proof. For $m \leq k+1$ this is Proposition 4.2; for $m \geq k+2$ it is Theorem 4.8. At $k=1$ the statement is Proposition 2.7, and at $k=0$ the empty subset dominates the 0×0 identity. $\square$

For $k \geq 2$ the equivalence is vacuous below $m = k$, where Parseval admits no design at all: the assertion holds there for want of anything to quantify over, not because selection succeeds.

4.6. **The sharp constants.** Theorem 4.9 settles the decision problem over $\mathbb{C}$ and leaves the quantitative one. By Proposition 4.1 and Theorem 4.8,

$$\frac{1}{k} \;\leq\; \alpha_k(\mathbb{C}) \;\leq\; \frac{10}{9}\alpha_{\mathrm{SIC}} \;<\; 1 \qquad (k \geq 2),$$

and at rank two both ends are known: $\alpha_2(\mathbb{C}) = 2 - 2/\sqrt{3}$, the upper bound by Lemma 4.5 and the lower by [23]. At rank three the two ends are far apart, and two designs push the upper end below the padded seed.

Example 4.10 (Lean)**.** Two coplanar trines of lines in $\mathbb{C}^3$ sharing an axis: for $j = 0, 1, 2$,

$$g_j := (\sqrt{2},\, 0,\, \omega^j), \qquad g_{3+j} := (0,\, \sqrt{2},\, \omega^j), \qquad t_c := \tfrac{1}{6}.$$

Every leverage is $2 + 1 = 3$, and $\sum_j g_j g_j^{\mathsf{H}} = \operatorname{diag}(6, 0, 3)$, $\sum_j g_{3+j} g_{3+j}^{\mathsf{H}} = \operatorname{diag}(0, 6, 3)$, the off-diagonal entries cancelling by $1 + \omega + \omega^2 = 0$; the total is $6I_3$, so (1) holds. This design has value exactly $3 - \sqrt{5}$.

To see the last assertion, and to prepare the rank-three record, we record the determinant of a triple once. Let $C = \{a, b, c\}$ and let A be the 3×3 matrix with rows $g_a^{\mathsf{H}}, g_b^{\mathsf{H}}, g_c^{\mathsf{H}}$. Then $S_C = A^{\mathsf{H}} A$ while the Gram matrix is $\operatorname{Gram}_C = AA^{\mathsf{H}}$, with entries $\langle g_i, g_j \rangle$; for square A the two have the same characteristic polynomial. Writing $\beta_{ij} := \langle g_i, g_j \rangle$ and expanding the 3×3 determinant of $\operatorname{Gram}_C - sI_3$ gives, with $d := 3 - s$ when all three leverages equal 3,

$$\text{(27)} \qquad \det(S_C - sI_3) \;=\; d^3 - d(|\beta_{ab}|^2 + |\beta_{bc}|^2 + |\beta_{ca}|^2) + 2 \operatorname{Re} \beta_{abc}, \qquad \beta_{abc} := \beta_{ab}\beta_{bc}\beta_{ca}.$$

The three moduli form a phase-free budget, and the phases enter through the single real number $\operatorname{Re} \beta_{abc}$, affinely.

For the trine, the two coplanar triples $\{g_0, g_1, g_2\}$ and $\{g_3, g_4, g_5\}$ have atom sums $\operatorname{diag}(6, 0, 3)$ and $\operatorname{diag}(0, 6, 3)$, of least eigenvalue 0. Every other triple consists of two vectors from one side and one from the other, say g_i, g_j, g_{3+l} with $i \neq j$; then

$$\beta_{ij} = 2 + \omega^{j-i}, \qquad \beta_{j,3+l} = \omega^{l-j}, \qquad \beta_{3+l,i} = \omega^{i-l},$$

so the budget is $|2 + \omega^{j-i}|^2 + 1 + 1 = 3 + 2 = 5$ and

$$\beta_{abc} = (2 + \omega^{j-i})\, \omega^{i-j} = 2\omega^{i-j} + 1, \qquad \operatorname{Re} \beta_{abc} = 1 + 2 \cdot (-\tfrac{1}{2}) = 0.$$

By (27), $\det(S_C - sI_3) = d^3 - 5d = d(d^2 - 5)$, so the spectrum of S_C is $\{3,\, 3 - \sqrt{5},\, 3 + \sqrt{5}\}$ and $\lambda_{\min}(S_C) = 3 - \sqrt{5} = 0.7639320225\ldots$ for all eighteen such triples. The value of the design is the maximum over the twenty triples, namely $3 - \sqrt{5}$. In particular $\mathrm{GTZ}_{\mathbb{C}}(6, 3)$ fails, at a level well below the one Theorem 4.8 provides.

Example 4.11 (Lean)**.** The Hesse configuration. Index nine directions of $\mathbb{C}^3$ by a slot $p \in \{0, 1, 2\}$ and a phase $q \in \{0, 1, 2\}$: the direction u_{3p+q} carries 0 in slot p, ω^q in slot $p + 1$ and $-\omega^{2q}$ in slot $p + 2$, indices modulo three. Explicitly, with ω written w,

$$\begin{aligned} u_0 &= (0, 1, -1), & u_3 &= (-1, 0, 1), & u_6 &= (1, -1, 0), \\ u_1 &= (0, w, -w^2), & u_4 &= (-w^2, 0, w), & u_7 &= (w, -w^2, 0), \\ u_2 &= (0, w^2, -w), & u_5 &= (-w, 0, w^2), & u_8 &= (w^2, -w, 0). \end{aligned}$$

Each u_c has $\|u_c\|^2 = 2$, and $\sum_c u_c u_c^{\mathsf{H}} = 6I_3$: the family p contributes 1 to each of the diagonal slots $p + 1$ and $p + 2$ for each of its three phases, so every slot receives 6 from the two families that do not vanish there, while its one off-diagonal entry, $\omega^q \overline{(-\omega^{2q})} = -\omega^{-q}$ in position $(p + 1, p + 2)$, sums to $-\sum_q \omega^{-q} = 0$ over the family. Put

$$g_c := \sqrt{\tfrac{3}{2}}\, u_c, \qquad t_c := \tfrac{1}{9}.$$

Then (1) holds, $\ell_c = 3$ for every c, and for $a \neq b$

$$\langle u_a, u_b \rangle^3 = -1, \qquad \text{hence} \qquad \beta_{ab}^3 = \langle g_a, g_b \rangle^3 = -\tfrac{27}{8}.$$

The overlap identity is two computations. Within the family p, with $r := q' - q \not\equiv 0$,

$$\langle u_{3p+q}, u_{3p+q'} \rangle = \omega^r + \omega^{2r} = -1;$$

and between the families p and $p + 1$ the only slot on which both are nonzero is $p + 2$, so

$$\langle u_{3p+q}, u_{3(p+1)+q'} \rangle = \overline{(-\omega^{2q})}\, \omega^{q'} = -\omega^{q+q'}.$$

Each value is -1 times a cube root of unity, hence a cube root of -1; the remaining pairs are the conjugates of these.

Since $|\beta_{ab}|^2$ is a nonnegative real whose cube is $|\beta_{ab}^3|^2 = (27/8)^2$, it equals $9/4$; the configuration is equiangular, and the budget in (27) is $3 \cdot \frac{9}{4} = \frac{27}{4}$ for every triple. The triangle invariant satisfies $\beta_{abc}^3 = (\beta_{ab}\beta_{bc}\beta_{ca})^3 = -(27/8)^3$, so β_{abc} is one of the three cube roots of $-(27/8)^3$, namely $-\frac{27}{8}$ and $\frac{27}{16}(1 \pm i\sqrt{3})$. Hence

$$\text{(28)} \qquad \det(S_C - sI_3) \;=\; (3-s)^3 - \tfrac{27}{4}(3-s) + 2\,\mathrm{Re}\,\beta_{abc}, \qquad \mathrm{Re}\,\beta_{abc} \in \left\{-\tfrac{27}{8},\ \tfrac{27}{16}\right\},$$

the first alternative occurring exactly when $\det S_C = \frac{27}{4} + 2\,\mathrm{Re}\,\beta_{abc}$ vanishes, i.e. for the twelve triples lying on a line of the Hesse configuration, and the second for the remaining seventy-two. With $\mathrm{Re}\,\beta_{abc} = \frac{27}{16}$ the right-hand side of (28) expands to

$$\text{(29)} \qquad \det(S_C - sI_3) \;=\; -\tfrac{1}{8}\,h(s), \qquad h(s) := 8s^3 - 72s^2 + 162s - 81,$$

and with $\mathrm{Re}\,\beta_{abc} = -\frac{27}{8}$ it is $-s(s-\frac{9}{2})^2$.

Proposition 4.12 (Lean)**.** *The roots of h are $3(1-\cos\frac{2\pi}{9})$, $3(1-\cos\frac{4\pi}{9})$ and $3(1-\cos\frac{8\pi}{9})$. The design of Example 4.11 has value exactly*

$$h_3 \;:=\; 3\Big(1 - \cos\tfrac{2\pi}{9}\Big) \;=\; 0.7018666706\ldots$$

Consequently $\alpha_3(\mathbb{C}) \le h_3$, and $\mathrm{GTZ}_{\mathbb{C}}(m,3)$ fails for every $m \ge 5$.

Proof. Substituting $s = 3(1-\gamma)$ in (29) gives

$$h(3(1-\gamma)) \;=\; -27(8\gamma^3 - 6\gamma + 1),$$

and for $\gamma = \cos\theta$ the identity $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ turns $8\gamma^3 - 6\gamma + 1 = 0$ into $\cos 3\theta = -\frac{1}{2}$. On $[0,\pi]$ the solutions are $\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$, which gives the three roots; they are distinct, and $s = 3(1-\cos\theta)$ increases with θ, so h_3 is the least. Exact evaluation isolates it: h is negative at $\frac{7}{10}$, where its value is $-\frac{136}{1000}$, and positive at $\frac{71}{100}$, where its value is $\frac{588088}{10^6}$, so

$$\text{(30)} \qquad \tfrac{7}{10} \;<\; h_3 \;<\; \tfrac{71}{100}.$$

By (29) the seventy-two nondegenerate triples all have characteristic polynomial $-\frac{1}{8}h$, hence $\lambda_{\min}(S_C) = h_3$ for each; the twelve degenerate triples have $\lambda_{\min}(S_C) = 0$. The value is the maximum, h_3. By (30), $h_3 < 1$, so no triple dominates, so $\mathrm{GTZ}_{\mathbb{C}}(9,3)$ fails; Lemma 4.7 propagates the failure to every $m \ge 9$, and Theorem 4.8 covers $5 \le m \le 8$. ☐

An explicit certificate can replace the eigenvalue computation, exhibiting positive semidefiniteness at the threshold without reference to the spectral theorem. Take the triple $\{u_0, u_1, u_3\}$. From $1 + \omega + \omega^2 = 0$,

$$u_0u_0^{\mathsf{H}} + u_1u_1^{\mathsf{H}} + u_3u_3^{\mathsf{H}} \;=\; \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & \omega \\ -1 & \omega^2 & 3 \end{pmatrix},$$

so that S_C is $\frac{3}{2}$ times this matrix. Writing $p := \frac{3}{2} - s$ and $q := 3 - s$ and $\mathrm{atom}(v) := vv^{\mathsf{H}}$, a direct expansion of both sides gives the polynomial identity

$$\text{(31)} \qquad pq\,(S_C - sI_3) \;=\; q \cdot \mathrm{atom}\big(p, 0, -\tfrac{3}{2}\big) + p \cdot \mathrm{atom}\big(0, q, \tfrac{3}{2}\omega^2\big) - \tfrac{1}{8}h(s)\,E_{33},$$

E_{33} being the matrix unit in the last diagonal slot: the entries other than $(3,3)$ agree as polynomials in s, and the residue at $(3,3)$ is

$$pq\big(\tfrac{9}{2} - s\big) - \tfrac{9}{4}(p+q) \;=\; \Big(-s^3 + 9s^2 - \tfrac{99}{4}s + \tfrac{81}{4}\Big) - \Big(\tfrac{81}{8} - \tfrac{9}{2}s\Big) \;=\; -\tfrac{1}{8}h(s).$$

At $s = h_3$ the cubic vanishes and both p and q are positive, since $h_3 < \frac{71}{100} < \frac{3}{2}$; dividing (31) by the positive scalar pq exhibits $S_C - h_3 I_3$ as a nonnegative combination of two rank-one positive semidefinite matrices. The threshold h_3 is therefore attained by this triple.

The three complex records at rank three are ordered

$$h_3 = 0.7018666706\ldots \;<\; 3-\sqrt{5} = 0.7639320225\ldots \;<\; \tfrac{10}{9}\alpha_{\mathrm{SIC}} = 0.9392216240\ldots,$$

both inequalities being exact. For the second, $12 < 7\sqrt{3}$ gives $\alpha_{\mathrm{SIC}} > \frac{5}{6}$ and hence $\frac{10}{9}\alpha_{\mathrm{SIC}} > \frac{25}{27}$, while $5 > \left(\frac{223}{100}\right)^2$ gives $3-\sqrt{5} < \frac{77}{100}$, and $2500 > 2079$. For the first,

$$h(3-\sqrt{5}) \;=\; (576-1008+486-81) + (-256+432-162)\sqrt{5} \;=\; 14\sqrt{5}-27 \;>\; 0,$$

since $980 > 729$. The three roots of h sum to 9, so they cannot all lie below 2; with $h(2) = 19 > 0$ this places 2 strictly between the two smallest roots. As $3-\sqrt{5} < 2$ and $h(3-\sqrt{5}) > 0$, the number $3-\sqrt{5}$ lies in the same interval, hence above h_3.

Remark 4.13. Identity (27) shows where the two fields differ. Over $\mathbb{R}$ the entries β_{ij} are real, so β_{abc} is real and $\operatorname{Re}\beta_{abc} = \pm|\beta_{abc}|$ is forced to an extreme of its range; over $\mathbb{C}$ it may sit anywhere in $[-|\beta_{abc}|, |\beta_{abc}|]$, and the separating quantity is the phase defect $|\beta_{abc}|^2 - (\operatorname{Re}\beta_{abc})^2 = (\operatorname{Im}\beta_{abc})^2$, which vanishes identically for every real configuration. The trine of Example 4.10 isolates the mechanism: its modulus budget 5 would permit $2\operatorname{Re}\beta_{abc} = 2\sqrt{3}$ and hence a positive determinant at $s = 1$, but its phase defect is 3 and its $\operatorname{Re}\beta_{abc}$ is exactly 0, leaving $\det(S_C - I_3) = -2$. No argument that sees only the moduli $|\beta_{ij}|$ can decide either problem; the corresponding no-go statement is recorded in Appendix A.

Remark 4.14. Collecting the two ends,

$$\tfrac{1}{3} \;\le\; \alpha_3(\mathbb{C}) \;\le\; 3\left(1-\cos\tfrac{2\pi}{9}\right) \;=\; 0.7018\ldots,$$

the lower bound from Proposition 4.1 and the upper from Proposition 4.12. The gap stays open: every rank-three statement above bounds $\alpha_3(\mathbb{C})$ from above, and such bounds move the record down without pinning it. Determining $\alpha_3(\mathbb{C})$ is Problem 3. The rank-inverse bound $1/k$ is sharp only at $k = 1$, where the one-atom design at $(1,1)$ has value 1.

5. The real problem: decided cells

The counting of §2 decides rank one and the square $m = k$, and duality converts corank one and corank two into rank one and rank two (Theorem 3.14). Rank two itself is thus the last small case outstanding, and nothing so far bounds the size at all. One device supplies both: an operation that produces from a design a smaller design whose dominating subsets transport back.

There are two such operations, and they pay for the reduction in different currencies. Deleting a vector of length at most one (§5.1) costs one from the size but moves every surviving vector by a congruence; the transport back is an identity between quadratic forms, and it works only because the deleted vector was short. Carathéodory reduction (§5.4) leaves the vectors where they are and cuts the size to $k(k+1)/2+1$, and there the transport back is immediate, since S_C in (2) never sees the weights. Deletion stalls when every vector is long; at rank two the stalled case can be settled outright (§5.2).

5.1. **Deleting a light vector.**

Definition 5.1. A vector of a design is *light* if $\ell_c \le 1$ and *heavy* if $\ell_c > 1$. The design is *all-heavy* if every one of its vectors is heavy.

By Lemma 2.3 the t-weighted mean of the ℓ_c equals k, so at rank $k \geq 2$ a design always has a heavy vector; whether it has a light one is the dichotomy exploited here. The elementary fact behind the deletion is the rank-one criterion

$$I_k - gg^\mathsf{T} \succeq 0 \quad \Longleftrightarrow \quad \|g\|^2 \leq 1, \tag{32}$$

whose forward direction is the value of the form at g, namely $\|g\|^2(1 - \|g\|^2)$, and whose converse is Cauchy–Schwarz: $\langle g, u\rangle^2 \leq \|g\|^2\|u\|^2 \leq \|u\|^2$.

Theorem 5.2 (Lean)**.** *Let $m \geq 2$ and let $\mathcal{D}$ be a design at (m, k) possessing a light vector. If* $\mathrm{GTZ}(m-1, k)$ *holds, then $\mathcal{D}$ admits a dominating k-subset.*

Proof. Let $\ell_d \leq 1$. Since $m \geq 2$ and all weights are positive with $\sum_c t_c = 1$, we have $t_d < 1$.

Step 1: the residual Parseval mass is positive definite. Put $\Sigma_d := I_k - t_d\, g_d g_d^\mathsf{T}$, which by (1) is $\sum_{c \neq d} t_c\, g_c g_c^\mathsf{T}$. For $u \neq 0$,

$$u^\mathsf{T}\Sigma_d u = \|u\|^2 - t_d\langle g_d, u\rangle^2 \;\geq\; (1 - t_d\ell_d)\|u\|^2 \;>\; 0,$$

using Cauchy–Schwarz and $t_d\ell_d \leq t_d < 1$. Write $\Sigma_d = LL^\mathsf{T}$ with L invertible and set $R := L^{-\mathsf{T}}$, so that

$$R^\mathsf{T}\Sigma_d R = I_k, \qquad \det R \neq 0. \tag{33}$$

Step 2: the deflated design. On the index set $E := \{1, \dots, m\} \setminus \{d\}$ put

$$g'_c := \sqrt{1 - t_d}\; R^\mathsf{T} g_c, \qquad t'_c := \frac{t_c}{1 - t_d} \qquad (c \in E).$$

The weights are positive and sum to one, because $\sum_{c \in E} t_c = 1 - t_d$. For the Parseval condition the factor $1 - t_d$ cancels against the reweighting, and the congruence identity $(R^\mathsf{T}g)(R^\mathsf{T}g)^\mathsf{T} = R^\mathsf{T}(gg^\mathsf{T})R$ gives

$$\sum_{c \in E} t'_c\,(g'_c)(g'_c)^\mathsf{T} \;=\; \sum_{c \in E} t_c\, R^\mathsf{T} g_c g_c^\mathsf{T} R \;=\; R^\mathsf{T}\Sigma_d R \;=\; I_k$$

by (33). So $((g'_c), (t'_c))$ is a design at $(m-1, k)$.

Step 3: transport. By hypothesis some $C \subseteq E$ with $|C| = k$ dominates in the deflated design. Its subset sum is $\sum_{c \in C}(g'_c)(g'_c)^\mathsf{T} = (1 - t_d)R^\mathsf{T} S_C R$, so domination reads

$$R^\mathsf{T}\big((1 - t_d)S_C - \Sigma_d\big)R \;\succeq\; 0,$$

and congruence by the invertible R gives $(1 - t_d)S_C - \Sigma_d \succeq 0$. The identity

$$S_C - I_k \;=\; \frac{1}{1 - t_d}\Big[\big((1 - t_d)S_C - \Sigma_d\big) \;+\; t_d(I_k - g_d g_d^\mathsf{T})\Big] \tag{34}$$

is verified by expanding Σ_d: the two occurrences of $t_d\, g_d g_d^\mathsf{T}$ cancel and the bracket equals $(1 - t_d)(S_C - I_k)$. Its first summand is positive semidefinite by the previous display, its second by (32) and $\ell_d \leq 1$, and the scalar is positive. Hence C dominates in $\mathcal{D}$. ☐

Lightness enters only in the second summand of (34); the rest is the congruence and its bookkeeping. Iterating the theorem removes light vectors one at a time:

Corollary 5.3 (Lean)**.** *Let $k \geq 1$. If every all-heavy design of rank k, of whatever size, admits a dominating k-subset, then* $\mathrm{GTZ}(m, k)$ *holds for every m.*

Proof. Strong induction on m. A design at (m, k) has $m \geq k$. If $m = k$, apply Proposition 2.6. If $m > k$ and some vector is light, then $m - 1 \geq k \geq 1$ and $m \geq 2$, so Theorem 5.2 applies with the induction hypothesis $\mathrm{GTZ}(m-1, k)$. Otherwise the design is all-heavy and the assumption applies. ☐

The tetrahedron of Example 2.11 has $\ell_c = 3$ for every c and is therefore all-heavy: the deletion is unavailable precisely at the design where, by Example 2.11, every dominating triple has $\lambda_{\min}(S_C) = 1$. Here a pattern begins that runs through the paper: the reductions consume the interior of the problem and leave the extremal locus of §6 standing.

5.2. **Rank two.** Fix a design at $(m, 2)$ and write $g_c = (x_c, y_c)$. Identify $\mathbb{R}^2$ with $\mathbb{C}$ by $(x, y) \mapsto x+iy$ (a device for the computation below, unrelated to the complex problem of §4), and let $\eta_c \in \mathbb{R}^2$ correspond to the square of the complex number attached to g_c:

$$\eta_c \;:=\; (x_c^2 - y_c^2,\; 2x_cy_c), \qquad \hat{\ell}_c \;:=\; \ell_c - 2. \tag{35}$$

Squaring doubles arguments and squares moduli, so $\|\eta_c\| = \ell_c$; and since the complex number attached to g_c times the conjugate of the one attached to g_d has real part $a_{cd} := \langle g_c, g_d\rangle$ and imaginary part $\pm b_{cd}$, where

$$b_{cd} \;:=\; x_cy_d - x_dy_c,$$

taking real parts of the square of that product gives

$$\langle \eta_c, \eta_d\rangle \;=\; a_{cd}^2 - b_{cd}^2, \qquad a_{cd}^2 + b_{cd}^2 \;=\; \ell_c\ell_d, \tag{36}$$

the second identity being Lagrange's. The three scalar components of (1) at $k = 2$ read $\sum_c t_cx_c^2 = \sum_c t_cy_c^2 = 1$ and $\sum_c t_cx_cy_c = 0$, so that

$$\sum_c t_c\,\eta_c \;=\; 0, \qquad \sum_c t_c\,\ell_c \;=\; 2, \qquad \sum_c t_c\,\hat{\ell}_c \;=\; 0. \tag{37}$$

The design is thus centred in the squared picture, and its centred lengths are centred as well. These three identities are the only use made of (1) in this subsection.

Example 5.4. Let $u_0, u_1, u_2 \in \mathbb{R}^2$ be unit vectors at mutual angles $2\pi/3$ and put $g_c = \sqrt{2}\,u_c$, $t_c = \frac{1}{3}$. From $\sum_c u_cu_c^{\mathsf{T}} = \frac{3}{2}I_2$ one gets $\sum_c t_c\,g_cg_c^{\mathsf{T}} = I_2$, so this is a design at $(3, 2)$, with $\ell_c = 2$ and $\hat{\ell}_c = 0$. It is the member at $k = 2$, $t_c = \frac{1}{3}$ of the extremal family of Theorem 3.19: (21) reads $2 \cdot \frac{1}{3} \cdot 2 = \frac{4}{3} = 2 - 1 + \frac{1}{3}$. Each η_c has length 2, and the three of them again make angles $2\pi/3$; every pair $C = \{c, d\}$ has

$$S_C \;=\; 3I_2 - 2\,u_eu_e^{\mathsf{T}}, \qquad \operatorname{spec} S_C = \{1, 3\},$$

where e is the third index. Every pair dominates, and every pair dominates with least eigenvalue exactly 1. We call this design the rank-two trine.

The remaining argument turns on one scalar invariant of a pair of indices.

Definition 5.5. For indices c, d of a design at $(m, 2)$ put

$$K_{cd} \;:=\; \langle \eta_c, \eta_d\rangle - \hat{\ell}_c\hat{\ell}_d + 2. \tag{38}$$

Lemma 5.6. *Let $c \neq d$ with $\ell_c + \ell_d > 2$. Then $\{c, d\}$ dominates if and only if $K_{cd} \leq 0$.*

Proof. Let M be the 2×2 matrix with columns g_c, g_d, so that $X := S_{\{c,d\}} - I_2 = MM^{\mathsf{T}} - I_2$. For a 2×2 matrix A one has $\det(A - I) = \det A - \operatorname{tr} A + 1$; with $A = MM^{\mathsf{T}}$, $\det A = (\det M)^2 = b_{cd}^2$ and $\operatorname{tr} A = \ell_c + \ell_d$, so

$$\det X \;=\; b_{cd}^2 - \ell_c - \ell_d + 1. \tag{39}$$

On the other hand, eliminating a_{cd}^2 from (36) in (38) and using $\hat{\ell} = \ell - 2$,

$$K_{cd} = \ell_c\ell_d - 2b_{cd}^2 - (\ell_c - 2)(\ell_d - 2) + 2 = -2\big(b_{cd}^2 - \ell_c - \ell_d + 1\big) = -2\det X.$$

Now $\operatorname{tr} X = \ell_c + \ell_d - 2 > 0$. A symmetric 2×2 matrix of positive trace is positive semidefinite exactly when its determinant is nonnegative: if $\det X \geq 0$ then $X_{00}X_{11} \geq X_{01}^2 \geq 0$, so X_{00} and X_{11} have the same sign and the positive trace makes both nonnegative, whereupon either $X_{00} = 0$, forcing $X_{01} =$

0 and leaving the diagonal matrix $\operatorname{diag}(0, X_{11})$, or $X_{00}\, x^{\mathsf{T}} X x = (X_{00}x_0 + X_{01}x_1)^2 + (\det X)x_1^2 \geq 0$; the converse is the 2×2 minor. Hence $X \succeq 0$ if and only if $K_{cd} \leq 0$. □

At the rank-two trine of Example 5.4 one has $\hat{\ell}_c = 0$ and $\langle \eta_c, \eta_d \rangle = 4\cos(4\pi/3) = -2$ for $c \neq d$, so $K_{cd} = 0$ for every pair: consistent with Lemma 5.6 and with $\det(S_C - I) = 0 \cdot 2 = 0$. The inequality $K_{cd} \leq 0$ is therefore an equality on a design, and no argument below may improve it to a strict inequality.

In an all-heavy design every pair has $\ell_c + \ell_d > 2$, so if no pair dominates then Lemma 5.6 gives

$$K_{cd} > 0 \qquad \text{for all } c \neq d. \tag{40}$$

The contradiction is produced in two steps: a linear identity satisfied by K, and a choice of test vector.

Lemma 5.7. *For every c one has $\sum_d t_d K_{cd} = 2$, and $K_{cd} = K_{dc}$. Consequently, for every $u\colon \{1, \dots, m\} \to \mathbb{R}$,*

$$\sum_{c,d} t_c t_d K_{cd} u_c u_d \;=\; 2\sum_c t_c u_c^2 \;-\; \tfrac{1}{2}\sum_{c,d} t_c t_d K_{cd}(u_c - u_d)^2. \tag{41}$$

If (40) holds and u is not constant, the left side of (41) is strictly smaller than $2\sum_c t_c u_c^2$.

Proof. Summing (38) against t_d and using the three balance identities (37),

$$\sum_d t_d K_{cd} = \Big\langle \eta_c, \sum_d t_d \eta_d \Big\rangle - \hat{\ell}_c \sum_d t_d \hat{\ell}_d + 2\sum_d t_d = 0 - 0 + 2.$$

Symmetry of K is symmetry of the inner product. For (41), expand $(u_c - u_d)^2$ and use the row sum twice:

$$\sum_{c,d} t_c t_d K_{cd} u_c^2 = \sum_c t_c u_c^2 \sum_d t_d K_{cd} = 2\sum_c t_c u_c^2,$$

and likewise for u_d^2 by symmetry, so the correction term equals $4\sum_c t_c u_c^2$ minus twice the left side. The diagonal terms of the correction vanish; if u is nonconstant there are $c \neq d$ with $u_c \neq u_d$, and (40) makes the corresponding summand strictly positive. □

Identity (41) is the only use of the balance identities (37), and it is sharp: the correction term vanishes identically when $K \equiv 0$, as at the rank-two trine. It remains to produce a nonconstant u for which the left side of (41) is at least $2\sum_c t_c u_c^2$. The test vectors are the linear functionals of the squared picture, $u_c = \langle \eta_c, r \rangle$, and the two lemmas below treat the two possible positions of the family (η_c) in the plane.

Lemma 5.8. *Let a design at $(m, 2)$ be all-heavy and suppose the vectors η_c all lie on one line through the origin. Then some pair dominates.*

Proof. Let $r \neq 0$ be orthogonal to that line and let $r^\perp$ be r rotated by $\pi/2$, so every η_c is a multiple of $r^\perp$. Put $\mu_c := \langle \eta_c, r^\perp \rangle$; then $|\mu_c| = \|\eta_c\|\|r\| = \ell_c\|r\| \neq 0$. By (37) $\sum_c t_c \mu_c = 0$, and the weights are positive, so the μ_c take both signs; choose $\mu_p > 0 > \mu_q$. Since $\eta_p = (\mu_p/\|r\|^2)\, r^\perp$ and likewise for q,

$$\langle \eta_p, \eta_q \rangle = \frac{\mu_p \mu_q}{\|r\|^2} < 0, \qquad |\langle \eta_p, \eta_q \rangle| = \|\eta_p\|\|\eta_q\| = \ell_p \ell_q,$$

whence $\langle \eta_p, \eta_q \rangle = -\ell_p \ell_q$ and

$$K_{pq} = -\ell_p \ell_q - (\ell_p - 2)(\ell_q - 2) + 2 = -2(\ell_p - 1)(\ell_q - 1) < 0$$

because both vectors are heavy. By Lemma 5.6 the pair $\{p, q\}$ dominates. □

Lemma 5.9. *Let a design at $(m, 2)$ be all-heavy and suppose the vectors η_c span $\mathbb{R}^2$. Then some pair dominates.*

Proof. Introduce the second moments of the squared picture,

$$\Omega := \sum_c t_c\, \eta_c\eta_c^{\mathsf T}, \qquad \zeta := \sum_c t_c\, \hat\ell_c\eta_c, \qquad \sigma := \sum_c t_c\, \hat\ell_c^2,$$

so that Ω is a symmetric 2×2 matrix with $r^{\mathsf T}\Omega r = \sum_c t_c\langle \eta_c, r\rangle^2$. This vanishes only if every $\langle\eta_c, r\rangle$ does, so the spanning hypothesis gives $\Omega \succ 0$. Since $\|\eta_c\| = \ell_c$, the second and third identities of (37) give

$$\sigma = \sum_c t_c(\ell_c-2)^2 = \sum_c t_c\ell_c^2 - 4\sum_c t_c\ell_c + 4 = \operatorname{tr}\Omega - 4. \tag{42}$$

Put

$$\Psi \;:=\; \Omega^2 - \zeta\zeta^{\mathsf T} - 2\Omega.$$

Some direction is nonnegative for Ψ. Let $\xi := \Omega^{-1}\zeta$. Expanding the square and using $\sum_c t_c\langle\eta_c,\xi\rangle^2 = \xi^{\mathsf T}\Omega\xi = \zeta^{\mathsf T}\Omega^{-1}\zeta$ and $\sum_c t_c\hat\ell_c\langle\eta_c,\xi\rangle = \langle\zeta,\xi\rangle = \zeta^{\mathsf T}\Omega^{-1}\zeta$,

$$\sum_c t_c\big(\langle\eta_c,\xi\rangle - \hat\ell_c\big)^2 \;=\; \sigma - \zeta^{\mathsf T}\Omega^{-1}\zeta, \tag{43}$$

so $\zeta^{\mathsf T}\Omega^{-1}\zeta \le \sigma$: the residual of the least-squares fit of $\hat\ell$ by a linear functional of η is nonnegative. Choose R invertible with $R^{\mathsf T}\Omega R = I_2$, that is $RR^{\mathsf T} = \Omega^{-1}$, and let r_1, r_2 be its columns. Then $\operatorname{tr}(R^{\mathsf T}\Omega^2R) = \operatorname{tr}(\Omega^2RR^{\mathsf T}) = \operatorname{tr}\Omega$ and $\operatorname{tr}(R^{\mathsf T}\zeta\zeta^{\mathsf T}R) = \zeta^{\mathsf T}\Omega^{-1}\zeta$, so by (42) and (43)

$$r_1^{\mathsf T}\Psi r_1 + r_2^{\mathsf T}\Psi r_2 = \operatorname{tr}(R^{\mathsf T}\Psi R) = \operatorname{tr}\Omega - \zeta^{\mathsf T}\Omega^{-1}\zeta - 4 = \sigma - \zeta^{\mathsf T}\Omega^{-1}\zeta \;\ge\; 0.$$

Hence $r^{\mathsf T}\Psi r \ge 0$ for $r := r_1$ or $r := r_2$, and $r \neq 0$ because R is invertible.

The test vector. Suppose no pair dominates, so (40) holds, and put $u_c := \langle\eta_c, r\rangle$. By (37), $\sum_c t_cu_c = 0$; were u constant it would be identically zero, contradicting the spanning hypothesis. So Lemma 5.7 applies to u. Compute both sides. First

$$\sum_d t_du_dK_{cd} = \Big\langle \eta_c, \sum_d t_d\langle\eta_d,r\rangle\eta_d\Big\rangle - \hat\ell_c\sum_d t_d\hat\ell_d\langle\eta_d,r\rangle + 2\sum_d t_d\langle\eta_d,r\rangle = \langle\eta_c,\Omega r\rangle - \hat\ell_c\langle\zeta,r\rangle,$$

the last term vanishing by (37); multiplying by t_cu_c and summing,

$$\sum_{c,d} t_ct_dK_{cd}u_cu_d = \langle\Omega r,\Omega r\rangle - \langle\zeta,r\rangle^2 = r^{\mathsf T}(\Omega^2 - \zeta\zeta^{\mathsf T})r, \qquad \sum_c t_cu_c^2 = r^{\mathsf T}\Omega r.$$

Lemma 5.7 now reads $r^{\mathsf T}(\Omega^2-\zeta\zeta^{\mathsf T})r < 2\,r^{\mathsf T}\Omega r$, that is $r^{\mathsf T}\Psi r < 0$, contradicting the choice of r. $\square$

Theorem 5.10 (Lean)**.** GTZ$(m,2)$ *holds for every m.*

Proof. By Corollary 5.3 it suffices to treat all-heavy designs. Such a design has $m \ge 2$, and its squared vectors η_c either lie on a line through the origin or span $\mathbb{R}^2$; Lemma 5.8 and Lemma 5.9 produce a dominating pair in the two cases. $\square$

No eigenvalue is computed anywhere in the proof. Lemma 5.6 trades the least eigenvalue of a 2×2 matrix for a determinant, and what follows is quadratic algebra in the moments of the squared picture, uniform in m and in the weights as the induction of Corollary 5.3 demands. The rank-two trine of Example 5.4 shows the argument has nothing to spare: there $\Omega = 2I_2$, $\zeta = 0$ and $\sigma = 0$, so $\Psi = \Omega^2 - 2\Omega = 0$, every direction is admissible in Lemma 5.9, and the strict inequality of Lemma 5.7 fails, as it must at a design with $K \equiv 0$.

For uniform weights $t_c = 1/m$, Theorem 5.10 is the theorem of Sengupta and Pautov [27], whose base case $m = 4$ is due to Nesterenko [20]; the substitution (35) and the light/heavy split are theirs. The passage to arbitrary weights is new; it costs the reweighting of Theorem 5.2 in the light branch, and it replaces the Perron–Frobenius step of [27] by Lemma 5.7 and the trace computation of Lemma 5.9 in the heavy branch. The designs with $K \equiv 0$ at uniform weights are classified in [24]: the squared vectors fall into three clusters of equal vectors, of sizes p, q, r with $p+q+r=m$, the cluster of size p having length $\frac{1}{2m}+\frac{1}{2p}$ and similarly for the other two. The rank-two trine is the case $p=q=r=1$ at $m=3$: there the common length is $\frac{1}{6}+\frac{1}{2}=\frac{2}{3}$, which is $\|\eta_c\|/m = 2/3$ for the vectors of Example 5.4, the factor m^{-1} being the passage from a design at uniform weights to a matrix with orthonormal columns.

5.3. **Corank at most two.** Duality now carries rank two to corank two.

Theorem 5.11 (Lean)**.** GTZ$(k+2,k)$ *holds for every* $k \geq 2$.

Proof. By Corollary 3.15 the statement is equivalent to GTZ$(k+2,2)$, which is a case of Theorem 5.10. □

Unwinding the duality at $r=m-k=2$ makes the reduction explicit. Let $\mathcal{D}$ be a design at $(k+2,k)$, and put $m := k+2$. Its kernel plane carries an orthonormal basis, arranged as the columns of an $m \times 2$ matrix Z with rows z_c^{H}, and the co-weight rescaling $w_c = z_c/\sqrt{1-t_c} \in \mathbb{R}^2$ satisfies

$$\sum_c (1-t_c)\, w_c w_c^{\mathsf{T}} \;=\; I_2.$$

A k-subset is the complement of a pair $\{a,b\}$, and by the chain in the proof of Theorem 3.14 that k-subset dominates in $\mathcal{D}$ if and only if

$$w_a w_a^{\mathsf{T}} + w_b w_b^{\mathsf{T}} \;\succeq\; S^{\perp} \;=\; \sum_c t_c\, w_c w_c^{\mathsf{T}}. \tag{44}$$

With $R^{\mathsf{T}} S^{\perp} R = I_2$ and $h_c := R^{\mathsf{T}} w_c$, condition (44) is $h_a h_a^{\mathsf{T}} + h_b h_b^{\mathsf{T}} \succeq I_2$, that is domination of the pair $\{a,b\}$ in the design (h_c, t_c) at $(k+2,2)$. So a dominating pair at rank two produces a dominating k-subset at corank two.

The substance of the step is the normalization in (44). The coefficients in (16) are the co-weights $1-t_c$, which sum to $k+1$, whereas the rank-two input requires weights summing to one. Rescaling the kernel rows by t_c rather than by $1-t_c$, or comparing against I_2 instead of against $S^{\perp}$ in (44), yields a false statement; Example 3.17 exhibits the first of these failures at corank one.

The reduction is field-independent, but its input is not: over $\mathbb{C}$ rank two fails at size four (Theorem 4.9), and accordingly GTZ$_{\mathbb{C}}(k+2,k)$ is false for every $k \geq 2$; the two are the same statement, read through Corollary 3.15.

Corollary 5.12 (Lean)**.** GTZ(m,k) *holds whenever* $m-k \leq 2$*, and in particular whenever* $m \leq 5$.

Proof. Corank zero, one and two are Proposition 2.6, Theorem 3.19 and Theorem 5.11; the last requires $k \geq 2$, and for $k=1$ every size is covered by Proposition 2.7. If $m \leq 5$ then either $k \leq 2$, whence Proposition 2.7 or Theorem 5.10 applies, or $k \geq 3$ and $m \leq 5 \leq k+2$ gives $m-k \leq 2$. □

The two arms of the decided region, small rank and small corank, therefore meet in rank two. Corank two is also where the description of the designs attaining the threshold ceases to be a single law. At corank one (21) determines the leverages from the weights; at $(5,3)$ there are extremal designs whose vectors are pairwise non-parallel and which satisfy no such relation (Example 6.9). The corank arm stops here: $(7,4)$ has corank three and is carried by Corollary 3.15 to $(7,3)$, which is open.

5.4. **Crystallization.** Write

$$N(k) \;:=\; \frac{k(k+1)}{2}+1,$$

one more than the dimension of the space Sym_k of real symmetric $k \times k$ matrices. Above that size a design carries a smaller design on a subfamily of its own vectors.

Lemma 5.13 (Lean)**.** *Let $(g_c)_{c\le m}$ be vectors in $\mathbb{R}^k$ with $m > N(k)$. There is $\delta \in \mathbb{R}^m$, $\delta \neq 0$, with*

$$\sum_c \delta_c\, g_c g_c^{\mathsf{T}} = 0 \qquad \textit{and} \qquad \sum_c \delta_c = 0. \tag{45}$$

Proof. The map $\mathbb{R}^m \to \mathrm{Sym}_k \oplus \mathbb{R}$ sending δ to $\left(\sum_c \delta_c\, g_c g_c^{\mathsf{T}}, \sum_c \delta_c\right)$ is linear, and its target has dimension $k(k+1)/2 + 1 = N(k) < m$. A linear map between real vector spaces of dimensions m and $N(k) < m$ has nonzero kernel. □

Theorem 5.14 (Lean)**.** *Let $\mathcal{D}$ be a design at (m,k) with $m > N(k)$. There is a proper subset S of its index set and a design $\mathcal{D}'$ at $(|S|,k)$ whose vectors are $(g_c)_{c\in S}$, indexed in increasing order.*

Proof. Take δ as in Lemma 5.13. Since $\sum_c \delta_c = 0$ and $\delta \neq 0$, the set $\{c : \delta_c > 0\}$ is nonempty; let

$$\theta \;:=\; \min\Big\{\frac{t_c}{\delta_c} \;:\; \delta_c > 0\Big\} \;>\; 0,$$

attained at c_0, and put $t'_c := t_c - \theta\,\delta_c$. These numbers are nonnegative: if $\delta_c \le 0$ then $t'_c \ge t_c > 0$, and if $\delta_c > 0$ then $\theta \le t_c/\delta_c$ gives $\theta\delta_c \le t_c$. They sum to 1 and satisfy $\sum_c t'_c\, g_c g_c^{\mathsf{T}} = I_k$, by the two identities (45). Finally $t'_{c_0} = 0$.

Let $S := \{c : t'_c > 0\}$, so $c_0 \notin S$ and S is proper. Restricting the two sums to S omits only vanishing terms, so the vectors $(g_c)_{c\in S}$ with the weights $(t'_c)_{c\in S}$, re-indexed by the increasing enumeration of S, form a design at $(|S|,k)$. □

Theorem 5.15 (Lean)**.** *Fix k. If* $\mathrm{GTZ}(m,k)$ *holds for every $m \le k(k+1)/2+1$, then* $\mathrm{GTZ}(m,k)$ *holds for every m.*

Proof. Strong induction on m; the case $m \le N(k)$ is the hypothesis. Let $m > N(k)$ and let $\mathcal{D}$ be a design at (m,k). Take S and $\mathcal{D}'$ as in Theorem 5.14 and put $m' := |S| < m$. By the induction hypothesis some $C \subseteq S$ with $|C| = k$ dominates in $\mathcal{D}'$. The subset sum S_C of (2) does not involve the weights and the vectors of $\mathcal{D}'$ are those of $\mathcal{D}$, so the subset sum of C in $\mathcal{D}$ is the subset sum of C in $\mathcal{D}'$, and C dominates in $\mathcal{D}$. □

The bound is Carathéodory's [4]: condition (1) exhibits I_k as a convex combination of the points $g_c g_c^{\mathsf{T}}$ of the $k(k+1)/2$-dimensional space Sym_k, and a point of a convex hull in dimension N is a convex combination of at most $N+1$ of the points ([26], §17). The proof of Theorem 5.14 is the standard support-reduction step of that theorem, carried out by hand because the representation must use a *subfamily of the given vectors* rather than merely few matrices: only then is the dominating subset produced downstairs a subset of the original index set. This step uses the weight-freeness of (2), and it separates the reduction from that of Theorem 5.2, where the vectors move and (34) must carry the conclusion back.

Domination never enters Lemma 5.13, so there is no reason to expect $N(k)$ to be the true threshold; see Problem 6.

5.5. **Monotonicity in the size.** The finitely many cells left by Theorem 5.15 are not independent: the statement at a size implies the statement at all smaller sizes. The mechanism is that a design can be padded by splitting one of its vectors into two equal copies of half the weight, and that a k-subset containing both copies never dominates.

Lemma 5.16 (Lean)**.** *Let C be a subset of the index set of a design at (m,k) with $|C| \le k$ containing two distinct indices carrying the same vector. Then C does not dominate.*

Proof. Delete one of the two indices from C, obtaining C_0 with $|C_0| \le k-1$. The linear map $\mathbb{R}^k \to \mathbb{R}^{C_0}$, $p \mapsto (\langle g_c, p\rangle)_{c\in C_0}$, has nonzero kernel; take $p \neq 0$ in it. Then $\langle g_c, p\rangle = 0$ for every $c \in C$, the deleted index included, since its vector is one of the retained ones. Hence

$$p^{\mathsf{T}}(S_C - I_k)p = \sum_{c\in C} \langle g_c, p\rangle^2 - \|p\|^2 = -\|p\|^2 < 0.$$ □

Proposition 5.17 (Lean)**.** $\mathrm{GTZ}(m+1,k) \Rightarrow \mathrm{GTZ}(m,k)$.

Proof. For $k=0$ there is nothing to prove. Let $\mathcal{D}$ be a design at (m,k), so $m \geq k$. Define a design $\mathcal{D}^+$ at $(m+1,k)$ by keeping the vectors $g_1,\dots,g_m$ and appending $g_{m+1} := g_1$, with weights $t_1^+ = t_{m+1}^+ = t_1/2$ and $t_c^+ = t_c$ for $2 \leq c \leq m$. Both $\sum_c t_c^+$ and $\sum_c t_c^+ g_c g_c^\mathsf{T}$ are unchanged, so $\mathcal{D}^+$ is a design. By hypothesis some C^+ with $|C^+| = k$ dominates in $\mathcal{D}^+$. By Lemma 5.16, C^+ does not contain both 1 and $m+1$, so the map fixing $1,\dots,m$ and sending $m+1$ to 1 is injective on C^+; its image C has k elements and the same subset sum, hence dominates in $\mathcal{D}$. □

Theorem 5.18 (Lean)**.** $\mathrm{GTZ}(N(k),k) \Rightarrow \mathrm{GTZ}(m,k)$ *for every* m*. Hence the conjecture at rank* k *is a single statement, at the single size* $k(k+1)/2+1$*.*

Proof. Proposition 5.17, iterated, gives $\mathrm{GTZ}(m,k)$ for every $m \leq N(k)$; Theorem 5.15 then gives every m. □

5.6. **Rank three.** At $k=3$ the bound of Theorem 5.15 is $M(3)=7$ and Corollary 5.12 disposes of every size up to five.

Corollary 5.19. $\mathrm{GTZ}(m,3)$ *for all* m *is equivalent to the conjunction of* $\mathrm{GTZ}(6,3)$ *and* $\mathrm{GTZ}(7,3)$*, and also to* $\mathrm{GTZ}(7,3)$ *alone.*

Proof. The first equivalence is Theorem 5.15 together with Corollary 5.12; the second is Theorem 5.18 at $k=3$, or equally Proposition 5.17 applied once to obtain $\mathrm{GTZ}(6,3)$ from $\mathrm{GTZ}(7,3)$. □

Both cells are carried through the rest of the paper even though the second implies the first, because the variational argument of §7 at a cell consumes the two cells one size below: at $(7,3)$ it consumes $(6,3)$ and $(6,2)$. The logical reduction to one cell and the analytic route to that cell run in opposite directions.

Neither cell is reducible by the symmetries at hand. The cell $(6,3)$ is self-dual, having rank and corank both equal to 3; and Theorem 3.14 carries $(7,3)$ to $(7,4)$, again of corank three. Since the corank arm stops at two (Theorem 5.11), duality cannot shorten the list.

The same pair of cells is reached from above. Padding a design at (m,k) with one vector along a new coordinate axis produces a design at $(m+1,k+1)$, and the padding reverses:

Proposition 5.20 (Lean)**.** *Let* $1 \leq k \leq m$*. Then* $\mathrm{GTZ}(m+1,k+1) \Rightarrow \mathrm{GTZ}(m,k)$.

Proof. Let $\mathcal{D}$ be a design at (m,k) in which no k-subset dominates. For each of the finitely many k-subsets C there is, by (2), a vector $p_C \neq 0$ with $\sum_{c\in C}\langle g_c,p_C\rangle^2 < \|p_C\|^2$; let $\sigma \in (0,1)$ be the average of 1 and the largest of the finitely many ratios $\sum_{c\in C}\langle g_c,p_C\rangle^2/\|p_C\|^2$, so that

$$\sum_{c\in C}\langle g_c,p_C\rangle^2 < \sigma\|p_C\|^2 \qquad \text{for every } k\text{-subset } C. \tag{46}$$

Build a design at $(m+1,k+1)$: the vectors $\hat{g}_c := (\sigma^{-1/2}g_c, 0)$ with weights σt_c for $c \leq m$, together with $\hat{g}_{m+1} := (0,\dots,0,(1-\sigma)^{-1/2})$ of weight $1-\sigma$. The weights are positive and sum to one, and (1) holds blockwise, the two scalings cancelling in each block. By hypothesis some C^+ with $|C^+| = k+1$ dominates there. If $m+1 \notin C^+$, every selected vector has last coordinate zero and the last standard basis vector gives $1 \leq 0$. So $m+1 \in C^+$ and $C := C^+ \setminus \{m+1\}$ has k elements; testing domination against $(p_C,0)$ gives

$$\|p_C\|^2 \;\leq\; \sigma^{-1}\sum_{c\in C}\langle g_c,p_C\rangle^2 \;<\; \|p_C\|^2$$

by (46), a contradiction. Hence some k-subset of $\mathcal{D}$ dominates after all. □

Combining Proposition 5.20 with Corollary 5.19, GTZ$(8,4)$ implies all of rank three. The family of statements GTZ(m,k) is therefore monotone in both parameters, and no implication in this section can be shown to be strict: were the conjecture true, all of them would be equivalences, so a strictness proof would refute it.

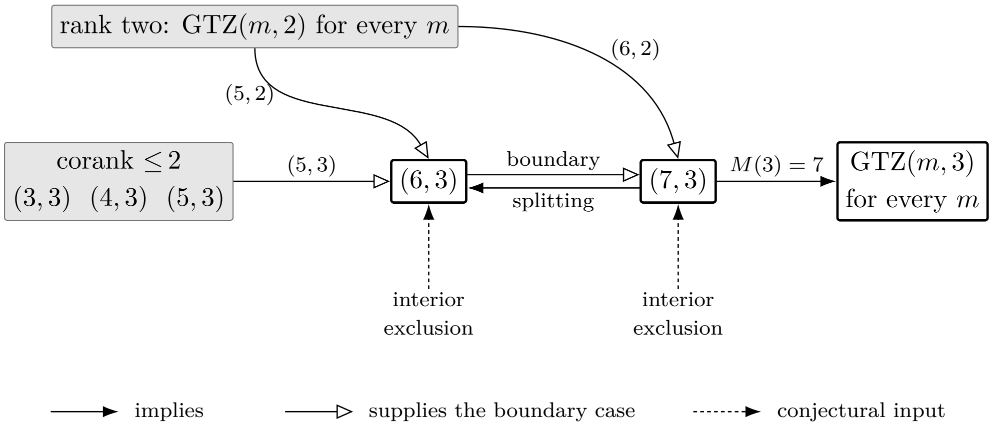


FIGURE 5. Rank three. Solid arrows are the reductions of Corollary 5.12 and Theorem 5.15; the boundary analysis of §7 at a cell consumes the two cells one size below. Dashed arrows carry the two unproved inputs of §8.

6. The equality locus

The inequality (2) is an equality on a large set, and how large that set is governs what a proof of the conjecture may look like. This section constructs the set at every size, recalls its classification at corank one, and computes its dimension: it meets every point of the open weight simplex. The consequence, used throughout the rest of the paper, is that no argument for GTZ can carry a margin.

A second theme runs alongside. Volume sampling is the probability measure on k-subsets adapted to (1), and under it the selection inequality holds on average, by a factor growing with the rank. Turning that average into a single subset is the derandomization the problem asks for; §6.5 says exactly how much of it the identities give and where the gap begins.

6.1. Extremal designs.

Definition 6.1. A design is *extremal* if some k-subset dominates and no k-subset dominates strictly: there is C with $S_C \succeq I_k$, and there is no C with $S_C \succ I_k$.

For a Hermitian matrix A one has $A \succeq I_k$ exactly when $\lambda_{\min}(A) \geq 1$, and $A \succ I_k$ exactly when $\lambda_{\min}(A) > 1$. The definition therefore reads as follows.

Lemma 6.2. *A design is extremal if and only if* $\max_{|C|=k} \lambda_{\min}(S_C) = 1$. *At an extremal design every dominating k-subset has* $\lambda_{\min}(S_C) = 1$ *exactly, so that* $S_C - I_k$ *is positive semidefinite and singular.*

Proof. The maximum is at least 1 precisely when some C dominates, and exceeds 1 precisely when some C dominates strictly. A dominating C has $\lambda_{\min}(S_C) \geq 1$, and the maximum being 1 forces equality. $\square$

The quantity in Lemma 6.2 will recur; for $m \geq k$ put

$$\alpha(m,k) \;:=\; \inf_{\mathcal{D}} \max_{|C|=k} \lambda_{\min}(S_C), \tag{47}$$

the infimum over real designs of size m and rank k; this is the real counterpart of the constant of Remark 4.14, refined by the size. By construction $\alpha_k(\mathbb{F}) = \inf_{m \geq k} \alpha(m,k)$, so the assertion $\alpha_k(\mathbb{R}) \geq 1$ of §4.1 is $\mathrm{GTZ}(m,k)$ for every m. Since a maximum is at least the infimum of maxima, $\mathrm{GTZ}(m,k)$ holds if and only if $\alpha(m,k) \geq 1$, and a design is extremal exactly when it realizes the value 1.

At corank zero there is nothing to construct. If $m = k$ and $k \geq 2$ then the unique k-subset satisfies

$$S_C - I_k \;=\; \sum_c (1 - t_c)\, g_c g_c^{\mathsf{H}} \;\succ\; 0,$$

because the g_c are then a basis of $\mathbb{F}^k$ and $m \geq 2$ forces $t_c < 1$ for every c. So no design at $m = k$, $k \geq 2$, is extremal. Extremal designs first occur at corank one, where by Theorem 3.19(iii)–(iv) they occur at every weight vector; the next subsection extends them to every larger size.

6.2. **Splitting, and existence at every size.** The construction uses that S_C does not see the weights, being by (2) the sum of the rank-one matrices attached to the selected vectors, and it uses the *gap form* behind Lemma 6.3: for every k-subset C and every $x \in \mathbb{F}^k$,

$$x^{\mathsf{H}}(S_C - I_k)x \;=\; \sum_{c \in C} |\langle g_c, x\rangle|^2 \;-\; \|x\|^2. \tag{48}$$

Lemma 6.3. *Let C be a k-subset containing two distinct indices c, c' with $g_{c'} = a\, g_c$ for some scalar a. Then S_C is singular and C does not dominate.*

Proof. The family $(g_c)_{c \in C}$ has k members and spans a subspace of dimension at most $k-1$, so some unit vector x is orthogonal to all of them. Then $S_C x = 0$, and by (48), $x^{\mathsf{H}}(S_C - I_k)x = -1 < 0$. $\square$

Now let $\mathcal{D}$ be a design at (m,k), let e be an index and let $0 < a < t_e$. Replace the pair (g_e, t_e) by the two pairs (g_e, a) and $(g_e, t_e - a)$, leaving everything else alone. The weights remain positive and still sum to one, and

$$a\, g_e g_e^{\mathsf{H}} + (t_e - a)\, g_e g_e^{\mathsf{H}} \;=\; t_e\, g_e g_e^{\mathsf{H}},$$

so (1) is unchanged and the result is a design at $(m+1,k)$. The vector is not rescaled: (1) is affected only through the total weight carried by a direction. This is the *split* at e with ratio a.

Iterating splits from a design at $(k+1,k)$ produces, at size m, a design whose m vectors take at most $k+1$ distinct values and whose weights are arbitrary subject to prescribed class totals. We construct these directly, which is shorter and displays the reweighting.

Theorem 6.4 (Lean)**.** *Let $k \geq 1$ and $m \geq k+1$, let t be a weight vector on m atoms, and let $\varphi : \{1, \ldots, m\} \to \{1, \ldots, k+1\}$ be a surjection. Put*

$$T_j \;:=\; \sum_{\varphi(c) = j} t_c \qquad (1 \leq j \leq k+1),$$

let $(h_j, T_j)_{j \leq k+1}$ be a design at $(k+1,k)$ with weights T in which every k-subset dominates and none dominates strictly, and set $g_c := h_{\varphi(c)}$. Then $(g_c, t_c)_{c \leq m}$ is an extremal design at (m,k) with weights exactly t. In particular extremal designs exist at (m,k) for every $m \geq k+1$, over either field, and at every weight vector.

Proof. The T_j are positive and sum to one, so Theorem 3.19(iii)–(iv) supplies the design (h_j, T_j) demanded. Grouping the sum (1) by classes,

$$\sum_{c=1}^{m} t_c\, g_c g_c^{\mathsf{H}} \;=\; \sum_{j=1}^{k+1} \Big(\sum_{\varphi(c) = j} t_c \Big) h_j h_j^{\mathsf{H}} \;=\; \sum_{j=1}^{k+1} T_j\, h_j h_j^{\mathsf{H}} \;=\; I_k,$$

so (g,t) is a design at (m,k), with weights t by construction.

Let C be a k-subset. If φ is not injective on C then C carries two equal vectors and does not dominate, by Lemma 6.3. If φ is injective on C then $\varphi(C)$ is a k-subset of $\{1,\dots,k+1\}$ and

$$S_C \;=\; \sum_{c\in C} h_{\varphi(c)} h_{\varphi(c)}^{\mathsf{H}} \;=\; S_{\varphi(C)},$$

which dominates and does not dominate strictly. Since φ is surjective there is at least one C of the second kind: choose k of the $k+1$ classes and one index from each. Hence some k-subset dominates and none dominates strictly. □

At $m = k+1$ the surjection is a bijection and the theorem returns Theorem 3.19(iii)–(iv). At $m \geq k+2$ some class contains two indices, so the design carries a pair of equal vectors; §6.4 shows that this feature is not forced.

Proposition 6.5. *For fixed φ the assignment $t \mapsto \mathcal{D}(t,\varphi)$ of Theorem 6.4 is a section of the weight map over the whole open simplex: every weight vector carries an extremal design. At $m = k+1$ it is onto, in the following sense: two extremal designs of size $k+1$ with the same weights differ by*

$$g_c \;\longmapsto\; \varepsilon_c U g_c, \qquad U \in O(k) \text{ resp. } U(k), \qquad |\varepsilon_c| = 1,$$

a transformation under which no S_C changes. Consequently the extremal locus at corank one has dimension exactly $m-1 = k$ modulo that group, and at every $m \geq k+1$ the extremal locus contains, for each partition of $\{1,\dots,m\}$ into $k+1$ nonempty blocks, a family of dimension $m-1$.

Proof. Only the corank-one assertion needs an argument. Let $m = k+1$. By Theorem 3.19 a design of that size with weights t is extremal if and only if its chart projection is $P = I - zz^{\mathsf{H}}$ with $|z_c|^2 = (1-t_c)/k$ for every c. One direction is part (iii) of that theorem. For the other, extremality forbids strict domination, so by part (i) no p_{c_0} exceeds the t-mean $\sum_c t_c p_c$; a family none of whose members exceeds its own weighted mean is constant, and by (19) the constant is $1/k$, which is the stated condition on z. These moduli determine z up to the individual phases ε_c, and P determines V up to $V \mapsto VU$, since two isometries with the same range differ by the unitary $U = V^{\mathsf{H}}V'$. Replacing z by $\operatorname{diag}(\varepsilon)z$ replaces P by $\operatorname{diag}(\varepsilon)P\operatorname{diag}(\varepsilon)^{\mathsf{H}}$, hence V by $\operatorname{diag}(\varepsilon)V$, hence $g_c = v_c/\sqrt{t_c}$ by $\varepsilon_c g_c$; and $g_c g_c^{\mathsf{H}}$ is invariant under that. So the fibre of the weight map over each t is a single orbit, the quotient is mapped bijectively onto the open simplex, and the dimension is $m-1=k$. □

The corank-one classification does not extend: the equations (21) that cut it out have no solutions above corank one.

Proposition 6.6 (Lean)**.** *Let $k \geq 2$. If a design of size m satisfies $k\,t_c\ell_c = k-1+t_c$ at every atom, then $m = k+1$.*

Proof. Sum (21) over c. The left side is $k\sum_c t_c\ell_c = k^2$ by Lemma 2.3; the right side is $m(k-1) + \sum_c t_c = m(k-1)+1$. Hence $(k-1)(k+1) = m(k-1)$, and $k \geq 2$ permits the cancellation. □

So above corank one the extremal designs are cut out by no such closed system, and indeed the designs of Theorem 6.4 violate (21) at most of their atoms. The following instances are used again in §8 and in Appendix B.

Example 6.7 (Lean)**.** Take for (h_j) the regular tetrahedron of Example 2.11, so that $k = 3$, $T_j = \frac{1}{4}$, $\ell(h_j) = 3$ and $\langle h_i, h_j\rangle = -1$ for $i \neq j$.

At $m = 6$, with $\varphi = (1,2,3,3,4,4)$ and weights $(\frac{1}{4},\frac{1}{4},a,\frac{1}{4}-a,b,\frac{1}{4}-b)$ for any $0 < a,b < \frac{1}{4}$, Theorem 6.4 produces a two-parameter family of extremal designs at $(6,3)$. Every leverage is 3, while (21) at $k=3$ reads $9t_c = 2+t_c$, that is $t_c = \frac{1}{4}$: the law fails at each of the four split atoms, in accordance with Proposition 6.6.

At $m = 7$, with $\varphi = (1, 2, 3, 4, 1, 2, 3)$ and weights $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$, one obtains an extremal design at $(7, 3)$ in which again every leverage is 3. Of its 35 triples, the 20 carrying three distinct classes dominate: for such a triple T, missing the class j, the computation of Example 2.11 gives

$$S_T \;=\; 4I_3 - h_j h_j^\mathsf{T}, \qquad x^\mathsf{T}(S_T - I_3)x \;=\; 3\|x\|^2 - \langle h_j, x\rangle^2,$$

so that the gap has spectrum $\{0, 3, 3\}$ and $\lambda_{\min}(S_T) = 1$. The remaining 15 triples repeat a class and do not dominate, by Lemma 6.3.

6.3. **No margin.**

Corollary 6.8 (Lean)**.** *Let $k \geq 1$, $m \geq k+1$ and $\varepsilon > 0$. For every weight vector t there is a design at (m, k) with weights t admitting no k-subset with $S_C \succeq (1+\varepsilon)I_k$. Consequently $\alpha(m, k) \leq 1$, and if* GTZ(m, k) *holds then $\alpha(m, k) = 1$ and the infimum in (47) is attained.*

Proof. Let $\mathcal{D}$ be the extremal design of Theorem 6.4 with weights t. If some k-subset had $S_C \succeq (1+\varepsilon)I_k$ then $S_C - I_k \succeq \varepsilon I_k \succ 0$, so that C would dominate strictly. The remaining assertions are Lemma 6.2. □

Corollary 6.8 constrains what a proof of the conjecture can look like.

There is no uniform margin. For every $\varepsilon > 0$ the assertion "every design admits a k-subset with $S_C \succeq (1+\varepsilon)I_k$" is false, at every cell with $m \geq k+1$. An argument whose output is a strict inequality — a positivity certificate carrying a strictly positive multiplier, or a bound $\lambda_{\min}(S_C) \geq 1 + f$ with f a positive function of the design data — therefore fails at the extremal designs themselves, whatever its degree or complexity.

There is no slack to transport. Suppose GTZ(m, k) is derived from a smaller cell (m', k') with $m' \geq k' + 1$. The slack $\max_C \lambda_{\min}(S_C) - 1$ available at the smaller cell is zero at the designs of Theorem 6.4, so no step may consume a positive amount of it: the reduction must be an identity and not an estimate. The compression lift of §7.2 is such an identity at vanishing weight, and Theorem 7.21 exhibits its failure at every positive weight, on the extremal locus itself.

The exactness, finally, occupies a set of full dimension. By Proposition 6.5 the extremal locus meets every point of the open weight simplex, at every size $m \geq k+1$, and has dimension $m-1$, which is the dimension of the weights. No restriction on the weights avoids it, and no compactness argument on a subregion of the design space yields a positive infimum for the slack unless the closure of that subregion misses the locus. The analysis at the extremal points must therefore be first-order, and §7 and §8 supply it. Appendix A collects the strategies that fail for these reasons, together with their refuting witnesses.

6.4. **The diamond.** Every design of Theorem 6.4 of size $m \geq k+2$ carries a pair of equal vectors and takes at most $k+1$ distinct values; neither feature is forced. We exhibit an extremal design at $(5, 3)$ whose five vectors are pairwise non-parallel. It is described by a graph, and its data are rational even though its vectors are not.

Let $\mathcal{G}$ be the complete graph on the vertices $1, 2, 3, 4$ with the edge 34 deleted, grounded at the vertex 4, and list its five edges as

$$e_0 = 12, \qquad e_1 = 13, \qquad e_2 = 14, \qquad e_3 = 23, \qquad e_4 = 24.$$

A potential is a vector $y = (y_1, y_2, y_3) \in \mathbb{R}^3$, the ground carrying $y_4 = 0$, and the *drop* across an edge is the difference of the potentials at its endpoints:

$$d(y) \;=\; (y_1 - y_2,\; y_1 - y_3,\; y_1,\; y_2 - y_3,\; y_2) \;=\; \big(\langle b_e, y\rangle\big)_{0 \leq e \leq 4}, \tag{49}$$

the reduced incidence vectors being

$$b_0 = (1, -1, 0), \quad b_1 = (1, 0, -1), \quad b_2 = (1, 0, 0), \quad b_3 = (0, 1, -1), \quad b_4 = (0, 1, 0).$$

Give the edges the conductances and weights

$$c \;=\; (2,3,3,3,3), \qquad t_e \;=\; \tfrac{1}{5} \quad (0 \le e \le 4),$$

and form the weighted Laplacian

$$L \;:=\; \sum_{e=0}^{4} c_e\, b_e b_e^{\mathsf T} \;=\; \begin{pmatrix} 8 & -2 & -3 \\ -2 & 8 & -3 \\ -3 & -3 & 6 \end{pmatrix}, \qquad y^{\mathsf T} L y \;=\; \sum_e c_e\, d_e(y)^2. \tag{50}$$

Its leading principal minors are 8, 60 and $\det L = 180$, so $L \succ 0$. Choose any R with $R^{\mathsf T} L R = I_3$ — for instance $R = L^{-1/2}$ — and put

$$g_e \;:=\; \sqrt{5c_e}\; R^{\mathsf T} b_e \qquad (0 \le e \le 4). \tag{51}$$

Then

$$\sum_e t_e\, g_e g_e^{\mathsf T} \;=\; \sum_e \tfrac{1}{5} \cdot 5c_e\, R^{\mathsf T} b_e b_e^{\mathsf T} R \;=\; R^{\mathsf T} L R \;=\; I_3,$$

so (g_e, t_e) is a design at $(5,3)$; call it $\mathcal{D}_\Diamond$. It is determined by the graph data up to the ambiguity $R \mapsto RU$ with U orthogonal, which replaces every g_e by $U^{\mathsf T} g_e$ and changes no S_C.

Since $RR^{\mathsf T} = L^{-1}$,

$$\ell_e \;=\; 5c_e\, b_e^{\mathsf T} L^{-1} b_e, \qquad L^{-1} \;=\; \frac{1}{180}\begin{pmatrix} 39 & 21 & 30 \\ 21 & 39 & 30 \\ 30 & 30 & 60 \end{pmatrix}, \tag{52}$$

whence $b_0^{\mathsf T} L^{-1} b_0 = \frac{36}{180} = \frac{1}{5}$ and $b_e^{\mathsf T} L^{-1} b_e = \frac{39}{180} = \frac{13}{60}$ for $e \ge 1$, so that

$$\ell_0 = 2, \qquad \ell_1 = \ell_2 = \ell_3 = \ell_4 = \tfrac{13}{4}.$$

As a check, $\sum_e t_e \ell_e = \frac{1}{5}(2 + 4 \cdot \frac{13}{4}) = 3 = k$, as Lemma 2.3 requires. The substitution $y = Rx$, a bijection of $\mathbb{R}^3$, gives for every 3-subset C

$$x^{\mathsf T} S_C x = \sum_{e \in C} 5c_e\, d_e(y)^2, \qquad x^{\mathsf T} x = x^{\mathsf T} R^{\mathsf T} L R x = \sum_e c_e\, d_e(y)^2,$$

so that domination becomes a statement about potentials alone, free of the whitening R:

$$x^{\mathsf T}(S_C - I_3)x \;=\; Q_C(y), \qquad Q_C(y) := \sum_{e=0}^{4} (5 \cdot [e \in C] - 1) c_e\, d_e(y)^2, \tag{53}$$

where $[\,\cdot\,]$ is 1 when the enclosed statement holds and 0 otherwise; and C dominates if and only if $Q_C \ge 0$ on $\mathbb{R}^3$.

Example 6.9 (Lean)**.** The design $\mathcal{D}_\Diamond$ is extremal at $(5,3)$; its five vectors are pairwise non-parallel; and it is not one of the designs of Theorem 6.4.

Proof. For the star at the vertex 1, that is $C = \{e_0, e_1, e_2\}$, expanding (53) by means of (49) gives

$$\begin{aligned} Q_C(y) &= 8(y_1 - y_2)^2 + 12(y_1 - y_3)^2 + 12y_1^2 - 3(y_2 - y_3)^2 - 3y_2^2 \\ &= 32y_1^2 + 2y_2^2 + 9y_3^2 - 16y_1y_2 - 24y_1y_3 + 6y_2y_3, \end{aligned}$$

and therefore

$$2\, Q_C(y) \;=\; \left(-8y_1 + 2y_2 + 3y_3\right)^2 + 9\, y_3^2. \tag{54}$$

So C dominates. The form (54) vanishes at $y = (1,4,0)$, so it is singular and C does not dominate strictly.

There are ten 3-subsets: eight spanning trees of $\mathcal{G}$, and its two triangles $\{e_0, e_1, e_3\}$ on the vertices $1, 2, 3$ and $\{e_0, e_2, e_4\}$ on $1, 2, 4$. For each, the potential listed below gives the stated value of Q_C:

$\{e_0, e_1, e_2\}$	$(1, 4, 0)$	0	$\{e_0, e_1, e_3\}$	$(1, 1, 1)$	-6
$\{e_0, e_1, e_4\}$	$(4, 1, 5)$	0	$\{e_0, e_2, e_3\}$	$(1, 4, 5)$	0
$\{e_0, e_2, e_4\}$	$(0, 0, 1)$	-6	$\{e_0, e_3, e_4\}$	$(4, 1, 0)$	0
$\{e_1, e_2, e_3\}$	$(2, 8, 5)$	0	$\{e_1, e_2, e_4\}$	$(3, -3, 5)$	0
$\{e_1, e_3, e_4\}$	$(8, 2, 5)$	0	$\{e_2, e_3, e_4\}$	$(-3, 3, 5)$	0

Each entry is one evaluation of (53). At $\{e_1, e_2, e_4\}$ and $y = (3, -3, 5)$, for instance, the drops (49) are $(6, -2, 3, -8, -3)$ and

$$Q_C(y) = -2 \cdot 36 + 12 \cdot 4 + 12 \cdot 9 - 3 \cdot 64 + 12 \cdot 9 = 0.$$

For each triangle the potential is constant on the triangle's vertex set, so its three drops vanish and Q_C reduces to minus the Dirichlet contribution of the two edges leaving that set, namely -6. Hence no 3-subset has $Q_C > 0$ off the origin, that is none dominates strictly, while by the first paragraph one dominates: $\mathcal{D}_\Diamond$ is extremal.

Proportionality of g_e and g_f is proportionality of b_e and b_f, since R^T is invertible and the factors $\sqrt{5c_e}$ do not vanish. Proportional nonzero vectors have equal support, and the five supports $\{1, 2\}, \{1, 3\}, \{1\}, \{2, 3\}, \{2\}$ are distinct. So no two vectors are parallel; in particular $\mathcal{D}_\Diamond$ carries five distinct directions, whereas a design of Theorem 6.4 carries at most $k + 1 = 4$. $\square$

Corollary 6.10 (Lean)**.** *The assertion "every extremal design contains two parallel vectors" is true for every design of Theorem 6.4 of size $m \geq k + 2$, and false at $(5, 3)$.*

Proof. If $m \geq k + 2$ then φ is not injective, so two indices carry equal vectors. Example 6.9 is the counterexample. $\square$

An argument branching on the presence of a parallel pair therefore requires a separate treatment of $\mathcal{D}_\Diamond$, and of every extremal design not obtained by splitting a smaller one. Whether the tetrahedron and the diamond generate the whole rank-three extremal locus is Problem 5.

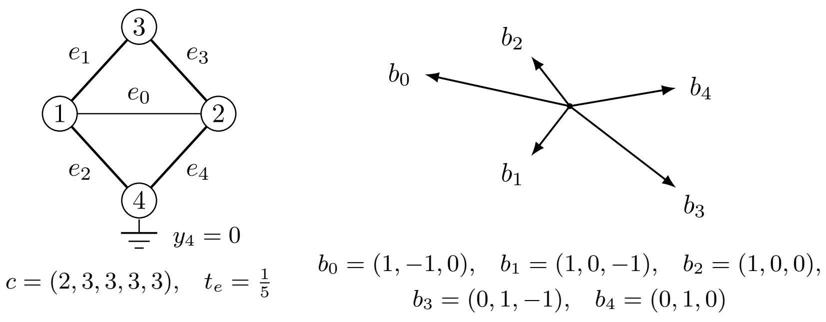


FIGURE 6. The diamond $\mathcal{D}_\Diamond$. Left: the graph K_4 with the edge 34 deleted and the vertex 4 grounded, its edges carrying the conductances $c = (2, 3, 3, 3, 3)$ and the uniform weights $t_e = \frac{1}{5}$; line weight is proportional to conductance. Right: the five reduced incidence vectors of (49), pairwise non-proportional, whence the five atoms (51) are pairwise non-parallel.

6.5. **Volume sampling.** Volume sampling is the probability measure on k-subsets adapted to (1), and under it the selection inequality holds on average. We work in the chart of §3: $v_c = \sqrt{t_c}\, g_c$, the matrix V has rows v_c^H and satisfies $V^\mathsf{H} V = I_k$, and $P = V V^\mathsf{H}$ by (13). For a k-subset C write V_C for the $k \times k$ submatrix of V on the rows C, so that $P[C] = V_C V_C^\mathsf{H}$.

Lemma 6.11. *Let X be $k \times m$ and Y be $m \times k$. Then*

$$\det(XY) \;=\; \sum_{|C|=k} \det(X^C)\det(Y_C),$$

where X^C is the submatrix of X on the columns C and Y_C the submatrix of Y on the rows C.

Proof. Write $x_1, \ldots, x_m$ for the columns of X. The j-th column of XY is $\sum_c Y_{cj} x_c$, so multilinearity of the determinant in its columns gives

$$\det(XY) \;=\; \sum_f \Big(\prod_{j=1}^k Y_{f(j)j}\Big) \det\big(x_{f(1)}, \ldots, x_{f(k)}\big),$$

the sum being over all maps $f : \{1, \ldots, k\} \to \{1, \ldots, m\}$. A non-injective f repeats a column and contributes zero. An injective f has an image C of size k; writing $f = \iota_C \circ \sigma$ with ι_C the increasing enumeration of C and σ a permutation, the determinant equals $\operatorname{sgn}(\sigma)\det(X^C)$, and summing $\operatorname{sgn}(\sigma)\prod_j Y_{\iota_C(\sigma(j))j}$ over σ gives $\det(Y_C)$. $\square$

Apply the lemma with $X = V^{\mathsf{H}}$ and $Y = V$. Each term is $\det((V^{\mathsf{H}})^C)\det(V_C) = |\det V_C|^2 = \det(V_C V_C^{\mathsf{H}}) = \det P[C]$, while the left side is $\det(V^{\mathsf{H}} V) = 1$. Setting

$$\pi_C \;:=\; \det P[C] \;=\; |\det V_C|^2 \;=\; \Big(\prod_{c \in C} t_c\Big) \det \operatorname{Gram}\big((g_c)_{c\in C}\big) \tag{55}$$

— the last equality because $V_C = \operatorname{diag}(\sqrt{t_c})_{c\in C} M_C$ with M_C the matrix of the selected vectors — we conclude that π is a probability measure on k-subsets, and that $\pi_C > 0$ exactly when $(g_c)_{c\in C}$ is a basis of $\mathbb{F}^k$. Call such a C a *basis* of the design. The measure π is volume sampling [9]; it is the determinantal measure carried by the bases of the underlying matroid [17].

Lemma 6.12. *For every atom c,*

$$\Pr_\pi[c \in C] \;=\; \sum_{C \ni c} \pi_C \;=\; P_{cc} \;=\; s_c.$$

Proof. Fix c and let $D_\sigma := \operatorname{diag}(1, \ldots, \sigma, \ldots, 1)$ carry σ in the position c. On one hand $V^{\mathsf{H}} D_\sigma V = V^{\mathsf{H}} V + (\sigma - 1) v_c v_c^{\mathsf{H}}$ is a rank-one perturbation of the identity, with

$$\det(V^{\mathsf{H}} D_\sigma V) \;=\; 1 + (\sigma - 1)\|v_c\|^2 \;=\; 1 + (\sigma - 1)s_c.$$

On the other hand Lemma 6.11 applied to $X = V^{\mathsf{H}} D_\sigma$ and $Y = V$ gives, since scaling the column e of V^{H} multiplies every minor containing it by that factor,

$$\det(V^{\mathsf{H}} D_\sigma V) \;=\; \sum_{|C|=k} \sigma^{[c\in C]}\,\pi_C \;=\; \sum_{C \not\ni c} \pi_C \;+\; \sigma \sum_{C\ni c} \pi_C.$$

Both expressions are affine in σ; comparing the coefficients of σ gives the claim. $\square$

Summing Lemma 6.12 over c returns $\sum_c s_c = k$ (Lemma 2.3), the expected cardinality of C, as it must.

Theorem 6.13 (Lean)**.** *For every design over either field,*

$$\sum_c s_c\, g_c g_c^{\mathsf{H}} \;\succeq\; I_k.$$

Proof. Let $x \in \mathbb{F}^k$ be arbitrary. Cauchy–Schwarz at each atom gives $|\langle g_c, x\rangle|^2 \le \ell_c \|x\|^2$, whence

$$\|x\|^2 \sum_c s_c |\langle g_c, x\rangle|^2 \;=\; \sum_c t_c\, \ell_c \|x\|^2\, |\langle g_c, x\rangle|^2 \;\ge\; \sum_c t_c |\langle g_c, x\rangle|^4.$$

Cauchy–Schwarz for the probability weights t bounds the right side below by

$$\Big(\sum_c t_c|\langle g_c, x\rangle|^2\Big)^2 \;=\; \|x\|^4$$

by (1). Dividing by $\|x\|^2 > 0$ gives $x^{\mathsf{H}}(\sum_c s_c g_c g_c^{\mathsf{H}})x \geq \|x\|^2$. Every step is homogeneous in x, so no normalization is used. ☐

The leverage-weighted sum is the π-average of the subset sum.

Theorem 6.14. *For every design over either field,*

$$\mathbb{E}_\pi[S_C] \;=\; \sum_c s_c\, g_c g_c^{\mathsf{H}} \;\succeq\; I_k.$$

Proof. Exchanging the two summations and applying Lemma 6.12,

$$\mathbb{E}_\pi[S_C] \;=\; \sum_{|C|=k} \pi_C \sum_{c\in C} g_c g_c^{\mathsf{H}} \;=\; \sum_c \Big(\sum_{C\ni c} \pi_C\Big) g_c g_c^{\mathsf{H}} \;=\; \sum_c s_c\, g_c g_c^{\mathsf{H}},$$

and the inequality is Theorem 6.13. ☐

The selection problem is thus the derandomization of Theorem 6.14: the π-average of S_C dominates, and one asks for a single C. The average dominates with a margin that grows with the rank.

Proposition 6.15 (Lean). *$\sum_c t_c \ell_c^2 \geq k^2$, with equality if and only if $\ell_c = k$ for every c.*

Proof. Cauchy–Schwarz for the weights t gives $\left(\sum_c t_c \ell_c\right)^2 \leq \sum_c t_c \ell_c^2$, and the left side is k^2 by Lemma 2.3. Equality holds exactly when ℓ is constant, and its t-mean is k. ☐

Since $\operatorname{tr}\mathbb{E}_\pi[S_C] = \sum_c s_c \ell_c = \sum_c t_c \ell_c^2 \geq k^2$ while $\operatorname{tr} I_k = k$, the average clears the threshold by a factor at least k in trace. No single subset need clear it at all: that is Corollary 6.8.

The measure sees the extremal locus through the mixture of characteristic polynomials

$$q(x) \;:=\; \mathbb{E}_\pi[\chi_{S_C}(x)] \;=\; \sum_{|C|=k} \pi_C \det(xI_k - S_C). \tag{56}$$

Call a basis C *tight* if $1 \in \operatorname{spec} S_C$, and a design *totally tight* if every basis of it is tight.

Proposition 6.16. *Every design of Theorem 6.4 is totally tight.*

Proof. Let C be a k-subset. If φ is not injective on C then S_C is singular by Lemma 6.3, so $\pi_C = 0$ by (55) and C is not a basis. Otherwise $S_C = S_{\varphi(C)}$ dominates and does not dominate strictly, so $\lambda_{\min}(S_C) = 1$ by Lemma 6.2. ☐

Proposition 6.17. *The mixture (56) is monic of degree k; its subleading coefficient is*

$$[x^{k-1}]\, q \;=\; -\,\mathbb{E}_\pi[\operatorname{tr} S_C] \;=\; -\sum_c t_c \ell_c^2 \;\leq\; -k^2;$$

and $q(1) = 0$ at every totally tight design.

Proof. Each χ_{S_C} is monic of degree k and $\sum_C \pi_C = 1$, so q is monic of degree k; and $[x^{k-1}]\chi_{S_C} = -\operatorname{tr} S_C$, which gives the first equality. The second is Lemma 6.12 again:

$$\mathbb{E}_\pi[\operatorname{tr} S_C] = \sum_C \pi_C \sum_{c\in C} \ell_c = \sum_c \ell_c \sum_{C\ni c} \pi_C = \sum_c s_c \ell_c = \sum_c t_c \ell_c^2,$$

and the bound is Proposition 6.15. The last assertion is Proposition 6.18. ☐

Proposition 6.18 (Lean). *$q(1) = 0$ at every totally tight design.*

Proof. $q(1) = \sum_C \pi_C \det(I_k - S_C)$, and every summand with $\pi_C > 0$ has C a basis, hence tight, hence $\det(I_k - S_C) = 0$. $\square$

We compute the mixture at the tetrahedron of Example 2.11, which by Theorem 3.19(iii)–(iv) is the corank-one extremal design at $k = 3$ and uniform weights, and is therefore totally tight by Proposition 6.16. Each of its four triples has Gram matrix $4I_3 - J$, with J the all-ones matrix, of determinant 16; so by (55)

$$\pi_C \;=\; \left(\tfrac{1}{4}\right)^3 \cdot 16 \;=\; \tfrac{1}{4}$$

for each of the four, confirming $\sum_C \pi_C = 1$ and giving $\Pr_\pi[c \in C] = 3 \cdot \frac{1}{4} = \frac{3}{4} = s_c$ as Lemma 6.12 requires. Every triple has spectrum $\{1, 4, 4\}$, so every $\chi_{S_C}(x) = (x-1)(x-4)^2$ and the mixture is that same polynomial:

$$q(x) \;=\; (x-1)(x-4)^2 \;=\; x^3 - 9x^2 + 24x - 16. \tag{57}$$

Its subleading coefficient is $-9 = -\sum_c t_c \ell_c^2 = -4 \cdot \frac{1}{4} \cdot 9$, so the bound of Proposition 6.15 is attained here; $q(1) = 0$, as Proposition 6.17 requires; and the least root of q sits at the threshold. Meanwhile $\mathbb{E}_\pi[S_C] = \frac{3}{4} \sum_c g_c g_c^\top = 3I_3$ while $\max_C \lambda_{\min}(S_C) = 1$: the average clears the threshold by a factor of three, and no triple clears it at all.

Remark 6.19. Only the identities above have been used; three nearby statements do not follow from them. First, q is not asserted to be real-rooted; for a projection determinantal measure that is available through the strong Rayleigh property [2, 1], and nothing here needs it. Second, real-rootedness of q would not imply that some C has $\lambda_{\min}(S_C)$ at least the least root of q: the least root of an average of real-rooted polynomials may exceed the least root of every member, and an exact two-polynomial witness is recorded in Appendix A. Third, the hypothesis under which the conclusion does follow is a common interlacing of the family $\{\chi_{S_C} : \pi_C > 0\}$, equivalently real-rootedness of every convex combination of it [18]; that is a property of the family, to be verified and not inferred, and it fails for the witness just cited. Establishing it here, and with it the derandomization of Theorem 6.14, is Problem 4; by Proposition 6.17 any such argument is exact at the extremal designs, where the least root of q is 1.

7. A variational principle

A counterexample to $\mathrm{GTZ}(m, k)$ is a design whose selection objective falls below the threshold. One would like to take the worst such design and read off what minimality forces; but the designs of a given size and rank form a space that is not compact, and on it the objective attains no minimum. This section supplies a compact space containing that one. The objective extends to it continuously, every point added at the boundary turns out to carry a dominating subset, and a counterexample may therefore be taken to minimize globally, hence to satisfy first-order conditions.

The method has a sharp limit, which §7.4 locates. The identity that discharges the added points is available at vanishing weight and at no positive weight whatever, and its failure occurs on the extremal locus of §6. That is why the argument is run at the limit point rather than along a minimizing sequence.

7.1. **The compactification.** Fix m and k. In a design the weights are confined to the open simplex and the vectors are not confined at all. Parseval controls only the product: by Lemma 2.3,

$$\sum_c s_c \;=\; \sum_c t_c \ell_c \;=\; k, \qquad \text{so} \qquad \ell_c \;\le\; \frac{k}{t_c}. \tag{58}$$

As a weight tends to zero the corresponding vector may therefore grow without bound, and it may do so while keeping its leverage score $s_c = t_c \ell_c$ fixed.

Example 7.1. Let $k = 1$, $m = 2$ and, for $0 < u \le \frac{1}{2}$,

$$t_1 = 1-u, \quad t_2 = u, \qquad g_1 = \frac{1}{\sqrt{2(1-u)}}, \quad g_2 = \frac{1}{\sqrt{2u}}.$$

Then $t_1 g_1^2 + t_2 g_2^2 = \frac{1}{2} + \frac{1}{2} = 1$, so this is a design at $(2,1)$ for every such u. Its leverages are $\ell_1 = 1/2(1-u)$ and $\ell_2 = 1/2u$, and $\ell_2 \to \infty$ as $u \to 0$. No subsequence converges in the design space. Its leverage scores are $s_1 = s_2 = \frac{1}{2}$, constant in u.

The chart of §3 is the change of variables that retains the leverage scores and discards the divergence. With $v_c = \sqrt{t_c}\, g_c$ and $P = VV^{\mathsf{H}}$ as in (13),

$$P_{cc} \;=\; \|v_c\|^2 \;=\; t_c \ell_c \;=\; s_c, \tag{59}$$

so (58) bounds the diagonal of P, and P with it. The domination criterion, by Lemma 3.6, is a statement about (P,t) alone. In Example 7.1 the chart is constant: $v_1 = v_2 = 1/\sqrt{2}$, so

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad t = (1-u,\, u) \;\longrightarrow\; (1,0) \quad (u \to 0),$$

and the family converges, in these coordinates, to a pair (P,t) whose second weight vanishes while the corresponding diagonal entry of P does not. Such pairs are the points to be added.

Definition 7.2. A *chart point* of size m and rank k is a pair (P,t) in which P is a real symmetric idempotent $m \times m$ matrix with $\operatorname{tr} P = k$ and t lies in the closed simplex $\Delta^{m-1} = \{t \in \mathbb{R}^m : t_c \ge 0,\ \sum_c t_c = 1\}$. Put $D := \operatorname{diag}(t)$, $W := P - D$, and

$$G(P,t) \;:=\; \max_{|C|=k} \lambda_{\min}(W[C]). \tag{60}$$

A k-subset C *dominates* at (P,t) if $W[C] \succeq 0$. Write $\mathrm{GTZ}^{\mathrm{ch}}(m,k)$ for the assertion that every chart point of size m and rank k admits a dominating k-subset.

Since a maximum over a finite set is attained,

$$G(P,t) \ge 0 \quad \Longleftrightarrow \quad \text{some } k\text{-subset dominates at } (P,t). \tag{61}$$

No condition ties P to t: the relation (59) is the definition of s_c, not a constraint, and on the compactification the vector lengths are recovered only where $t_c > 0$, by $\ell_c = P_{cc}/t_c$. That formula is the boundary's only singularity, and it does not appear in Definition 7.2.

Proposition 7.3. *The chart points of size m and rank k with $t_c > 0$ for every c are exactly the chart points of designs of size m and rank k over $\mathbb{R}$, and the design is determined by the chart point up to a rotation of $\mathbb{R}^k$. Under this correspondence a k-subset dominates in the design if and only if it dominates at the chart point.*

Proof. A design produces (VV^{T}, t), which satisfies Definition 7.2: VV^{T} is symmetric, idempotent because $V^{\mathsf{T}}V = I_k$, and of trace $\operatorname{tr}(V^{\mathsf{T}}V) = k$; and t lies in the open simplex. Conversely let (P,t) be a chart point with all weights positive. A symmetric idempotent is the orthogonal projection onto its range, so $\operatorname{rank} P = \operatorname{tr} P = k$; let the columns of the $m \times k$ matrix V be an orthonormal basis of that range, so that $P = VV^{\mathsf{T}}$ and $V^{\mathsf{T}}V = I_k$. Write v_c^{T} for the c-th row of V and put $g_c := v_c/\sqrt{t_c}$. Then

$$\sum_c t_c\, g_c g_c^{\mathsf{T}} \;=\; \sum_c v_c v_c^{\mathsf{T}} \;=\; V^{\mathsf{T}}V \;=\; I_k,$$

which is (1); the weights are positive and sum to one; so this is a design, and its chart point is (P,t). Two orthonormal bases of the range differ by an element of the orthogonal group, which acts on the g_c by a rotation and leaves (2) invariant. The last assertion is Lemma 3.6. ☐

Lemma 7.4. *The set of chart points of size m and rank k is compact, and nonempty when $k \le m$.*

Proof. It is the product of Δ^{m-1}, which is compact, with the set Π_k of symmetric idempotents of trace k. The three conditions defining Π_k are polynomial equations in the entries, so Π_k is closed; and for $P \in \Pi_k$,

$$\|P\|_F^2 \;=\; \operatorname{tr}(P^\mathsf{T} P) \;=\; \operatorname{tr}(P^2) \;=\; \operatorname{tr} P \;=\; k,$$

so Π_k lies on a sphere of the Frobenius norm and is bounded. It is the Grassmannian of k-planes in $\mathbb{R}^m$, nonempty for $k \le m$. $\square$

Lemma 7.5. *G is continuous. Indeed, for chart points (P,t) and (Q,t') of the same size and rank,*

$$|G(P,t) - G(Q,t')| \;\le\; \|P-Q\|_2 + \max_c |t_c - t'_c|,$$

where $\|\cdot\|_2$ is the operator norm.

Proof. For symmetric A one has $\lambda_{\min}(A) = \min_{\|x\|=1} x^\mathsf{T} A x$, an infimum of functions of A each of which satisfies $|x^\mathsf{T} A x - x^\mathsf{T} B x| \le \|A-B\|_2$ for $\|x\| = 1$; hence $|\lambda_{\min}(A) - \lambda_{\min}(B)| \le \|A-B\|_2$. Restriction to a principal block does not increase the operator norm, and the difference of the two gap matrices is $(P-Q) - \operatorname{diag}(t-t')$, whose operator norm is at most $\|P-Q\|_2 + \max_c |t_c - t'_c|$; so each of the finitely many functions $(P,t) \mapsto \lambda_{\min}(W[C])$ obeys the stated bound; a maximum of finitely many such functions obeys it as well. $\square$

The interior of the compactification is dense in it, by a mixing of the weights which will be used twice more.

Lemma 7.6. *Let (P,t) be a chart point of size m and rank k, let $\bar t := (1/m, \dots, 1/m)$ and, for $0 \le \theta \le 1$, put $t^\theta := (1-\theta)t + \theta \bar t$. Then (P, t^θ) is a chart point, its weights are strictly positive for $\theta > 0$, and for every $x \in \mathbb{R}^m$*

$$x^\mathsf{T} W^\theta x \;=\; x^\mathsf{T} W x - \theta \sum_c \Big(\frac{1}{m} - t_c\Big) x_c^2, \qquad W^\theta := P - \operatorname{diag}(t^\theta). \tag{62}$$

Consequently the chart points with strictly positive weights are dense, and

$$\min G \;=\; \inf\{\, G(P,t) : (P,t) \text{ the chart point of a design} \,\}.$$

Proof. The simplex is convex, so $t^\theta \in \Delta^{m-1}$, and $t^\theta_c \ge \theta/m > 0$ for $\theta > 0$; P is untouched. Identity (62) is $\operatorname{diag}(t^\theta) = \operatorname{diag}(t) + \theta \operatorname{diag}(\bar t - t)$ evaluated against x. Letting $\theta \to 0$ gives $t^\theta \to t$, which is the density statement; the displayed equality of extrema then follows from Lemma 7.5 together with Proposition 7.3, which identifies the points with positive weights as the chart points of designs, and from Lemma 7.4, which supplies the minimum. $\square$

The design space itself admits no such statement. It is the locus $t_c > 0$, which is open in the compactification and, by Example 7.1, not closed in it; a minimizing sequence of designs need not converge to a design. Hence the added points must be treated, and no argument confined to designs produces a minimizer.

Remark 7.7. The objective (60) is not the design objective $F(\mathcal{D}) = \max_C \lambda_{\min}(S_C)$ of §8. With $D[C] = \operatorname{diag}(t_c)_{c \in C}$ the congruence (15) reads $W[C] = D[C]^{1/2}(\operatorname{Gram}_C - I_k) D[C]^{1/2}$ — the Gram matrix, not the subset sum — so that, substituting $x = D[C]^{-1/2} y$,

$$\frac{x^\mathsf{T} W[C] x}{x^\mathsf{T} D[C] x} \;=\; \frac{y^\mathsf{T}(\operatorname{Gram}_C - I_k) y}{\|y\|^2}.$$

For $|C| = k$ the matrix M_C is square, so $\operatorname{Gram}_C = M_C M_C^\mathsf{H}$ and $S_C = M_C^\mathsf{H} M_C$ have the same characteristic polynomial and hence the same least eigenvalue; minimizing the last display over

$y \neq 0$ therefore gives $\lambda_{\min}(S_C) - 1$, and

$$F(\mathcal{D}) - 1 \;=\; \max_{|C|=k} \; \min_{x \neq 0} \frac{x^\top W[C] x}{x^\top D[C] x},$$

against $G = \max_C \min_{\|x\|=1} x^\top W[C] x$. The two agree in sign, by (61) and Lemma 3.6. Beyond the sign they are related by a congruence which is not a scalar unless the selected weights are equal, so they do not share their minimizers. The generalized form degenerates exactly where $D[C]$ becomes singular, that is, exactly on the points the compactification adds; the absolute form does not.

7.2. **The boundary.** Let (P, t) be a chart point with $t_z = 0$ for some index z. Two identities dispose of it; the diagonal entry P_{zz} decides which applies, and the dichotomy is exhaustive by the following lemma.

Lemma 7.8. *For every chart point and every index z,*

$$\|Pe_z\|^2 \;=\; e_z^\top P^\top P e_z \;=\; e_z^\top P^2 e_z \;=\; e_z^\top P e_z \;=\; P_{zz}. \tag{63}$$

Hence $P_{zz} \geq 0$; and $P_{zz} = 0$ forces $Pe_z = 0$, that is, the z-th column and, by symmetry, the z-th row of P vanish identically.

Proof. The chain (63) uses symmetry and idempotency in that order. A vanishing sum of squares $\sum_c P_{cz}^2 = P_{zz} = 0$ has all terms zero. $\square$

By (59) the quantity P_{zz} is the limit of the leverage score s_z. The dichotomy is therefore the distinction between the two ways an atom can leave a design: its Parseval mass vanishes, or its mass survives while its weight vanishes and its vector escapes to infinity. Example 7.1 is of the second kind, with $P_{zz} = \frac{1}{2}$.

Case A: the atom vanishes.

Lemma 7.9. *Let (P, t) be a chart point of size m and rank k and let z satisfy $t_z < 1$ and $Pe_z = 0$. Delete the index z: let $\widehat{P}$ be the principal submatrix of P on the remaining indices and $\widehat{t}_c := t_c/(1 - t_z)$. Then $(\widehat{P}, \widehat{t})$ is a chart point of size $m - 1$ and rank k, and for every $y \in \mathbb{R}^m$ with $y_z = 0$, writing $\widehat{y}$ for its restriction,*

$$y^\top W y \;=\; \widehat{y}^\top \widehat{W} \widehat{y} + \frac{t_z}{1 - t_z} \sum_{c \neq z} t_c y_c^2 \;\geq\; \widehat{y}^\top \widehat{W} \widehat{y}. \tag{64}$$

Consequently a subset dominating at $(\widehat{P}, \widehat{t})$ dominates, under the inclusion of index sets, at (P, t).

Proof. Symmetry is inherited. Since the z-th row and column of P vanish, the deleted index contributes nothing to any product, so $\widehat{P}\widehat{P}$ is the corresponding submatrix of $P^2 = P$, and $\operatorname{tr} \widehat{P} = \operatorname{tr} P - P_{zz} = k$. The weights $\widehat{t}_c$ are nonnegative and sum to $(1 - t_z)/(1 - t_z) = 1$. For the identity, $y_z = 0$ gives $y^\top P y = \widehat{y}^\top \widehat{P} \widehat{y}$, while

$$\widehat{y}^\top \operatorname{diag}(\widehat{t}) \widehat{y} - y^\top D y \;=\; \sum_{c \neq z} \Big(\frac{t_c}{1 - t_z} - t_c \Big) y_c^2 \;=\; \frac{t_z}{1 - t_z} \sum_{c \neq z} t_c y_c^2,$$

which is (64) after subtraction. The right-hand side is nonnegative because $t_z \geq 0$ and $t_z < 1$. $\square$

The reweighting is favourable at every admissible t_z, not only at $t_z = 0$; the hypothesis $t_z = 0$ is needed only at the pivot of the next case, where §7.4 locates the failure.

Case B: the direction escapes.

Lemma 7.10. *Let (P,t) be a chart point of size m and rank $k \geq 1$ and let $P_{zz} > 0$. Then*

$$P' \;:=\; P - \frac{(Pe_z)(Pe_z)^{\mathsf{T}}}{P_{zz}} \tag{65}$$

is symmetric and idempotent with $\operatorname{tr} P' = k-1$ and $P'e_z = 0$. Hence (P',t) is a chart point of size m and rank $k-1$ whose z-th row and column vanish.

Proof. Write $r := Pe_z$ and $R := rr^{\mathsf{T}}$, so that $P' = P - R/P_{zz}$; symmetry is clear. Idempotency of P gives $Pr = P^2 e_z = Pe_z = r$, hence $PR = R$ and, transposing, $RP = R$. Lemma 7.8 gives $RR = r(r^{\mathsf{T}}r)r^{\mathsf{T}} = P_{zz}R$. Therefore

$$\Big(P - \frac{R}{P_{zz}}\Big)^2 \;=\; P^2 - \frac{PR + RP}{P_{zz}} + \frac{RR}{P_{zz}^2} \;=\; P - \frac{2R}{P_{zz}} + \frac{R}{P_{zz}} \;=\; P - \frac{R}{P_{zz}}.$$

The trace drops by $\operatorname{tr} R/P_{zz} = \|r\|^2/P_{zz} = 1$, again by (63). Finally $r^{\mathsf{T}}e_z = P_{zz}$, so $P'e_z = r - r\,P_{zz}/P_{zz} = 0$. □

Deflation is the compression of the surviving vectors onto the orthogonal complement of the escaping direction, carried out in the chart: P' is the Schur complement of the pivot P_{zz} in P. The computation chooses no basis of that complement and takes no square root; it performs one division, by the pivot. Since $P = P' + rr^{\mathsf{T}}/P_{zz}$ with $r = Pe_z$, one has for every $x \in \mathbb{R}^m$

$$x^{\mathsf{T}}Wx \;=\; x^{\mathsf{T}}(P' - D)x + \frac{\langle Pe_z, x\rangle^2}{P_{zz}}. \tag{66}$$

The upstairs gap form exceeds the deflated one by the rank-one pivot term; the lift below consumes that excess in full, so it is not slack. Splitting the probe along the escaping coordinate turns (66) into the identity that the rest of the section uses.

Lemma 7.11. *Let (P,t) be a chart point with $P_{zz} > 0$ and let P' be as in (65). Write $x = y + x_z e_z$ with $y_z = 0$. Then*

$$x^{\mathsf{T}}Wx \;=\; y^{\mathsf{T}}(P' - D)\,y \;+\; \frac{\big(\langle Pe_z, y\rangle + x_z P_{zz}\big)^2}{P_{zz}} \;-\; t_z\,x_z^2. \tag{67}$$

Proof. Expand $x^{\mathsf{T}}Px$ against the coordinate spike. With $a := \langle Pe_z, y\rangle$ and $e_z^{\mathsf{T}}Pe_z = P_{zz}$,

$$x^{\mathsf{T}}Px \;=\; y^{\mathsf{T}}Py + 2x_z a + x_z^2 P_{zz},$$

and (66) applied to y gives $y^{\mathsf{T}}Py = y^{\mathsf{T}}P'y + a^2/P_{zz}$. Since $y_z = 0$, $x^{\mathsf{T}}Dx = y^{\mathsf{T}}Dy + t_z x_z^2$. Subtracting,

$$x^{\mathsf{T}}Wx \;=\; y^{\mathsf{T}}(P' - D)y + \frac{a^2}{P_{zz}} + 2x_z a + x_z^2 P_{zz} - t_z x_z^2,$$

and the three middle terms are $(a + x_z P_{zz})^2/P_{zz}$, since $P_{zz} \neq 0$. □

Identity (67) is exact at every weight. Its three terms are, in order: the deflated gap, a square, and a deficit proportional to the weight at the pivot. At $t_z = 0$ the deficit is absent and the lift follows.

Theorem 7.12. *Let (P,t) be a chart point of size m and rank $k \geq 1$, let z satisfy $P_{zz} > 0$ and $t_z = 0$, and let C be a subset of the remaining indices dominating at the chart point (P',t) of rank $k-1$. Then $C \cup \{z\}$ dominates at (P,t).*

Proof. Let x be supported on $C \cup \{z\}$ and split $x = y + x_z e_z$ with $y_z = 0$; then y is supported on C, so $y^{\mathsf{T}}(P' - D)y \geq 0$ by hypothesis. In (67) the last term vanishes because $t_z = 0$, and the middle term is nonnegative because $P_{zz} > 0$. Hence $x^{\mathsf{T}}Wx \geq 0$. □

The pivot matches because the weight there is zero: the deflation (65) subtracts P_{zz} from the diagonal at z, and (67) restores the same amount.

Example 7.13. Return to Example 7.1. At its limit point, $z = 2$, $P_{zz} = \frac{1}{2}$ and $t_z = 0$. Here $Pe_2 = (\frac{1}{2}, \frac{1}{2})^\mathsf{T}$ with $\|Pe_2\|^2 = \frac{1}{2} = P_{22}$, as (63) requires, and (65) gives $P' = P - 2 \cdot \frac{1}{4}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right) = 0$, of trace $0 = k - 1$. Deleting z leaves the chart point of size 1 and rank 0, at which the empty selection dominates; Theorem 7.12 returns $\{2\}$, and indeed $W_{22} = P_{22} - t_2 = \frac{1}{2} \geq 0$.

The dichotomy.

Theorem 7.14 (Lean)**.** *Let $m \geq 2$ and $k \geq 1$, and assume $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k)$ and $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k-1)$. Then every chart point of size m and rank k with $t_z = 0$ for some z admits a dominating k-subset.*

Proof. By Lemma 7.8 either $P_{zz} = 0$ or $P_{zz} > 0$.

If $P_{zz} = 0$ then $Pe_z = 0$, and $t_z = 0 < 1$, so Lemma 7.9 produces a chart point of size $m - 1$ and rank k. By $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k)$ some k-subset dominates there, and by (64) its image dominates at (P, t).

If $P_{zz} > 0$, deflate: by Lemma 7.10, (P', t) is a chart point of rank $k - 1$ whose z-th column vanishes, so Lemma 7.9 applies to it and produces a chart point of size $m - 1$ and rank $k - 1$. By $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k-1)$ some $(k-1)$-subset C dominates there; by (64) it dominates at (P', t); and by Theorem 7.12, $C \cup \{z\}$, of cardinality k, dominates at (P, t). $\square$

Both branches reduce the size by one, so an induction on the dichotomy is well founded on the size alone; the rank window is a consequence rather than an extra hypothesis. At $k = 1$ the second hypothesis is $\mathrm{GTZ}^{\mathrm{ch}}(m-1, 0)$, which holds because the empty selection dominates. The reduction transports neither margin nor rate nor subsequence, which keeps it compatible with Corollary 6.8.

Corollary 7.15 (Lean)**.** *Under the hypotheses of Theorem 7.14, every chart point of size m and rank k admitting* no *dominating k-subset has $t_c > 0$ for every c.*

Proof. Contrapositive of Theorem 7.14. $\square$

7.3. **The principle.**

Corollary 7.15 speaks about a given chart point. Before using it, we record that the closed statement and the statement about designs are the same statement, so the hypotheses of Theorem 7.14 are ordinary smaller instances of the problem.

Proposition 7.16. *Write $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ for the assertion that every chart point of size m and rank k with strictly positive weights admits a dominating k-subset. Then $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ implies $\mathrm{GTZ}^{\mathrm{ch}}(m, k)$, at the same size and rank.*

Proof. Let (P, t) be a chart point at which no k-subset dominates. For each of the finitely many k-subsets C choose x^C supported on C with $(x^C)^\mathsf{T} W x^C < 0$, scaled so that $(x^C)^\mathsf{T} W x^C = -1$. Put $\mu_C := \sum_c (1/m - t_c)(x^C_c)^2$ and $\theta := \left(1 + \sum_C |\mu_C|\right)^{-1} \in (0, 1]$. By (62),

$$(x^C)^\mathsf{T} W^\theta x^C \;=\; -1 - \theta\mu_C \;<\; 0 \qquad \text{for every } C,$$

since $\theta|\mu_C| < 1$. So no k-subset dominates at (P, t^θ), whose weights are strictly positive by Lemma 7.6, contradicting $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$. $\square$

Corollary 7.17. *For every m and k the three statements $\mathrm{GTZ}(m, k)$, $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ and $\mathrm{GTZ}^{\mathrm{ch}}(m, k)$ are equivalent.*

Proof. By Proposition 7.3 the designs of size m and rank k correspond to the chart points with positive weights, and domination corresponds to domination; so $\mathrm{GTZ}(m, k)$ and $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ are the same assertion. Proposition 7.16 gives $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k) \Rightarrow \mathrm{GTZ}^{\mathrm{ch}}(m, k)$, and the converse is trivial, a chart point with positive weights being a chart point. $\square$

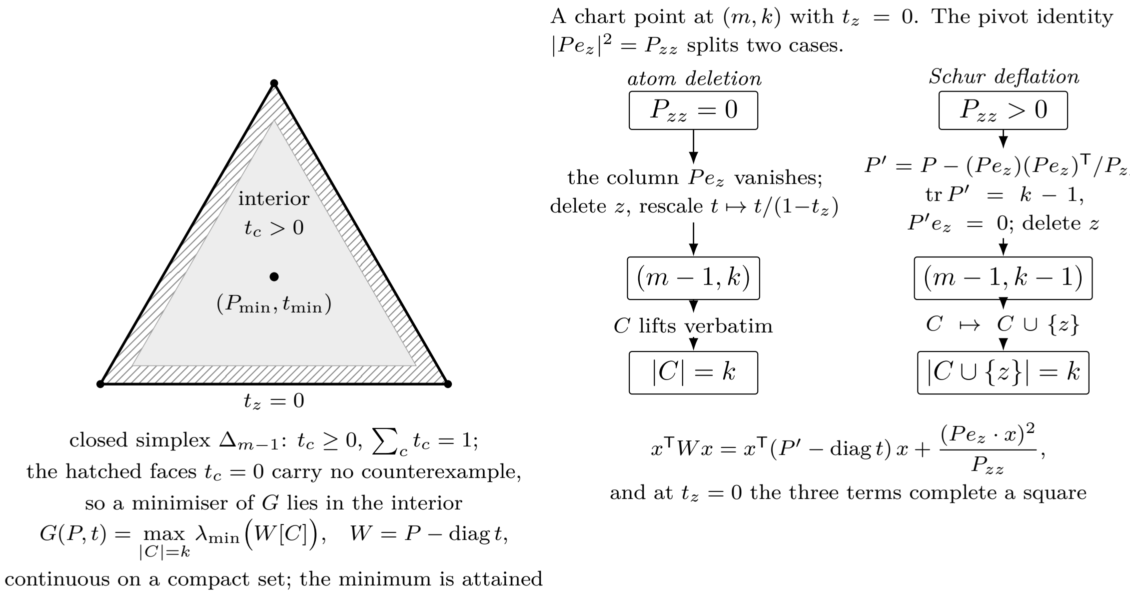


FIGURE 7. The compactification and the boundary dichotomy. Left: the objective G of (60) is continuous on a compact space, so it attains its minimum; Theorem 7.14 empties the faces $t_z = 0$ of counterexamples, which places the minimum in the interior. Right: the two mechanisms, separated by the pivot identity (63). At $P_{zz} = 0$ the atom is deleted and the weights rescaled; at $P_{zz} > 0$ the point is deflated by (65) and the selected subset regains z, the deficit in (67) being absent exactly because $t_z = 0$.

Theorem 7.18. *Let $m > k \geq 1$ and assume* $\mathrm{GTZ}(m-1,k)$ *and* $\mathrm{GTZ}(m-1,k-1)$*. If* $\mathrm{GTZ}(m,k)$ *fails, then G attains its minimum over the compact space of chart points of size m and rank k at a point $(P_{\min}, t_{\min})$ with*

$$G(P_{\min}, t_{\min}) < 0 \qquad \text{and} \qquad (t_{\min})_c > 0 \ \text{ for every } c.$$

In particular $(P_{\min}, t_{\min})$ is the chart point of a design $\mathcal{D}_{\min}$ of size m and rank k; no k-subset dominates in $\mathcal{D}_{\min}$; and $\mathcal{D}_{\min}$ minimizes G over all designs of size m and rank k.

Proof. By Corollary 7.17 the hypotheses are $\mathrm{GTZ}^{\mathrm{ch}}(m-1,k)$ and $\mathrm{GTZ}^{\mathrm{ch}}(m-1,k-1)$, so Theorem 7.14 and Corollary 7.15 are available at size m and rank k.

A counterexample to $\mathrm{GTZ}(m,k)$ is a design in which no k-subset dominates; by Proposition 7.3 its chart point admits no dominating k-subset, so $G < 0$ there by (61), and the minimum of G is negative. By Lemmas 7.4 and 7.5 that minimum is attained, at $(P_{\min}, t_{\min})$ say. Then $G(P_{\min}, t_{\min}) < 0$, so no k-subset dominates at $(P_{\min}, t_{\min})$, so $(t_{\min})_c > 0$ for every c by Corollary 7.15, and $(P_{\min}, t_{\min})$ is the chart point of a design $\mathcal{D}_{\min}$ by Proposition 7.3. Finally $\min G$ equals the infimum of G over designs, by Lemma 7.6, so $\mathcal{D}_{\min}$ attains that infimum. □

At $(6,3)$ the hypotheses are $\mathrm{GTZ}(5,3)$ and $\mathrm{GTZ}(5,2)$, both instances of Corollary 5.12. At $(7,3)$ they are $\mathrm{GTZ}(6,3)$ and $\mathrm{GTZ}(6,2)$; the second is again Corollary 5.12, and the first is the cell $(6,3)$ itself. The two open cells of Corollary 5.19 are therefore treated in order: $(6,3)$ first, and its resolution supplies the hypothesis needed at $(7,3)$.

Nothing has been weakened along the way: by Corollary 7.17 the statement about chart points with positive weights is the original problem. What the theorem adds is the extremality of the counterexample, and no cheaper device produces it. The mixing of Lemma 7.6 does move a counterexample into the interior, but it destroys whatever distinguished the original point, so it cannot be applied to a minimizer. Minimizing over the design space directly fails as well: that space is not compact, and a minimizing sequence of designs need not converge to a design (Example 7.1). The minimum exists only on the compactification, and Corollary 7.15 puts it back inside.

> Transport stationarity across the congruence $D[C]^{1/2}$ of Remark 7.7, or restate the system of §8 for G; either removes the hypothesis of Definition 8.2 from Theorem 8.23.

7.4. **Sharpness of the boundary step.** Theorem 7.12 requires $t_z = 0$ exactly. The composite of Lemma 7.9 and Theorem 7.12, which deflates at z, deletes z, renormalizes, and lifts a dominating subset back, is meaningful at any $t_z < 1$; a minimizing-sequence argument would need this composite rather than the boundary statement, since along such a sequence the weight at the escaping label is positive and merely tends to zero. The composite fails at every positive weight, no threshold repairs it, and a single determinant shows why.

Lemma 7.19. *Let (P, t) be a chart point, let z satisfy $P_{zz} > 0$ and $t_z < 1$, and let $c \neq z$. Let P' be the deflation (65) and let $\varepsilon := P'_{cc} - t_c/(1 - t_z)$ be the gap of the compressed point at c. Then*

$$\det W[\{c, z\}] \;=\; \varepsilon\, P_{zz} \;-\; t_z\Big(P_{cc} - t_c - \frac{t_c P_{zz}}{1 - t_z}\Big). \tag{68}$$

Proof. By (65), $P'_{cc} = P_{cc} - P_{cz}^2/P_{zz}$, so

$$P_{cz}^2 \;=\; P_{zz}(P_{cc} - P'_{cc}) \;=\; P_{zz}\Big(P_{cc} - \frac{t_c}{1 - t_z} - \varepsilon\Big).$$

Since $W[\{c, z\}] = \left(\begin{smallmatrix} P_{cc} - t_c & P_{cz} \\ P_{cz} & P_{zz} - t_z \end{smallmatrix}\right)$,

$$\begin{aligned}
\det W[\{c, z\}] &= (P_{cc} - t_c)(P_{zz} - t_z) - P_{zz}\Big(P_{cc} - \frac{t_c}{1 - t_z} - \varepsilon\Big) \\
&= P_{cc}P_{zz} - t_z P_{cc} - t_c P_{zz} + t_z t_c - P_{zz}P_{cc} + \frac{t_c P_{zz}}{1 - t_z} + \varepsilon P_{zz} \\
&= \varepsilon\, P_{zz} - t_z\Big(P_{cc} - t_c - \frac{t_c P_{zz}}{1 - t_z}\Big),
\end{aligned}$$

using $t_c P_{zz}/(1 - t_z) - t_c P_{zz} = t_z\, t_c P_{zz}/(1 - t_z)$ in the last step. □

Identity (68) separates the two contributions. The compressed slack ε enters undamped, with coefficient $P_{zz} > 0$; the weight at the pivot enters linearly, with a coefficient tending to $P_{cc} - t_c$ as $t_z \to 0$. Hold the chart and the survivor fixed. If $\varepsilon > 0$ then the determinant tends to $\varepsilon\, P_{zz} > 0$, and the lift survives for all small t_z. If $\varepsilon = 0$ then the determinant is $-t_z(P_{cc} - t_c - t_c P_{zz}/(1 - t_z))$, negative for every $t_z > 0$ at which the bracket is positive, however small t_z is. The lift therefore fails wherever the compressed slack is zero, no matter how small t_z; it remains to exhibit a family on which ε is frozen at zero.

Example 7.20. On the index set $\{0, 1, 2, 3\}$ let

$$P \;=\; \frac{1}{3}\begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \qquad t_1 = \tau, \quad t_0 = t_2 = t_3 = \frac{1 - \tau}{3} \qquad (0 \le \tau \le 1),$$

and let $z = 1$ be the pivot and $c = 0$ the survivor. Write $M := 3P$. The ten inner products of the rows $m_0, \ldots, m_3$ of M are

$\langle m_i, m_j \rangle$	$j = 0$	1	2	3
$i = 0$	6	3	0	3
1	3	3	3	0
2	0	3	6	-3
3	3	0	-3	3

that is, $M^2 = 3M$; hence $P^2 = P$, and P is symmetric with $\operatorname{tr} P = (2+1+2+1)/3 = 2$. So (P, t) is a chart point of size 4 and rank 2 for every τ. Its leverages are $\ell_c = P_{cc}/t_c = \frac{8}{3}, \frac{4}{3}, \frac{8}{3}, \frac{4}{3}$ at $\tau = \frac{1}{4}$, all exceeding 1; the point is not one that the deletion of a light atom disposes of, as in the first branch of the proof of Theorem 5.10.

At the pivot, $P_{11} = \frac{1}{3}$ and $Pe_1 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0)^{\mathsf{T}}$, so $\|Pe_1\|^2 = \frac{1}{3} = P_{11}$, in accordance with (63). Deflation (65) subtracts $\frac{1}{3}$ from every entry indexed in $\{0, 1, 2\}$:

$$P' = \frac{1}{3}\begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \qquad \operatorname{tr} P' = 1.$$

Deleting the index 1 leaves the three survivors with weights $((1-\tau)/3)/(1-\tau) = \frac{1}{3}$ each, independently of τ. The compressed gap at the survivor 0 is therefore

$$\varepsilon = P'_{00} - \tfrac{1}{3} = \tfrac{1}{3} - \tfrac{1}{3} = 0 \qquad \text{at every } \tau,$$

so $\{0\}$ dominates the compressed point — with slack exactly zero, at every parameter value. Upstairs, by (68) with $\varepsilon = 0$, $P_{zz} = \frac{1}{3}$, $P_{cc} = \frac{2}{3}$, $t_c = (1-\tau)/3$ and $t_z = \tau$,

$$\det W[\{0,1\}] = -\tau\Big(\frac{2}{3} - \frac{1-\tau}{3} - \frac{1-\tau}{3} \cdot \frac{1/3}{1-\tau}\Big) = -\tau \cdot \frac{6 - 3(1-\tau) - 1}{9} = -\frac{\tau(2+3\tau)}{9},$$

negative for every $\tau > 0$. Explicitly $W[\{0,1\}] = \begin{pmatrix} (1+\tau)/3 & 1/3 \\ 1/3 & (1-3\tau)/3 \end{pmatrix}$, and the probe $x = (-1, 1+\tau, 0, 0)$ gives

$$\begin{aligned} x^{\mathsf{T}} W x &= \frac{1+\tau}{3} - \frac{2(1+\tau)}{3} + \frac{(1-3\tau)(1+\tau)^2}{3} \\ &= \frac{1+\tau}{3}\big(-1 + (1-3\tau)(1+\tau)\big) = -\frac{(1+\tau)\tau(2+3\tau)}{3}. \end{aligned}$$

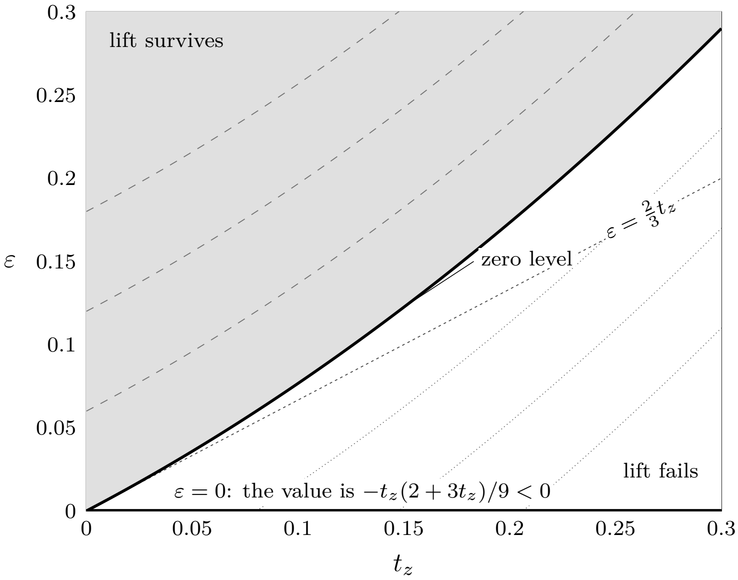
At $\tau = \frac{1}{4}$, where all four weights equal $\frac{1}{4}$, this reads $W[\{0,1\}] = \begin{pmatrix} 5/12 & 1/3 \\ 1/3 & 1/12 \end{pmatrix}$ with determinant $\frac{5}{144} - \frac{1}{9} = -\frac{11}{144}$, and the integral probe $(1, -4)$ gives $\frac{5}{12} - \frac{8}{3} + \frac{16}{12} = -\frac{11}{12}$.

Theorem 7.21 (Lean)**.** *There is no $\tau_0 > 0$ such that the compression lift holds at every chart point with $t_z < \tau_0$. Explicitly: for every $\tau_0 > 0$ there are a chart point of size 4 and rank 2, a pivot z with $P_{zz} > 0$ and $0 < t_z < \tau_0$, and a subset dominating the compressed point of size 3 and rank 1, whose image together with z does not dominate.*

Proof. Take the family of Example 7.20 at $\tau = \min\{\tau_0, 1\}/2$. Then $0 < \tau < \tau_0$ and $\tau < 1$; the compressed point is dominated by $\{0\}$, since its gap there is 0; and $\{0, 1\}$ does not dominate upstairs, the probe $(-1, 1+\tau, 0, 0)$ giving $-(1+\tau)\tau(2+3\tau)/3 < 0$. ☐

The obstruction sits on the equality locus of §6. Shrinking t_z does shrink the deficit, but along this family it shrinks nothing else: the compressed slack is frozen at zero, so the deficit is never overtaken. By (68) a positive compressed slack would overtake it, and no positive lower bound on ε is to be had. The compressed point is a chart point one size and one rank lower, hence by Proposition 7.3 the chart point of a design wherever its weights are positive; and by Theorem 6.4

designs whose dominating subsets all have $\lambda_{\min}(S_C) = 1$, that is with chart slack zero, exist at corank one for every rank and at rank three for every size. This is Corollary 6.8 again, in local form. It is why the boundary is discharged at $t_z = 0$ by the identity (67) instead of at small t_z by an estimate, and why Theorem 7.18 speaks of the limit point rather than of a minimizing sequence.



On the chart of Example 7.20 identity (68) reads $\det W[\{c, z\}] = \left(3\varepsilon - 2t_z - 3t_z^2\right)/9$, with zero level $\varepsilon = \frac{2}{3}t_z + t_z^2$. Dotted: the levels $-\frac{1}{50}, -\frac{1}{75}, -\frac{1}{150}$. Dashed: their positives.

FIGURE 8. Sharpness of the compression lift. The compressed slack ε enters (68) undamped, the pivot weight t_z linearly. The family of Example 7.20 is frozen on the axis $\varepsilon = 0$. The zero level leaves the origin with slope $\frac{2}{3}$, so that axis lies strictly inside the failure region for every $t_z > 0$, and no threshold in t_z repairs the lift.

8. Interior critical points

§7 upgrades a failure of $\mathrm{GTZ}(m, k)$ to an extremal failure: not merely some design below the threshold, but one minimizing the objective. What does minimality cost such a design? The constraint set is a smooth manifold, which gives the question a tangent space to be answered in. The objective is a maximum of least-eigenvalue functions, locally Lipschitz but in general nowhere differentiable, so the answer comes from nonsmooth first-order theory rather than from a gradient. The two together yield a finite system of algebraic identities, and from there on everything is algebra.

A design at (m, k) over $\mathbb{R}$ is written as a pair (g, t) with $g = (g_1, \ldots, g_m) \in (\mathbb{R}^k)^m$ and $t = (t_1, \ldots, t_m)$. Put

$$F(g, t) := \max_{|C|=k} \lambda_{\min}(S_C), \qquad S_C = \sum_{c \in C} g_c g_c^{\mathsf{T}}. \tag{69}$$

By Definition 2.1 and the eigenvalue reading recorded in Lemma 6.2, a k-subset dominates exactly when $\lambda_{\min}(S_C) \geq 1$, so $\mathrm{GTZ}(m, k)$ asserts $F \geq 1$ on every design, and a failure of $\mathrm{GTZ}(m, k)$ is a design with $F < 1$.

Lemma 8.1. *$F(g, t) > 0$ for every design.*

Proof. By (1) the vectors span $\mathbb{R}^k$, so some k-subset C is a basis; equivalently, $\sum_{|C|=k} \pi_C = 1$ in (55) while $\pi_C > 0$ only for bases. For a basis and $x \neq 0$, $x^\top S_C x = \sum_{c \in C} \langle g_c, x \rangle^2 > 0$, so $\lambda_{\min}(S_C) > 0$. □

The first-order theory below is applied at a design where F is minimal, and one statement is needed to place such a design where the variational principle put its own.

Definition 8.2. The cell (m, k) *has an extremal counterexample* if F attains a local minimum, over the designs of size m and rank k, at a design whose value is below 1.

Suppose $\mathrm{GTZ}(m, k)$ fails. The designs with $F < 1$ then form a nonempty set, open in the design manifold, and a local minimum of F on that set is a local minimum over all designs, since $F \geq 1$ off it. By Lemma 3.6 those designs are exactly the designs whose chart points carry no dominating k-subset, and, granting $\mathrm{GTZ}(m-1, k)$ and $\mathrm{GTZ}(m-1, k-1)$, Corollary 7.15 places all of that locus, within the compactification of §7, strictly inside the open weight simplex. It is not automatic that the infimum of F over the locus is attained rather than approached at the boundary of the compactification, where by Theorem 7.14 some subset dominates. Theorem 7.18 settles that question for the chart objective G; and by Remark 7.7 G and F agree in sign but are related by a congruence which is not a scalar, so a minimizer of one need not minimize the other. Definition 8.2 names what the passage requires, and it is carried as a hypothesis in Theorem 8.23 and nowhere else.

8.1. **The design manifold.** Fix m and k, write $\mathbb{S}^k$ for the space of real symmetric $k \times k$ matrices, and let

$$\Phi(g, t) \;:=\; \sum_c t_c\, g_c g_c^\top - I_k \;\in\; \mathbb{S}^k, \qquad \sigma(t) \;:=\; \sum_c t_c - 1 \;\in\; \mathbb{R}. \tag{70}$$

The designs are the points of $\{\Phi = 0\} \cap \{\sigma = 0\}$ satisfying the open condition $t \succ 0$. Both maps are polynomial, with differentials

$$\Phi'(g, t)[\dot g, \dot t] = \sum_c \Big(\dot t_c\, g_c g_c^\top + t_c (\dot g_c g_c^\top + g_c \dot g_c^\top) \Big), \qquad \sigma'(t)[\dot t] = \sum_c \dot t_c. \tag{71}$$

Lemma 8.3. *At every design the pair (Φ', σ') is surjective onto $\mathbb{S}^k \times \mathbb{R}$. Hence the designs form a smooth manifold of dimension $mk + m - \frac{1}{2}k(k+1) - 1$, whose tangent space $\mathcal{T}$ at (g, t) consists of the pairs $(\dot g, \dot t)$ annihilated by (71).*

Proof. Let $(Y, a) \in \mathbb{S}^k \times \mathbb{R}$ be given. Choose $\dot t$ with $\sum_c \dot t_c = a$, put $Y' := Y - \sum_c \dot t_c\, g_c g_c^\top$, and take $\dot g_c := \frac{1}{2} Y' g_c$. Then by (1)

$$\sum_c t_c (\dot g_c g_c^\top + g_c \dot g_c^\top) = \tfrac{1}{2} Y' \sum_c t_c\, g_c g_c^\top + \tfrac{1}{2} \Big(\sum_c t_c\, g_c g_c^\top \Big) Y' = Y',$$

so $\Phi'[\dot g, \dot t] = Y$ and $\sigma'[\dot t] = a$. The dimension count and the description of $\mathcal{T}$ are the submersion theorem. □

For an antisymmetric X the infinitesimal rotation

$$\dot g_c \;=\; X g_c \quad (1 \leq c \leq m), \qquad \dot t = 0 \tag{72}$$

lies in $\mathcal{T}$, since $\sum_c t_c (X g_c g_c^\top + g_c g_c^\top X^\top) = X - X = 0$ by (1); and $X \mapsto (X g_c)_c$ is injective, because the g_c span $\mathbb{R}^k$ and a matrix killing a spanning set is zero. Write $\mathcal{O} \subseteq \mathcal{T}$ for the resulting subspace, of dimension $\frac{1}{2}k(k-1)$. The objective (69) is constant along $\mathcal{O}$.

8.2. **The first-order system.** The objective is a maximum of finitely many least-eigenvalue functions of matrices depending polynomially on g. It is locally Lipschitz, not differentiable, and not Clarke regular: each $\lambda_{\min}(S_C)$ is a *concave* function of S_C, and for a maximum of functions that are not regular the subdifferential is contained in, and in general not equal to, the convex hull of the subdifferentials of the active pieces [7, Prop. 2.3.12]. The inclusion runs in the direction a necessary condition needs.

The function $\lambda_{\min}$ is concave on $\mathbb{S}^k$, and its superdifferential at A is the *spectraplex*

$$\begin{aligned} \partial\lambda_{\min}(A) \;&:=\; \{\,\Theta \succeq 0 :\ \operatorname{tr}\Theta = 1,\ \operatorname{ran}\Theta \subseteq \ker(A - \lambda_{\min}(A)I)\,\} \\ &=\; \operatorname{conv}\{uu^{\mathsf T} :\ Au = \lambda_{\min}(A)u,\ \|u\| = 1\} \end{aligned} \tag{73}$$

[25, §2], [14, Vol. I, Ch. VI]. The Clarke subdifferential of a locally Lipschitz function composed with a smooth map is, in turn, contained in the image of the subdifferential under the adjoint of the derivative [7, Thm. 2.3.10]. Applied to $\lambda_{\min} \circ S_C$, and using that Θ is symmetric so that $\langle \Theta, \dot g_c g_c^{\mathsf T} + g_c \dot g_c^{\mathsf T}\rangle = 2\langle \Theta g_c, \dot g_c\rangle$, these give

$$\partial(\lambda_{\min} \circ S_C)(g,t) \;\subseteq\; \Big\{\, \phi_{C,\Theta} : (\dot g, \dot t) \mapsto 2\sum_{c\in C} \langle \Theta g_c, \dot g_c\rangle \;:\; \Theta \in \partial\lambda_{\min}(S_C) \,\Big\}. \tag{74}$$

The functionals $\phi_{C,\Theta}$ do not involve $\dot t$.

Theorem 8.4. *Let (g,t) be a design at which F attains a local minimum over the designs. Put $\nu := F(g,t)$ and $\mathcal{A} := \{C : |C| = k,\ \lambda_{\min}(S_C) = \nu\}$. Then there are a finite index set A, subsets $C_i \in \mathcal{A}$, reals $\lambda_i \geq 0$ with $\sum_{i\in A} \lambda_i = 1$, unit vectors u_i with $S_{C_i} u_i = \nu u_i$, a symmetric matrix Λ and a scalar μ such that*

$$\sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle\, u_i \;=\; t_c\, \Lambda g_c \qquad (1 \leq c \leq m), \tag{75}$$

$$g_c^{\mathsf T} \Lambda\, g_c \;=\; \mu \qquad (1 \leq c \leq m). \tag{76}$$

Proof. Compose F with a smooth local parametrization of the manifold of Lemma 8.3. The composition has an unconstrained local minimum, so zero lies in its Clarke subdifferential [7, Prop. 2.3.2]; by the chain rule [7, Thm. 2.3.10] there is $\psi \in \partial F(g,t)$ vanishing on $\mathcal{T}$. Inactive subsets remain inactive near (g,t), so the maximum rule [7, Prop. 2.3.12] together with (74) exhibits ψ as a convex combination

$$\psi \;=\; \sum_{C\in\mathcal{A}} \lambda_C\, \phi_{C,\Theta_C}, \qquad \lambda_C \geq 0, \quad \sum_C \lambda_C = 1, \quad \Theta_C \in \partial\lambda_{\min}(S_C). \tag{77}$$

Decompose each Θ_C spectrally, $\Theta_C = \sum_j w_{C,j} u_{C,j} u_{C,j}^{\mathsf T}$ with $w_{C,j} \geq 0$, $\sum_j w_{C,j} = 1$ and each $u_{C,j}$ a unit eigenvector of S_C at ν; this is possible by (73). Index by the pairs $i = (C,j)$, setting $C_i := C$, $u_i := u_{C,j}$ and $\lambda_i := \lambda_C w_{C,j}$. The new multipliers are nonnegative and sum to one, and (77) becomes

$$\psi(\dot g, \dot t) \;=\; 2\sum_c \Big\langle \sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle\, u_i,\ \dot g_c \Big\rangle. \tag{78}$$

By Lemma 8.3 the map (Φ', σ') is surjective with kernel $\mathcal{T}$, and ψ vanishes on $\mathcal{T}$; hence ψ factors through (Φ', σ'). That is, there are $\Lambda \in \mathbb{S}^k$ and $\mu \in \mathbb{R}$ with

$$\psi(\dot g, \dot t) \;=\; \langle \Lambda,\ \Phi'(g,t)[\dot g, \dot t]\rangle \;-\; \mu\, \sigma'(t)[\dot t], \tag{79}$$

the sign of μ being a normalization. Expanding the right side by (71), and using $\langle \Lambda, \dot g_c g_c^{\mathsf T} + g_c \dot g_c^{\mathsf T}\rangle = 2\langle \Lambda g_c, \dot g_c\rangle$ as before,

$$\langle \Lambda, \Phi'[\dot g, \dot t]\rangle - \mu \sum_c \dot t_c \;=\; 2\sum_c t_c \langle \Lambda g_c, \dot g_c\rangle \;+\; \sum_c \dot t_c (g_c^{\mathsf T} \Lambda g_c - \mu).$$

The coordinates $\dot g$ and $\dot t$ are independent, so comparison with (78) may be made term by term. The coefficient of $\dot g_c$ gives (75) after division by 2, and the coefficient of $\dot t_c$ gives (76). $\square$

The maximum rule in (77) is only an inclusion, because the pieces are concave and therefore not Clarke regular; the sharper first-order statement available at such a point is the directional one, namely $\max_{C\in\mathcal{A}} \lambda_{\min}(Q_C^\mathsf{T}\dot S_C Q_C) \ge 0$ for every $(\dot g, \dot t) \in \mathcal{T}$, where Q_C is an orthonormal basis of the $\lambda_{\min}$-eigenspace of S_C and $\dot S_C$ is the derivative of S_C [15, 8]. Nothing below uses more than the multiplier form. The infinitesimal rotations (72), for their part, impose no condition: for $\Theta \in \partial\lambda_{\min}(S_C)$ one has $\Theta S_C = \nu\Theta = S_C\Theta$, so

$$\phi_{C,\Theta}\left[(Xg_c)_c, 0\right] = 2\sum_{c\in C}\langle \Theta g_c, Xg_c\rangle = 2\,\mathrm{tr}(\Theta X S_C) = 2\nu\ \mathrm{tr}(\Theta X) = 0, \tag{80}$$

because Θ is symmetric and X antisymmetric.

8.3. **Stationarity data.** The conclusion of Theorem 8.4 is a finite algebraic object, and it is convenient to detach it from its provenance.

Definition 8.5. A *stationarity datum* for a design (g, t) at (m, k), with value $\nu \in \mathbb{R}$, consists of a finite index set A, k-subsets C_i ($i \in A$), reals $\lambda_i \ge 0$ with $\sum_i \lambda_i = 1$, vectors u_i with $\|u_i\| = 1$ and $S_{C_i}u_i = \nu u_i$, and a symmetric matrix Λ, satisfying (75) and (76). Write $A^+ := \{C_i : \lambda_i > 0\}$ for the *active family*, a set of k-subsets. The datum is *admissible* if in addition

$$\nu \;\ge\; \lambda_{\min}(S_C) \qquad \text{for every } k\text{-subset } C, \tag{81}$$

which says $\nu \ge F(g, t)$. It says $\nu = F(g, t)$ exactly when in addition $\nu = \lambda_{\min}(S_{C_i})$ for some i, which the definition does not require; Theorem 8.4 produces data for which it holds.

The subsets C_i need not be distinct: a tight eigenvalue of multiplicity r contributes r indices carrying one subset, the form the spectral decomposition in the proof of Theorem 8.4 delivers. Nothing below assumes that $i \mapsto C_i$ is injective, and this is why the active family is defined as a set of subsets rather than as a set of indices.

Definition 8.5 asks only that ν be *an* eigenvalue of each S_{C_i}. It does not ask that $\nu = \lambda_{\min}(S_{C_i})$, and it does not ask for (81). Theorem 8.4 supplies both; the algebra of the next two subsections supplies neither, and §8.7 shows that the difference is not a formality.

8.4. **The quadric law and the coverage law.** Pairing (75) with g_c gives, for every c,

$$\sum_{i\,:\,c\in C_i} \lambda_i\langle u_i, g_c\rangle^2 \;=\; t_c\,(g_c^\mathsf{T}\Lambda g_c). \tag{82}$$

Both laws of the next proposition come from this contraction. Throughout we use the polarized form of the subset sum,

$$y^\mathsf{T} S_C x \;=\; \sum_{c\in C}\langle g_c, y\rangle\langle g_c, x\rangle, \tag{83}$$

immediate from (2).

Proposition 8.6 (Lean)**.** *Let a stationarity datum be given. Then $\mu = \nu$, so that every vector lies on one central quadric of the multiplier,*

$$g_c^\mathsf{T}\Lambda\, g_c \;=\; \nu \qquad (1 \le c \le m); \tag{84}$$

for every c the coverage law

$$\sum_{i\,:\,c\in C_i} \lambda_i\langle u_i, g_c\rangle^2 \;=\; t_c\,\nu \tag{85}$$

holds; the multiplier is determined by the remaining data,

$$\Lambda \;=\; \nu \sum_{i\in A} \lambda_i\, u_i u_i^{\mathsf T}; \tag{86}$$

and consequently $\nu \geq 0$, $\Lambda \succeq 0$ *and* $\operatorname{tr}\Lambda = \nu$.

Proof. Sum (82) over c. On the right, (76) makes $g_c^{\mathsf T}\Lambda g_c$ independent of c, so the right side is $\mu\sum_c t_c = \mu$. On the left, exchange the two summations; for fixed i the inner sum runs over the k vectors of C_i and equals, by (83) and $S_{C_i}u_i = \nu u_i$,

$$\sum_{c\in C_i} \langle u_i, g_c\rangle^2 \;=\; u_i^{\mathsf T} S_{C_i} u_i \;=\; \nu\|u_i\|^2 \;=\; \nu.$$

Hence the left side is $\nu\sum_i \lambda_i = \nu$, so $\mu = \nu$. Substituting $\mu = \nu$ into (76) gives (84), and into (82) gives (85).

For (86), fix $x \in \mathbb{R}^k$ and resolve it through (1): $x = \sum_c t_c\langle g_c, x\rangle g_c$. Then

$$\begin{aligned}
\Lambda x &= \sum_c \langle g_c, x\rangle\,(t_c\Lambda g_c) = \sum_c \langle g_c, x\rangle \sum_{i\,:\,c\in C_i} \lambda_i\langle u_i, g_c\rangle\, u_i && \text{by (75)}\\
&= \sum_{i\in A}\lambda_i\Big(\sum_{c\in C_i}\langle g_c, u_i\rangle\langle g_c, x\rangle\Big)u_i = \sum_{i\in A}\lambda_i(u_i^{\mathsf T}S_{C_i}x)u_i && \text{by (83)}\\
&= \nu\sum_{i\in A}\lambda_i\langle u_i, x\rangle\, u_i,
\end{aligned}$$

the last step by symmetry of S_{C_i} together with $S_{C_i}u_i = \nu u_i$. A matrix is determined by its action, which is (86). Finally $\nu = u_i^{\mathsf T}S_{C_i}u_i = \sum_{c\in C_i}\langle g_c, u_i\rangle^2 \geq 0$ for any i, and (86) then exhibits Λ as a nonnegative multiple of a convex combination of rank-one projectors, so $\Lambda \succeq 0$ and $\operatorname{tr}\Lambda = \nu\sum_i\lambda_i = \nu$. □

The coverage law (85) is the contraction (82) rewritten by the quadric law, and the quadric law is that same contraction summed over c. The two are therefore not independent constraints, and no count of the equations of the system may treat them as such: the content of a stationarity datum is exactly (75) and (76).

Reading the quadric law through (86) removes the restriction $c \in C_i$ from the sum.

Lemma 8.7 (Lean)**.** *Let a stationarity datum with* $\nu \neq 0$ *be given. Then*

$$\sum_{i\in A}\lambda_i\langle u_i, g_c\rangle^2 \;=\; 1 \qquad (1 \leq c \leq m), \tag{87}$$

and consequently

(i) $\ell_c \geq 1$ *for every* c;
(ii) $t_c\nu \leq 1$ *for every* c, *and hence* $\nu \leq m$;
(iii) *for every* k*-subset* C, *active or not, there is an index* $i \in A$ *with* $u_i^{\mathsf T}S_C u_i \geq k$;
(iv) *for each* c *there is* i *with* $\lambda_i > 0$, $c \in C_i$ *and* $\langle u_i, g_c\rangle \neq 0$; *in particular* A^+ *covers* $\{1, \ldots, m\}$.

Proof. By (86), $g_c^{\mathsf T}\Lambda g_c = \nu\sum_i\lambda_i\langle u_i, g_c\rangle^2$; comparison with (84) and cancellation of $\nu \neq 0$ give (87).

(i) Cauchy–Schwarz and $\|u_i\| = 1$ give $\langle u_i, g_c\rangle^2 \leq \ell_c$; substituting in (87) and using $\sum_i\lambda_i = 1$ gives $1 \leq \ell_c$.

(ii) The sum in (85) is a subsum of the sum in (87) with nonnegative terms, so $t_c\nu \leq 1$. Summing over c against $\sum_c t_c = 1$ gives $\nu \leq m$.

(iii) Sum (87) over the k vectors of C and use (83): $\sum_i\lambda_i\, u_i^{\mathsf T}S_C u_i = k$. A convex combination does not exceed its largest term.

(iv) By (85) the sum at c equals $t_c\nu \neq 0$, so one of its terms is nonzero, and that forces $\lambda_i > 0$, $c \in C_i$ and $\langle u_i, g_c\rangle \neq 0$ for the corresponding i. □

Part (iii) names a unit vector; it implies $\lambda_{\max}(S_C) \geq k$ for every k-subset, but the vector is the useful half. Part (i) says every vector of a stationary design is heavy in the weak sense; the strict inequality $\ell_c > 1$ would require excluding equality in Cauchy–Schwarz at every positively weighted index simultaneously, and is not available.

8.5. **The tetrahedron.** Before drawing exclusions we verify the entire system on the design of Example 2.11, where every constant in it can be displayed.

Example 8.8. Take the tetrahedron: $g_c = \sqrt{3}\,u_c$ with u_c the four unit vertex vectors, $t_c = \frac{1}{4}$, so that $\ell_c = 3$ and $\langle g_i, g_j\rangle = -1$ for $i \neq j$. There are four triples, each the complement of one index; for $C = \{0,1,2,3\} \setminus \{c_0\}$ one has $S_C = 4I_3 - g_{c_0}g_{c_0}^{\mathsf{T}}$, with spectrum $\{1,4,4\}$. Take

$$A = \{0,1,2,3\}, \qquad C_i = \{0,1,2,3\} \setminus \{i\}, \qquad \lambda_i = \tfrac{1}{4}, \qquad u_i = \tfrac{1}{\sqrt{3}}\,g_i, \qquad \nu = 1.$$

Each u_i is a unit vector, and it is the tight eigenvector: $\langle g_i, u_i\rangle = 3/\sqrt{3} = \sqrt{3}$, so $S_{C_i}u_i = 4u_i - \sqrt{3}\,g_i = 4u_i - 3u_i = u_i$, the eigenvalue being $\nu = 1$. The multiplier predicted by (86) is

$$\Lambda \;=\; 1 \cdot \sum_i \tfrac{1}{4} \cdot \tfrac{1}{3}\, g_i g_i^{\mathsf{T}} \;=\; \tfrac{1}{12} \sum_i g_i g_i^{\mathsf{T}} \;=\; \tfrac{1}{12} \cdot 4I_3 \;=\; \tfrac{1}{3} I_3,$$

using $\sum_i g_i g_i^{\mathsf{T}} = 4I_3$, which is (1) at $t_i = \frac{1}{4}$. Then $\operatorname{tr}\Lambda = 1 = \nu$, as Proposition 8.6 requires, and the quadric law (84) reads $g_c^{\mathsf{T}}\Lambda g_c = \ell_c/3 = 1 = \nu$.

For the coverage law, the triples containing c are the three with $i \neq c$, and for those $\langle u_i, g_c\rangle = \langle g_i, g_c\rangle/\sqrt{3} = -1/\sqrt{3}$, so

$$\sum_{i\,:\,c \in C_i} \lambda_i \langle u_i, g_c\rangle^2 \;=\; 3 \cdot \tfrac{1}{4} \cdot \tfrac{1}{3} \;=\; \tfrac{1}{4} \;=\; t_c\nu.$$

Identity (87) restores the missing index $i = c$, for which $\langle u_c, g_c\rangle^2 = 3$, and indeed $\frac{1}{4} + \frac{1}{4} \cdot 3 = 1$. For (75) itself, using $\sum_i g_i = 0$,

$$\sum_{i\,:\,c \in C_i} \lambda_i \langle u_i, g_c\rangle u_i \;=\; \sum_{i \neq c} \tfrac{1}{4}\Big(-\tfrac{1}{\sqrt{3}}\Big)\tfrac{g_i}{\sqrt{3}} \;=\; -\tfrac{1}{12} \sum_{i \neq c} g_i \;=\; \tfrac{g_c}{12} \;=\; \tfrac{1}{4} \cdot \tfrac{1}{3}\, g_c \;=\; t_c \Lambda g_c.$$

The datum is admissible: every triple has $\lambda_{\min} = 1$, so $F = 1 = \nu$. The bounds of Lemma 8.7 read $\ell_c = 3 \geq 1$, $t_c\nu = \frac{1}{4} \leq 1$ and $\nu = 1 \leq 4 = m$. Part (iii) of that lemma is not realized at the tight index of the subset itself, since $u_i^{\mathsf{T}} S_{C_i} u_i = \nu = 1 < 3$; it is realized at any other index, for which

$$u_j^{\mathsf{T}} S_{C_i} u_j = 4 - \langle g_i, u_j\rangle^2 = 4 - \tfrac{\langle g_i, g_j\rangle^2}{3} = 4 - \tfrac{1}{3} = \tfrac{11}{3} \;\geq\; 3 \qquad (j \neq i).$$

8.6. **Exclusions.** Each exclusion below is a statement about the datum alone; admissibility is nowhere used.

Proposition 8.9. *Let a stationarity datum with $\nu \neq 0$ have $\Lambda = a\,ww^{\mathsf{T}}$ for some unit vector w and some $a \in \mathbb{R}$. Then $\nu = k$. Since $\Lambda \succeq 0$, this covers every datum with* $\operatorname{rank}\Lambda \leq 1$.

Proof. Taking traces in (86) gives $a = \operatorname{tr}\Lambda = \nu$. Evaluating (86) as a quadratic form at an arbitrary x gives $\nu\langle w, x\rangle^2 = \nu \sum_i \lambda_i \langle u_i, x\rangle^2$, so after cancelling $\nu \neq 0$,

$$\sum_{i \in A} \lambda_i \langle u_i, x\rangle^2 \;=\; \langle w, x\rangle^2 \qquad \text{for every } x. \tag{88}$$

Put $x = w$. Then $\sum_i \lambda_i \langle u_i, w\rangle^2 = 1 = \sum_i \lambda_i$, so $\sum_i \lambda_i (1 - \langle u_i, w\rangle^2) = 0$; every summand is nonnegative, because $\|u_i\| = \|w\| = 1$ forces $\langle u_i, w\rangle^2 \leq 1$. Hence every summand vanishes. Fix i with $\lambda_i > 0$. Then $\langle u_i, w\rangle^2 = 1$ and

$$\big\|u_i - \langle u_i, w\rangle w\big\|^2 \;=\; 1 - 2\langle u_i, w\rangle^2 + \langle u_i, w\rangle^2 \;=\; 0,$$

so $u_i = \pm w$. Now read (84) through $\Lambda = \nu\, ww^{\mathsf T}$: it says $\nu\langle w, g_c\rangle^2 = \nu$, so $\langle w, g_c\rangle^2 = 1$ for every c. Therefore, by (83),

$$\nu \;=\; u_i^{\mathsf T} S_{C_i} u_i \;=\; w^{\mathsf T} S_{C_i} w \;=\; \sum_{c\in C_i} \langle w, g_c\rangle^2 \;=\; |C_i| \;=\; k. \qquad \square$$

Corollary 8.10. *No stationarity datum with $0 < \nu < 1$ has a multiplier of rank at most one.*

Proof. Otherwise $\nu = k$ by Proposition 8.9, and $\nu \neq 0$ forces $k \geq 1$, whence $\nu \geq 1$. $\square$

The second exclusion concerns the tight directions. Say that a datum has *independent tight support* if the only relation $\sum_{i\in A} \lambda_i \kappa_i\, u_i = 0$ with $\kappa : A \to \mathbb{R}$ has $\lambda_i\kappa_i = 0$ for every i; linear independence of the family $(u_i)_{\lambda_i>0}$ implies this, and the stated form makes the zero-multiplier indices cost nothing.

Proposition 8.11. *Let a stationarity datum with $\nu \neq 0$ have independent tight support. Then $t_c\nu = 1$ for every c; hence $\nu = m$ and $t_c = 1/m$ for every c. In particular no stationarity datum with $0 < \nu < 1$ has independent tight support: below the threshold the tight directions carried by positive multipliers are always dependent.*

Proof. Substituting (86) into the right side of (75) turns it into a combination of the u_i,

$$t_c \Lambda g_c \;=\; t_c \nu \sum_{i\in A} \lambda_i \langle u_i, g_c\rangle\, u_i,$$

so subtraction gives the relation

$$\sum_{i\in A} \lambda_i\, \langle u_i, g_c\rangle\, \big(\,[\,c \in C_i\,] - t_c\nu\,\big)\, u_i \;=\; 0, \tag{89}$$

where $[\,\cdot\,]$ is 1 when the condition holds and 0 otherwise. Independence kills every coefficient:

$$\lambda_i\, \langle u_i, g_c\rangle\, \big(\,[\,c \in C_i\,] - t_c\nu\,\big) \;=\; 0 \qquad (i \in A).$$

Fix c. By Lemma 8.7(iv) there is i with $\lambda_i > 0$, $c \in C_i$ and $\langle u_i, g_c\rangle \neq 0$; for that i the last factor is $1 - t_c\nu$, so $t_c\nu = 1$. Summing over c against $\sum_c t_c = 1$ gives $\nu = m$, and then $t_c = 1/m$. A design has at least one vector, so $\nu = m \geq 1$. $\square$

The conclusion identifies the datum rather than contradicting it: one might expect independence to force $\langle u_i, g_c\rangle = 0$, against coverage, but the coefficient in (89) vanishes through its last factor instead.

Counting gives a third exclusion: by Lemma 8.7(iv) the active family covers $\{1, \ldots, m\}$, and each of its members has k elements.

Proposition 8.12. *For a stationarity datum with $\nu \neq 0$,*

$$m \;\leq\; k\,|A^+|. \tag{90}$$

If $k < m$ then $|A^+| \geq 2$. If $|A^+| = 1$ then $m = k$ and $\nu = k$. If $m = 2k$ and $|A^+| = 2$, the two members of A^+ are disjoint and partition $\{1, \ldots, m\}$.

Proof. The union of the members of A^+ is all of $\{1, \ldots, m\}$ and has at most $k|A^+|$ elements, which is (90); the second assertion is (90) read backwards. If $A^+ = \{C\}$ then $m \leq k$, while $m \geq k$ always, so $m = k$; moreover every index with $\lambda_i > 0$ carries the same subset C, so summing (87) over $c \in C$ gives, as in Lemma 8.7(iii), $\sum_i \lambda_i u_i^{\mathsf T} S_C u_i = k$, while each such u_i is a unit eigenvector of S_C at ν, whence $\nu = k$. If $m = 2k$ and $A^+ = \{C, C'\}$ then $C \cup C'$ has $2k$ elements and $|C| + |C'| = 2k$, so $C \cap C' = \emptyset$. $\square$

Corollary 8.13 (Lean)**.** *A stationarity datum with $\nu \neq 0$ whose index set A is a singleton has $\nu = k$. In particular no such datum has $0 < \nu < 1$.*

Proof. A singleton $A = \{i_0\}$ has $\lambda_{i_0} = 1 > 0$, so $A^+ = \{C_{i_0}\}$ and Proposition 8.12 gives $\nu = k$; and $\nu \neq 0$ forces $k \geq 1$, whence $\nu \geq 1$. $\square$

Corollary 8.14. *Let a stationarity datum at $(6, 3)$ have $0 < \nu < 1$. Then either $|A^+| \geq 3$, or A^+ consists of two disjoint triples partitioning $\{1, \dots, 6\}$ and the tight directions carried by positive multipliers are linearly dependent. At $(7, 3)$ one has $|A^+| \geq 3$ unconditionally.*

Proof. At $(6, 3)$, Proposition 8.12 gives $|A^+| \geq 2$; if it is exactly 2 then $m = 2k$ forces the partition, and Proposition 8.11 forbids independent tight support at $\nu < 1$. At $(7, 3)$, (90) reads $7 \leq 3|A^+|$. $\square$

Whether the partition branch at $(6, 3)$ is inhabited is open.

A fourth exclusion follows from the same subsum comparison, and it acts on the opposite end of the range: on families so large that a single vector meets all of them.

Proposition 8.15 (Lean)**.** *Let a stationarity datum with $\nu \neq 0$ be given, and suppose some index c lies in C_i for every $i \in A$. Then $t_c \nu = 1$, and hence $\nu \geq 1$.*

Proof. The sum in (85) runs over $\{i : c \in C_i\}$ and the sum in (87) over all of A. Under the hypothesis the two index sets coincide, so the inequality $t_c \nu \leq 1$ of Lemma 8.7(ii) is an equality: $t_c \nu = 1$. Since $t_c \leq 1$ this gives $\nu = 1/t_c \geq 1$. $\square$

Corollary 8.16. *At a datum with $0 < \nu < 1$, no index lies in every member of A; in particular, for each index some active subset omits it.*

Propositions 8.9, 8.11, 8.12 and 8.15 are the exclusions available at $0 < \nu < 1$. Their combined reach is measured in §8.8.

8.7. **The reach of the system.** The exclusions bite on small active families. At the frontier size the system is satisfiable at the threshold itself, with a large active family.

Example 8.17. Split the tetrahedron of Example 2.11 at three of its four vertices. Let $h_0, \dots, h_3$ be the sign vectors $(1, 1, 1)$, $(1, -1, -1)$, $(-1, 1, -1)$, $(-1, -1, 1)$, so that $\sum_d h_d h_d^\top = 4I_3$ and $\langle h_d, h_e \rangle = -1$ for $d \neq e$; these are the vectors of Example 2.11 in coordinates. Give the seven vectors $g_0, \dots, g_6$ the directions $h_0, h_1, h_2, h_3, h_0, h_1, h_2$ and the weights $\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}$. Each direction carries total weight $\frac{1}{4}$, so (1) holds. This is one of the extremal designs of Theorem 6.4 at $(7, 3)$, and $F = 1$.

Call a triple *rainbow* if its three vectors point in three distinct directions. There are $2^3 = 8$ rainbow triples on the directions h_0, h_1, h_2, one for each choice of vector in each direction; and a rainbow triple containing the unsplit vector g_3 uses two of the three split directions, which gives $3 \cdot 2 \cdot 2 = 12$ more, twenty in all. For a rainbow triple C missing the direction h_d one has $S_C = 4I_3 - h_d h_d^\top$, exactly as in Example 8.8, so $u_C := h_d/\sqrt{3}$ is a unit eigenvector at $\nu = 1$. Give multiplier $\frac{1}{32}$ to the eight triples omitting h_3 and $\frac{1}{16}$ to the twelve containing g_3; these are positive and $8 \cdot \frac{1}{32} + 12 \cdot \frac{1}{16} = \frac{1}{4} + \frac{3}{4} = 1$. Each of the four directions is the missed one for total multiplier mass $\frac{1}{4}$, so (86) gives

$$\Lambda \;=\; 1 \cdot \sum_d \tfrac{1}{4} \cdot \tfrac{1}{3}\, h_d h_d^\top \;=\; \tfrac{1}{12} \cdot 4I_3 \;=\; \tfrac{1}{3} I_3,$$

whence $\operatorname{tr} \Lambda = 1 = \nu$ and $g_c^\top \Lambda g_c = 3/3 = 1 = \nu$, which is (84). A triple containing g_c misses a direction other than that of g_c, so $\langle u_C, g_c \rangle^2 = \frac{1}{3}$ there, and the coverage law reduces to $\frac{1}{3} \sum_{C \ni c} \lambda_C = t_c$. For $c = 3$ the twelve triples of multiplier $\frac{1}{16}$ all contain g_3, giving $\frac{1}{3} \cdot \frac{12}{16} = \frac{1}{4} = t_3$. For $c = 0$ the triples containing g_0 are four on the directions h_0, h_1, h_2, of multiplier $\frac{1}{32}$, and two each on h_0, h_1, h_3 and on h_0, h_2, h_3, of multiplier $\frac{1}{16}$, giving

$$\tfrac{1}{3}\left(\tfrac{4}{32} + \tfrac{2}{16} + \tfrac{2}{16}\right) = \tfrac{1}{3} \cdot \tfrac{3}{8} = \tfrac{1}{8} = t_0.$$

The datum is admissible, and $|A^+| = 20$.

Example 8.17 puts twenty triples in the active family at the cell to which rank three reduces. It does not conflict with Proposition 8.19 below, which asserts only that the multipliers *may be taken* supported on at most nineteen indices.

The next statement is the reason the conjectures of §8.8 carry an admissibility hypothesis.

Theorem 8.18. *There is a design at* $(4,2)$ *carrying a stationarity datum with* $\nu = 2 - \sqrt{2} < 1$. *That design satisfies* $\mathrm{GTZ}(4,2)$ *with margin: some* 2*-subset* C *has* $S_C - I = I \succ 0$, *so* $F = 2$. *Consequently the implication "a stationarity datum with* $\nu \neq 0$ *has* $\nu \geq 1$*" is false, and no strengthening of the identities* (75) *and* (76) *can make it true.*

Proof. Take

$$g_0 = (\sqrt{2}, 0), \quad g_1 = (-1, 1), \quad g_2 = (1,1), \quad g_3 = (0, \sqrt{2}), \qquad t_c = \tfrac{1}{4}.$$

Then $\ell_c = 2$ for every c and

$$\sum_c t_c\, g_c g_c^{\mathsf{T}} = \tfrac{1}{4}\left[\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}\right] = \tfrac{1}{4}\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = I_2,$$

so this is a design. Take the four pairs $C_0 = \{0,1\}$, $C_1 = \{0,2\}$, $C_2 = \{1,3\}$, $C_3 = \{2,3\}$, each with multiplier $\frac{1}{4}$, and $\Lambda := \frac{2-\sqrt{2}}{2} I_2$. Now

$$S_{C_0} = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \operatorname{tr} S_{C_0} = 4, \qquad \det S_{C_0} = 2,$$

so the eigenvalues of S_{C_0} are $2 \pm \sqrt{2}$, and the same holds for C_1, C_2, C_3 by symmetry. Put $\nu := 2 - \sqrt{2}$. Unit eigenvectors at ν are the normalizations $u_i := r_i/\|r_i\|$ of

$$r_0 = (1, 1+\sqrt{2}), \quad r_1 = (1, -(1+\sqrt{2})), \quad r_2 = (1, \sqrt{2}-1), \quad r_3 = (1, 1-\sqrt{2}),$$

with $\|r_0\|^2 = \|r_1\|^2 = 4 + 2\sqrt{2}$ and $\|r_2\|^2 = \|r_3\|^2 = 4 - 2\sqrt{2}$; for instance

$$(S_{C_0} - \nu I) r_0 = \begin{pmatrix} 1+\sqrt{2} & -1 \\ -1 & \sqrt{2}-1 \end{pmatrix}\begin{pmatrix} 1 \\ 1+\sqrt{2} \end{pmatrix} = \begin{pmatrix} (1+\sqrt{2}) - (1+\sqrt{2}) \\ -1 + (\sqrt{2}-1)(1+\sqrt{2}) \end{pmatrix} = 0.$$

Condition (76) holds because Λ is a multiple of the identity and all leverages are equal: $g_c^{\mathsf{T}} \Lambda g_c = \frac{2-\sqrt{2}}{2} \cdot 2 = 2 - \sqrt{2} = \nu$, which is already (84). For (75) take $c = 0$. The pairs containing 0 are C_0 and C_1, and $\langle r_0, g_0 \rangle = \langle r_1, g_0 \rangle = \sqrt{2}$, so, since normalizing r_i twice contributes $\|r_i\|^{-2}$,

$$\sum_{i\,:\,0 \in C_i} \lambda_i \langle u_i, g_0 \rangle u_i = \tfrac{1}{4} \cdot \frac{\sqrt{2}}{4 + 2\sqrt{2}}\,(r_0 + r_1) = \tfrac{1}{4} \cdot \frac{\sqrt{2}}{4 + 2\sqrt{2}}\,(2, 0) = \frac{\sqrt{2}-1}{4}\,(1,0),$$

because $\dfrac{\sqrt{2}}{4+2\sqrt{2}} = \dfrac{\sqrt{2}\,(4 - 2\sqrt{2})}{8} = \dfrac{4\sqrt{2} - 4}{8} = \dfrac{\sqrt{2}-1}{2}$, while

$$t_0 \Lambda g_0 = \tfrac{1}{4} \cdot \tfrac{2-\sqrt{2}}{2}\,(\sqrt{2}, 0) = \frac{2\sqrt{2}-2}{8}\,(1,0) = \frac{\sqrt{2}-1}{4}\,(1,0).$$

The three remaining vectors are handled in the same way. Finally, the two pairs omitted from the active family are orthogonal: $S_{\{0,3\}} = S_{\{1,2\}} = 2I_2$, so $\lambda_{\min} = 2$ there, $F = 2$, and $S_{\{0,3\}} - I = I \succ 0$. □

The datum of Theorem 8.18 is stationary for the maximum taken over the four chosen pairs, which minimize $\lambda_{\min}(S_C)$, and not for F. It violates (81), a disjunctive semialgebraic condition rather than an equation and therefore beyond the algebra of §8.4. It also violates the hypothesis of every exclusion: its multiplier is isotropic rather than of rank one, its four tight directions lie in $\mathbb{R}^2$ and are dependent, and its active family is four distinct pairs, neither a singleton nor a partition.

8.8. **The residue.** The residue is bounded below by (90) and above by Carathéodory's theorem applied to the convex combination (77).

Proposition 8.19. *A stationarity datum produced by Theorem 8.4 may be chosen with*

$$|A| \;\le\; m(k+1) - k^2. \tag{91}$$

At $(6,3)$ *this is* 15*, and at* $(7,3)$ *it is* 19*.*

Proof. After the spectral decomposition carried out in the proof of Theorem 8.4, identity (78) reads $\psi = \sum_{i\in A} \lambda_i \, \phi_{C_i, u_i u_i^\top}$, a convex combination of the rank-one functionals of (74); and ψ vanishes on $\mathcal{T}$. So the origin lies in the convex hull of the restrictions to $\mathcal{T}$ of those functionals. By (80) each restriction annihilates the subspace $\mathcal{O}$ of infinitesimal rotations, which by (72) lies in $\mathcal{T}$ and has dimension $\frac{1}{2}k(k-1)$. The combination therefore takes place in the annihilator of $\mathcal{O}$ inside the dual of $\mathcal{T}$, a space of dimension

$$\dim \mathcal{T} - \dim \mathcal{O} = \Big(mk + m - \tfrac{1}{2}k(k+1) - 1\Big) - \tfrac{1}{2}k(k-1) = mk + m - k^2 - 1,$$

by Lemma 8.3. Carathéodory's theorem [4], [26, §17] writes a point of a convex hull in a space of dimension d as a combination of at most $d+1$ of the points; here $d+1 = mk+m-k^2 = m(k+1)-k^2$. Keep those indices, discard the rest, and read the last two steps of the proof of Theorem 8.4 again from the reduced combination: the value, the subsets and the tight directions are unchanged, the multipliers are the new ones, and Λ is recomputed by (86). □

The dimension in that proof is the dimension of the design manifold modulo the orthogonal group: mk coordinates for the vectors and m for the weights, less the $\frac{1}{2}k(k+1)+1$ constraints, less the $\frac{1}{2}k(k-1)$ rotations. At $(6,3)$ it is $18+6-6-1-3 = 14$, and at $(7,3)$ it is $21+7-6-1-3 = 18$.

Combining, at $(6,3)$ the active family covers $\{1,\dots,6\}$ and has between 2 and 15 members, the value 2 occurring only in the partition branch of Corollary 8.14; at $(7,3)$ it covers $\{1,\dots,7\}$ and has between 3 and 19 members. The number of possibilities is finite, and at $(6,3)$ small enough to be listed.

Proposition 8.20. *Up to relabelling, the families of* 3*-subsets of* $\{1,\dots,6\}$ *whose union is* $\{1,\dots,6\}$ *number* 2102*, distributed by cardinality as in Table 1. Exactly one of them has two members, namely a partition into two triples; so* 2101 *have at least three members, and* 2069 *have at most fifteen. Up to relabelling, the families of* 3*-subsets of* $\{1,\dots,7\}$ *whose union is* $\{1,\dots,7\}$ *number* 7 011 184*; every one has at least three members, and* 5 249 381 *have at most nineteen.*

Against this census the exclusions reach only a corner. Of the 2136 classes of families of 3-subsets of $\{1,\dots,6\}$, Lemma 8.7(iv) removes the 34 that fail to cover; Proposition 8.15 removes 23 more, those in which some index lies in every member; and Corollary 8.14 leaves the single partition class undecided. Fifty-seven classes are excluded, 2079 are not. The reason is visible in Table 1: the exclusions bite at the two ends of the range, on families too small to cover and on families large enough to be saturated, while the distribution peaks at cardinality ten, where they say nothing. Conjecture 8.21 is not a finite verification away from what is proved here.

Proof. Orbit enumeration under the symmetric group. For $n = 6$ the 2^{20} families of triples were partitioned into orbits directly. Independently, for $n = 6$ and $n = 7$ the number of orbits of covering families of cardinality j was computed from Burnside's lemma as

$$\frac{1}{n!} \sum_{\varsigma \in S_n} \sum_{R} (-1)^{r-|R|} [x^j] \prod_{\kappa \subseteq T_R} (1 + x^{|\kappa|}),$$

where ς has r cycles on $\{1, \ldots, n\}$, the index R runs over sets of those cycles, T_R is their union, and κ runs over the cycles of the permutation induced by ς on the 3-subsets contained in T_R. A family fixed by ς is a union of such cycles, and its union of points is ς-invariant, so the alternating sum is inclusion–exclusion over the sets a fixed family can avoid. The two computations agree at $n = 6$. ☐

j	2	3	4	5	6	7	8	9	10	11
N_j	1	3	15	37	88	157	247	311	351	312
j	12	13	14	15	16	17	18	19	20	
N_j	249	161	94	43	21	7	3	1	1	

TABLE 1. The number N_j of families of j triples covering $\{1, \ldots, 6\}$, up to relabelling. The total is 2102; the Carathéodory bound (91) discards the last five columns, leaving 2069.

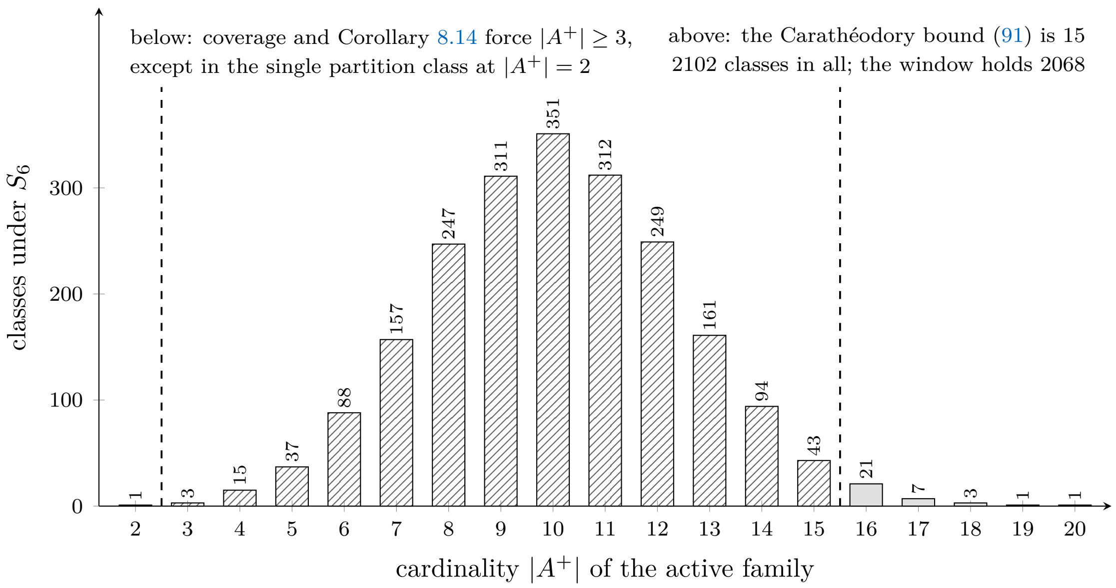


FIGURE 9. The census of Proposition 8.20 at $(6, 3)$. The active family covers $\{1, \ldots, 6\}$, so it has at least two members, and the one class of cardinality two is the partition that Corollary 8.14 leaves undecided; (91) removes the classes above fifteen.

Conjecture 8.21. *No design at* $(6, 3)$ *carries an admissible stationarity datum with* $0 < \nu < 1$.

Conjecture 8.22. *No design at* $(7, 3)$ *carries an admissible stationarity datum with* $0 < \nu < 1$.

Admissibility may not be dropped: Theorem 8.18 refutes the corresponding statement at $(4, 2)$ without it. With it, each conjecture is a finite conjunction of semialgebraic emptiness questions, one for each of the classes counted in Proposition 8.20. For a fixed active family the unknowns are the mk coordinates, the m weights, the multipliers, the tight directions, the entries of Λ and the value ν; the conditions are (75), (76), $\|u_i\| = 1$, $S_{C_i} u_i = \nu u_i$, $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, (1), $t \succ 0$, $\sum_c t_c = 1$, $0 < \nu < 1$, and (81) — the last a disjunction of sign conditions on the leading principal minors of $S_C - \nu I$, one disjunction for each k-subset C.

Theorem 8.23 (conditional)**.** *Assume Conjectures 8.21 and 8.22, and assume that a failure of* GTZ$(6,3)$*, respectively of* GTZ$(7,3)$*, produces an extremal counterexample at that cell in the sense of Definition 8.2. Then* GTZ$(m,3)$ *holds for every* m*.*

Proof. By Corollary 5.19 it suffices to prove GTZ$(6,3)$ and GTZ$(7,3)$. Suppose GTZ$(6,3)$ fails. By hypothesis F attains a local minimum over the designs at $(6,3)$ at a design of value $\nu < 1$, and $\nu > 0$ by Lemma 8.1. Theorem 8.4 produces at that design a stationarity datum with value ν, and it is admissible because $\nu = F$ there. This contradicts Conjecture 8.21. The same argument at $(7,3)$ with Conjecture 8.22 gives GTZ$(7,3)$. □

The extremality hypothesis is all this section takes from §7; Theorem 7.18 supplies it for the chart objective.

At $(6,3)$ the conjecture is an enumeration. At $(7,3)$ it is not, and an argument uniform in the active family is required. The exclusions of §8.6 clear away the small and the transversal configurations; what is left is where Example 8.17 lives, a system realized at the frontier size, at the threshold, by an active family of twenty triples.

> Decide the partition branch of Corollary 8.14: either exclude a datum whose active family is a two-block partition of $\{1,\dots,6\}$ with $0 < \nu < 1$, or exhibit one.

9. Formalization

The development is carried out in Lean 4 over Mathlib, and a statement whose head carries the tag *Lean* is verified there. Table 2 names the declaration behind each statement it lists, naming all of them where a statement is assembled from several; a tagged statement absent from the table is an elementary step of the same development, verified but not tabulated on its own. The formal statements are the originals, and the statements in the text are their transcriptions. What follows says what was formalized, under what axiom discipline, and where the formalization stops.

9.1. **What is formalized.** The formal definitions are those of §2 verbatim. A design is a structure carrying the vectors, the weights, positivity of the weights, their normalization, and (1); domination of a subset is positive semidefiniteness of $S_C - I$; and GTZ(m,k) is the assertion that every design of size m and rank k admits a dominating subset of cardinality k. No nondegeneracy is assumed: repeated atoms and zero atoms are admitted, and the results of §6 depend on that.

There is no separate real and complex development. The structure and the elementary theory are written once for scalars in an `RCLike` field with real weights, and $\mathbb{R}$ and $\mathbb{C}$ are instantiations, definitionally so over $\mathbb{R}$; Theorems 3.19 and 3.14 and Proposition 2.6 are proved at that generality and used at both. Where a statement is genuinely field-dependent, as Theorem 4.9 and Theorem 5.10 are, it is proved at the field where it holds.

The chart of §3 is formalized as the projection $P = VV^{\mathsf{H}}$ of (13) together with the equivalence of Lemma 3.6, and the compactification of §7 as the predicate of Definition 7.2 on pairs (P, t) with P a symmetric idempotent of trace k and t in the closed simplex. Graphic designs, used for the diamond of Example 6.9, are formalized through (97), so the whitener is an existential and the atoms are determined only up to the orthogonal group; what is rational, and what the formal statements quantify over, is the graph data. Finally the stationarity system of §8 is a structure bundling (75)–(76) with the active family and the multipliers; the results of that section are implications from it.

9.2. **Axiom discipline.** A ledger module lists, for the declarations of the development, the axioms on which each depends, printed by the kernel rather than asserted by the author. The permitted set is Mathlib's {`propext`, `Classical.choice`, `Quot.sound`}, and nothing outside it

Statement	Declaration
Lemma 2.3	`Gtz.trace_identity`
Lemma 2.5	`Gtz.coParsevalOperator_posDef`
Proposition 2.6	`Gtz.gtzWeighted_square`
Proposition 2.7	`Gtz.gtz_rank_one`
Lemma 2.9	`Gtz.exists_atom_covering_direction`
Example 2.11	`Gtz.tetraDesign_isTie`
Lemma 3.6	`Gtz.fieldDominates_iff_posSemidef_chartBlock_finset`
Remark 3.11	`Gtz.inertiaNoGoMatrix_positive_direction` and companions
Theorem 3.14	`Gtz.weighted_naimark_duality`
Theorem 3.19	`Gtz.corank_one_dominating_erasure`, `Gtz.fieldGtzWeighted_corank_one`, `Gtz.isTie_iff_forall_pivot_eq_one`, `Gtz.exists_realDesign_exactlyTied`, `Gtz.exists_complexDesign_exactlyTied`
Theorem 4.9	`Gtz.complexGtzWeighted_iff_size_le_rank_add_one`
(negative half)	`Gtz.not_complexGtzWeighted_of_rank_add_two_le_size`
Theorem 5.2	`Gtz.dominating_of_light_atom`
Theorem 5.10	`Gtz.gtz_rank_two`
Theorem 5.11	`Gtz.gtzWeighted_corank_two`
Corollary 5.12	`Gtz.gtzWeighted_of_le_five`
Theorem 5.15	`Gtz.crystallization`
Theorem 6.4	`Gtz.exists_isTie_of_weights_of_classes`, `Gtz.splitClassDesign_isTie`, `Gtz.splitTetraDesign_isTie`, `Gtz.splitSevenDesign_isTie`
Corollary 6.8	`Gtz.not_exists_card_posSemidef_sub_smul_one_of_isTie`, `Gtz.not_gtzWeightedFloor_rankThree_of_one_lt`
Example 6.9	`Gtz.diamondDesign_isTie`[a], `Gtz.not_hasParallelPair_diamondDesign`
Corollary 6.10	`Gtz.not_hingeHoldsAtSize_five_three`, `Gtz.diamondEdgeVector_eq`
Theorem 6.13	`Gtz.posSemidef_leverageWeightedAtomSum_sub_one`
Proposition 6.15	`Gtz.sq_rank_le_expectedElementary_one`
Proposition 6.18	`Gtz.mixedCharPoly_eval_one_eq_zero_of_totalTieSupported`
Theorem 7.14	`Gtz.exists_chartDominates_of_weight_eq_zero`
Corollary 7.15	`Gtz.weight_pos_of_forall_not_chartDominates`
Theorem 7.21	`Gtz.not_eventualCompressionLiftClaim`
Proposition 8.6	`Gtz.quadricLaw_of_isQuadricStationaryData`, `Gtz.coverageLaw_of_isQuadricStationaryData`, `Gtz.multiplierMatrix_eq_of_isQuadricStationaryData`
Corollary 8.13	`Gtz.value_eq_rank_of_singleActive`, `Gtz.not_isQuadricStationaryData_of_singleActive_of_value_lt_one`
Proposition A.1	`Gtz.not_hasUniformTieAggregateSeven`, `Gtz.splitSevenDesign_weightedTieAggregate_nonpos`, `Gtz.closedTieFailure_inhabited_seven`
Proposition A.7	`Gtz.dominates_of_twoMomentGap_nonneg`
Proposition 8.15	`Gtz.weight_mul_value_eq_one_of_saturatedAtom`, `Gtz.one_le_value_of_saturatedAtom`
Proposition A.9	`Gtz.icosa_twoMomentGap_eq`, `Gtz.icosa_no_twoMoment_certificate`, `Gtz.icosaDesign_strictly_dominates`

TABLE 2. Correspondence between the numbered statements and the formal development. The declaration marked a is proved but does not appear in the axiom ledger described in §9.2.

occurs. In particular no declaration depends on `sorryAx`: the development contains no incomplete proof. It contains no `axiom` declaration. Closed arithmetic over concrete finite index sets is discharged by kernel evaluation of decidable propositions, which introduces no axiom; no proof is closed by compiled evaluation.

The ledger covers exactly what it lists: one declaration, the entry marked a in Table 2, is absent from it, and is proved in a module duplicating one that the ledger does cover. Its proof is checked when that module is elaborated, but its axiom use is not certified by the ledger. The development also uses classical logic freely; nothing in it is a constructive claim.

No numerical assertion in the paper rests on floating-point evaluation. The witnesses of §7.4 and Appendix A are exact rationals in the formal statements and are checked as such; the icosahedral and graphic designs of Appendix B carry radicals, and the formal statements use the algebraic relations (100) and (98) that eliminate them. Remark A.11 records the conditioning phenomenon that makes this discipline necessary rather than decorative.

9.3. **Where the formalization stops.** Two steps of the argument are not formal. Both lie in §7–§8, and they are missing for different reasons.

The first is the attainment in Theorem 7.18: a continuous function on a compact space attains its minimum. The space of chart points is compact — a product of a Grassmannian and a closed simplex — and the objective G of Definition 7.2 is continuous, being a maximum of finitely many least-eigenvalue functions of continuous matrix-valued maps. We use this standard fact without formalizing it. Its two nontrivial inputs, Theorem 7.14 and Corollary 7.15, which place the minimizer at a design rather than at a boundary point, are formal.

The second is the passage from a local minimum to the system (75)–(76). The objective is a maximum of finitely many functions each of which is concave in S_C, so its directional derivative is the maximum over the active family of the directional derivatives of its members, and each of those is the least eigenvalue of the derivative compressed to the corresponding eigenspace. Nonnegativity of that quantity in every admissible direction, together with the spectraplex description of the superdifferential of $\lambda_{\min}$, is what produces the multipliers λ_C and the matrix Λ. Each ingredient is classical [8, 25, 15], and the framework, with the warning that for a maximum of concave functions the subdifferential rule is an inclusion rather than an equality, is [7]. None of it is in the ambient library, so §8 takes the system as its hypothesis and formalizes its consequences (Proposition 8.6, Proposition 8.9 and the exclusions) rather than deriving the system from minimality.

Theorem 8.23 should be read with that distinction in mind. Its two conjectural inputs speak about the system, not about minimality, so a proof of either closes the corresponding cell only together with the two steps above.

10. Problems

(1) *Prove Conjecture 8.21: the stationarity system* (75)–(76) *has no solution at* $(6, 3)$ *with* $0 < \nu < 1$*.*

By Theorem 7.18, whose hypotheses at this cell are the two instances $\mathrm{GTZ}^{\mathrm{ch}}(5, 3)$ and $\mathrm{GTZ}^{\mathrm{ch}}(5, 2)$ of Corollary 5.12, this closes $\mathrm{GTZ}(6, 3)$. The question is finite: the active family $\mathcal{A}^+$ covers the six indices by (85), has at least three members by Corollary 8.14, and may be taken of cardinality at most 15 by Carathéodory's theorem [4]; there are 2101 such families up to relabelling, of which 2068 lie within the Carathéodory bound (§B.7). The exclusions of §8.6, however, settle only 57 of the 2136 classes, and the remainder is concentrated where they are silent. For each, the task is to show that the ideal generated by (75), (76) and the equations $\lambda_{\min}(S_C) = \nu$ on the active family has no real point with $t \succ 0$, $\lambda \succeq 0$ and $0 < \nu < 1$.

Any elimination must respect the system's multiplicities and its redundancy. If S_C has ν as a multiple eigenvalue at an active C, then u_C ranges over an eigenspace and the multiplier attached to C is a positive semidefinite matrix on it, so the system is larger than its generic shape. And (85) is (75) contracted, with $\mu = \nu$ following from it, so the constraints must not be counted as independent.

(2) *Prove Conjecture 8.22: the same at* $(7, 3)$*.*

With Problem 1 this decides the real problem at rank three, by Theorem 8.23. Enumeration is not available: the census has 7 011 184 classes, of which 5 249 381 lie within the Carathéodory bound 19 (§B.7). An argument uniform in the active family is wanted, one using only the laws (76) and (85), the positivity of the weights, and the combinatorics of a covering family, never the family's identity. The known extremal designs at this cell, the split tetrahedra of §B.5, have active family of cardinality 20 and value $\nu = 1$; a uniform argument must therefore exclude $\nu < 1$ without excluding those.

(3) *Determine* $\alpha_3(\mathbb{C})$*.*

Remark 4.14 gives $\frac{1}{3} \le \alpha_3(\mathbb{C}) \le 3(1 - \cos\frac{2\pi}{9}) = 0.7019\ldots$, the upper bound realized by a binding triple of the Hesse configuration of nine equiangular lines in $\mathbb{C}^3$. We conjecture that the upper bound is the truth. More generally the sequence $\alpha_k(\mathbb{C})$ is unknown for every $k \ge 2$ beyond the bound $\alpha_k(\mathbb{C}) \ge 1/k$; over $\mathbb{R}$ the same question is empty below rank three and, if Problems 1 and 2 go as conjectured, empty at rank three, so it first has content at $\alpha_4(\mathbb{R})$.

(4) *Derandomize Theorem 6.14.*

Volume sampling assigns $\pi_C = \det P[C]$ to a k-subset by (55); the measure is a determinantal measure on the bases of the design [9], and it is strongly Rayleigh [2], so the mixture $q = \mathbb{E}_\pi[\chi_{S_C}]$ is real-rooted [1]. Two statements would together prove GTZ(m, k) at every rank at once:

(a) some k-subset C has $\lambda_{\min}(S_C)$ at least the least root of q;

(b) that least root is at least 1.

Neither is available. Statement (a) does not follow from real-rootedness of q alone (Proposition A.6), and the interlacing method of [18] requires the whole convex family to be real-rooted, a property of volume sampling that has not been verified. Statement (b) must be exact on the equality locus, by Proposition 6.17: at a design where every basis has 1 in the spectrum of S_C one has $q(1) = 0$, with no margin, and at the tetrahedron $q = (x-1)(x-4)^2$ (§B.1). A route through this problem would be the first argument in the subject that does not proceed cell by cell. Nothing in this paper manages that.

(5) *Classify the extremal designs.*

Theorem 3.19 classifies them at corank one: over each weight vector there is one, given in §B.2, and (21) is a complete description. Beyond corank one the constructed families are the split-class designs of §B.5, which have at most $k+1$ distinct directions, and the splittings of the diamond of §B.4, which has $k+2$. Is that list complete at $(6, 3)$ and $(7, 3)$? Is it finite, as a list of families? Does every extremal design have every basis active, that is $\lambda_{\min}(S_C) = 1$ for every independent k-subset C? Every known extremal design does: at the tetrahedron all four triples are bases and all are active, and at the diamond the eight spanning trees are bases and all eight have positive semidefinite singular gaps, the two circuits being dependent and carrying no volume-sampling mass. An affirmative answer is exactly the hypothesis under which Proposition 6.17 gives $q(1) = 0$, and hence exactly what Problem 4 would have to face at the equality locus.

(6) *Determine the true window in Theorem 5.15.*

The bound $k(k+1)/2+1$ is one more than the dimension of the space of symmetric $k \times k$ matrices and comes from Carathéodory's theorem applied to the representation (1) of I_k as a convex combination of the $g_c g_c^{\mathsf{T}}$. Nothing in the proof of Lemma 5.13 refers to domination, so no lower bound on the window follows from it. At $k=2$ the problem is decided at every size by Theorem 5.10, so the window is vacuous there; at $k=3$ the bound is 7 and the two cells $(6,3)$ and $(7,3)$ survive. Is the window ever larger than $k+3$? Equivalently: is there a rank k and a size $m > k+3$ at which $\mathrm{GTZ}(m,k)$ is strictly harder than at every smaller size? Corollary 5.12 settles corank at most two, so the question is first posed at corank three.

(7) *Find the subset.*

The selection principle asserts that a subset exists; neither known route exhibits one. Volume sampling dominates only on average (Theorem 6.14). The maximal-volume principle [10] does name a subset, and the guarantee it comes with is too weak. At uniform weights that principle bounds a locally maximal-volume choice by $\|A_C^{-1}\|^2 \le 1 + k(m-k)$ for the matrix A of rows $u_c = g_c/\sqrt{m}$, which in the normalization of (2) reads

$$\lambda_{\min}(S_C) \;\ge\; \frac{m}{1+k(m-k)}.$$

The right side equals 1 at $m = k+1$ and is strictly smaller for every $m > k+1$, so the principle reproduces Theorem 3.19 and nothing beyond it. Computing a maximal-volume subset is itself out of reach: the problem is NP-hard and NP-hard to approximate within a constant factor [6], and the rank-revealing factorizations that approximate it [13] inherit the weaker guarantee. Two questions therefore stand apart from the conjecture itself: does a subset of maximal volume dominate, and, granting $\mathrm{GTZ}(m,k)$, is there an algorithm polynomial in m and k that exhibits a dominating subset?

Appendix A. Strategies excluded, with witnesses

Every approach collected here fails, and each failure is recorded as an impossibility statement with an explicit certificate, not as an absence of proof. One cause accounts for nearly all of them. By Corollary 6.8 the selection inequality (2) is an equality on a set of full dimension in the weights; a device whose output is a strictly positive quantity is therefore asserting something false at every point of that set.

For a k-subset C of a design write Gram_C for the Gram matrix $(\langle g_c, g_d\rangle)_{c,d\in C}$. By (11), $\mathrm{Gram}_C = M_C M_C^{\mathsf{H}}$ and $S_C = M_C^{\mathsf{H}} M_C$ with M_C square, so the two have the same spectrum and

$$\delta(C) \;:=\; \det(\mathrm{Gram}_C - I_k) \;=\; \det(S_C - I_k). \tag{92}$$

Domination of C implies $\delta(C) \ge 0$; we call δ the *determinant leg.* At rank three we also use

$$\gamma(C) \;:=\; 4\det \mathrm{Gram}_C - (\operatorname{tr} \mathrm{Gram}_C - 1)^2, \tag{93}$$

a function of the first two moments of the spectrum of S_C alone.

A.1. Certificates with a uniform floor.

The natural request is a nonnegative combination of the determinant legs, at each design, bounded below by a positive constant. No such combination exists, and the obstruction has nothing to do with degree, polynomiality or equivariance: the coefficients below may be arbitrary nonnegative functions of the design.

Proposition A.1 (Lean). *There is no $\theta > 0$ with the following property: for every design $\mathcal{D}$ at $(7,3)$ with $\ell_c > 1$ for all c there are numbers $w_C(\mathcal{D}) \ge 0$, indexed by the 35 triples, with*

$$\sum_{|C|=3} w_C(\mathcal{D})\,\delta(C) \;\ge\; \theta.$$

Proof. Let $\mathcal{D}_\triangle$ be the split tetrahedron at $(7,3)$ of §B.5: seven atoms carrying the four tetrahedron directions with multiplicities $2,2,2,1$ and weights $(\frac{1}{8},\frac{1}{8},\frac{1}{8},\frac{1}{4},\frac{1}{8},\frac{1}{8},\frac{1}{8})$. Every leverage equals 3, so $\mathcal{D}_\triangle$ lies in the stated class. A triple of $\mathcal{D}_\triangle$ carries either three distinct directions or two equal ones. In the first case Gram_C has diagonal 3 and off-diagonal -1, so $\mathrm{Gram}_C - I_3 = 3I_3 - J_3$, of eigenvalues $0,3,3$ and determinant 0. In the second, after ordering so that the first two atoms agree,

$$\mathrm{Gram}_C - I_3 = \begin{pmatrix} 2 & 3 & -1 \\ 3 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \det = 2(4-1) - 3(6-1) + (-1)(-3+2) = -8.$$

Hence $\delta(C) \le 0$ for all 35 triples, and every nonnegative combination of the legs at $\mathcal{D}_\triangle$ is at most 0. $\square$

Any Positivstellensatz certificate in Putinar or Schmüdgen form yields, on evaluation at a design, exactly such a combination, its coefficients being the sums of squares evaluated there and its floor the constant of the representation. So:

Corollary A.2. *No representation $\sum_C \sigma_C\, \delta(C) = 1 + \sigma_0$, valid on the all-heavy locus at $(7,3)$ with σ_0 and the σ_C nonnegative there, exists — at any degree.*

Proof. Evaluate at $\mathcal{D}_\triangle$: the left side is at most 0 and the right side at least 1. $\square$

Consider the two systems on the all-heavy locus,

$$\text{(strict)} \quad \delta(C) < 0 \quad \text{for all } C, \qquad\qquad \text{(closed)} \quad \delta(C) \le 0 \quad \text{for all } C.$$

Failure of GTZ$(7,3)$ would give a point of the strict system; the design $\mathcal{D}_\triangle$ is a point of the closed one. So the closed system is inhabited and carries no certificate of emptiness of any kind, while the strict system is the one that must be certified empty. The problem lives in the gap between the two, which is the equality locus.

Remark A.3. Stengle's form, in which the multiplicative monoid part is nonempty, survives; one can bound its size. A Stengle certificate of emptiness of the strict system is an identity $f + g^2 + h = 0$ with f in the cone generated by the defining inequalities, h in the ideal generated by the equations, and g a product of powers of the strict generators $-\delta(C)$. At any point of the closed system all generators are nonnegative, so $f \ge 0$ and $h = 0$ there, forcing $g = 0$. Now the closed system contains one split tetrahedron for each partition of the seven indices into classes of sizes $2,2,2,1$, of which there are $7!/(2!^3\,3!) = 105$; at the partition Π the leg δ vanishes exactly on the 20 triples meeting three distinct classes of Π. So the set of triples occurring as factors of g must meet, for each of the 105 partitions, that partition's own family of 20. No set of at most three triples meets all 105 families; 5355 sets of four do, among them $\{123, 124, 135, 145\}$. Since δ has degree 3 in the entries of Γ and degree 6 in the atom coordinates, the monoid part of every Stengle certificate has degree at least 12 in the 28 Gram entries and at least 24 in the 21 coordinates. A sum of squares of degree 12 in 28 variables has a half-degree monomial basis of $\binom{34}{6} = 1\,344\,904$ elements.

A.2. **Margin transfer across a deletion.** Induction on the size deletes an atom, applies GTZ$(m-1,k)$, and lifts. The lift is exact, and one computation settles what it costs, once and for all.

Let $\mathcal{D}$ be a design at (m,k) and e an index with $s_e < 1$, equivalently $I - t_e\, g_e g_e^\top \succ 0$. Choose R with $R^\top (I - t_e g_e g_e^\top) R = I$ and set

$$g'_c := \sqrt{1-t_e}\; R^\top g_c, \qquad t'_c := \frac{t_c}{1-t_e} \qquad (c \ne e).$$

The weights are positive and sum to one, and

$$\sum_{c\neq e} t'_c\, g'_c g'^{\mathsf{T}}_c = R^{\mathsf{T}}\Big(\sum_{c\neq e} t_c\, g_c g_c^{\mathsf{T}}\Big)R = R^{\mathsf{T}}(I - t_e g_e g_e^{\mathsf{T}})R = I,$$

so $\mathcal{D}'$ is a design at $(m-1, k)$.

Proposition A.4. *Let $C \subseteq \{c : c \neq e\}$ have $|C| = k$ and satisfy $S'_C \succeq (1+\varepsilon)I$ in $\mathcal{D}'$, with $\varepsilon \geq 0$. Then C dominates in $\mathcal{D}$ provided*

$$1+\varepsilon \;\geq\; \kappa_e^{-1}, \qquad \textit{where} \qquad \kappa_e := \frac{1-s_e}{1-t_e}, \tag{94}$$

that is provided $\varepsilon \geq t_e(\ell_e - 1)/(1-s_e)$. The threshold is at most 0 if and only if $\ell_e \leq 1$.

Proof. From $R^{\mathsf{T}}(I - t_e g_e g_e^{\mathsf{T}})R = I$ one reads $RR^{\mathsf{T}} = (I - t_e g_e g_e^{\mathsf{T}})^{-1}$, hence for $y := R^{-1}w$,

$$y^{\mathsf{T}} S'_C v = (1-t_e)\, w^{\mathsf{T}} S_C w, \qquad y^{\mathsf{T}} y = w^{\mathsf{T}}(I - t_e g_e g_e^{\mathsf{T}})w = \|w\|^2 - t_e\langle g_e, w\rangle^2,$$

the first because $S'_C = (1-t_e)R^{\mathsf{T}} S_C R$. Applying the hypothesis at y and taking $\|w\| = 1$,

$$w^{\mathsf{T}} S_C w \;\geq\; \frac{(1+\varepsilon)(1 - t_e\langle g_e, w\rangle^2)}{1-t_e} \;\geq\; \frac{(1+\varepsilon)(1-s_e)}{1-t_e} \;=\; (1+\varepsilon)\,\kappa_e,$$

by Cauchy–Schwarz, $\langle g_e, w\rangle^2 \leq \ell_e$. This is at least 1 exactly under (94). Finally $\kappa_e^{-1} - 1 = (s_e - t_e)/(1-s_e) = t_e(\ell_e - 1)/(1-s_e)$, whose sign is that of $\ell_e - 1$. □

The proposition is an identity rather than an estimate, and it splits the induction into the two branches already in use. At $\ell_e \leq 1$ the threshold is nonpositive and the deletion needs nothing downstairs; this is the light-atom step that reduces $\mathrm{GTZ}(m,k)$ to all-heavy designs. At $\ell_e > 1$ the threshold is strictly positive.

Corollary A.5. *At rank three no deletion of a heavy atom transports domination without a margin, and no such margin is available: by Corollary 6.8, for every $m \geq 5$ and every $\varepsilon > 0$ there is a design at $(m-1, 3)$ with no k-subset satisfying $S_C \succeq (1+\varepsilon)I$.*

No choice of the deleted atom repairs this, because the threshold depends on e only through ℓ_e and by Lemma 2.3 some atom has $\ell_e \geq k$.

A.3. **Restricting the region.** Cutting the degenerate part out of the design space and running a compactness argument on the rest requires a cut that misses the equality locus. The two cuts that suggest themselves, a floor on the weights and the exclusion of parallel atoms, both leave the locus intact.

By Theorem 6.4 every weight vector on $m \geq k+1$ indices carries an extremal design, one for each partition of the indices into $k+1$ nonempty classes; the construction is displayed in §B.5. So no restriction of the weight simplex — to a compact subset, to a neighbourhood of the uniform point, to any nonempty relatively open set — excludes the equality locus, and the infimum of the selection objective over any such region is at most 1.

The second cut fails for a different reason. Every design produced by splitting has two parallel atoms, so a region defined by excluding parallel pairs might be hoped to miss the equality locus. It does not: the diamond of Example 6.9, whose data is in §B.4, is an extremal design at $(5,3)$ whose five atoms are pairwise non-parallel. Its splittings give extremal designs with a parallel pair at $(6,3)$ and $(7,3)$, and perturbing a split pair apart moves the design off the parallel locus while moving the value by an arbitrarily small amount, since the value is continuous.

A.4. **The compression lift at positive weight.** Theorem 7.21. The lift (67) used in Theorem 7.14 holds at $t_z = 0$ by an exact identity and fails at every positive t_z, however small; the witness is the chart point and one-parameter weight family of §B.3, whose relevant 2×2 block has determinant $-\tau(2+3\tau)/9$ at pivot weight τ. The failure is not a matter of size: the deficit is of first order in τ against a downstairs slack which the family freezes at 0. This is again Corollary 6.8, in its sharpest local form; hence the boundary analysis of §7.2 is carried out at the limit point and not along a minimizing sequence.

A.5. **Real-rootedness of the mixture.** Theorem 6.14 makes the selection problem a derandomization: $\mathbb{E}_\pi[S_C] \succeq I$, and one wants a single C. The interlacing method supplies derandomizations of just this shape, and the mixture $q = \mathbb{E}_\pi[\chi_{S_C}]$ of Proposition 6.17 is the object it would act on. Real-rootedness of q alone, however, buys nothing.

Proposition A.6. *There are real-rooted monic quadratics f_1, f_2 whose average is real-rooted and whose average has least root strictly larger than the least roots of both.*

Proof. Take

$$f_1 = X^2 - X, \qquad f_2 = X^2 - 23X + 60,$$

with roots $\{0,1\}$ and $\{3,20\}$, so least roots 0 and 3. Their average is $X^2 - 12X + 30$, of discriminant $144 - 120 = 24 > 0$, with roots $6 \pm \sqrt{6}$; the least is $6 - \sqrt{6} = 3.5505\ldots > 3$. □

The hypothesis the interlacing method actually requires is that the whole convex family be real-rooted — a common interlacing — and the pair above fails it:

$$\lambda f_1 + (1-\lambda) f_2 = X^2 + (22\lambda - 23)X + 60(1-\lambda), \qquad \mathrm{disc} = 484\lambda^2 - 772\lambda + 289,$$

which at $\lambda = \frac{4}{5}$ equals $-\frac{471}{25} < 0$. So the conclusion of [18] is unavailable for this pair, and no contradiction arises. Nor is real-rootedness preserved by averaging in general: $X^2 - 1$ and $X^2 + 10X + 25$ are real-rooted and their average $X^2 + 5X + 12$ has discriminant -23. A derandomization of Theorem 6.14 must therefore verify the interlacing hypothesis for volume sampling, not assume real-rootedness of the mixture. This is Problem 4.

A.6. **Two moments.** At rank three there is a genuine criterion in the first two moments of the spectrum; it is sharp among such criteria, and it does not decide the open cells. Three statements follow: the criterion, its sharpness among two-moment criteria, and a design at $(6,3)$ on which it is silent.

Proposition A.7 (Lean)**.** *Let C be a 3-subset of a design at rank three with $\ell_C := \operatorname{tr} \mathrm{Gram}_C > 3$. If $\gamma(C) \geq 0$ then C dominates.*

Proof. Let $x \leq y \leq z$ be the eigenvalues of S_C, all nonnegative, with $x + y + z = \ell_C$ and $xyz = \det \mathrm{Gram}_C$. By the arithmetic–geometric mean inequality applied to y, z,

$$4 \det \mathrm{Gram}_C = 4xyz \leq 4x\Big(\frac{y+z}{2}\Big)^2 = x(\ell_C - x)^2.$$

The hypothesis $\gamma(C) \geq 0$ reads $4 \det \mathrm{Gram}_C \geq (\ell_C - 1)^2$, so $x(\ell_C - x)^2 - (\ell_C - 1)^2 \geq 0$. Expanding both sides,

$$x(\ell_C - x)^2 - (\ell_C - 1)^2 = (x-1)\big(x^2 + (1 - 2\ell_C)x + (\ell_C - 1)^2\big),$$

an identity in x and ℓ_C. Write $Q(x)$ for the quadratic factor. Its vertex is at $x = \ell_C - \frac{1}{2} > 1$, so Q decreases on $[0,1]$, and $Q(1) = 2 - 2\ell_C + (\ell_C - 1)^2 = (\ell_C - 1)(\ell_C - 3) > 0$; hence $Q > 0$ on $[0,1]$. If $x < 1$ then $(x-1)Q(x) < 0$, contradicting the displayed inequality. So $x \geq 1$, that is $S_C \succeq I$. □

The quadratic factor Q is the multiplier of the certificate; no semidefinite programming enters anywhere. At the tetrahedron (§B.1) one has $\ell_C = 9$, $\det \mathrm{Gram}_C = 16$ and $\gamma(C) = 0$, so the criterion is tight on the very locus it was built to reach: the arithmetic–geometric step is an equality exactly when the two upper eigenvalues coincide, as they do at $\{1, 4, 4\}$.

Proposition A.8. *Let $\ell_C > 3$ and $0 \le \varepsilon$ satisfy $4\varepsilon < (\ell_C - 1)^2$. There is a positive semidefinite 3×3 matrix of trace ℓ_C and determinant ε whose least eigenvalue is smaller than 1. Hence no criterion depending on $(\mathrm{tr}\, \mathrm{Gram}_C, \det \mathrm{Gram}_C)$ alone is stronger than Proposition A.7.*

Proof. On $[0, 1]$ the function $x \mapsto x(\ell_C - x)^2$ has derivative $(\ell_C - x)(\ell_C - 3x) > 0$, so it increases from 0 to $(\ell_C - 1)^2$; let $x \in [0, 1)$ be the unique point with $x(\ell_C - x)^2 = 4\varepsilon$. The matrix $\mathrm{diag}(x, \frac{\ell_C - x}{2}, \frac{\ell_C - x}{2})$ is positive semidefinite because $\ell_C > x$, has trace ℓ_C and determinant $x(\ell_C - x)^2/4 = \varepsilon$, and its least eigenvalue is $x < 1$ since $3x < 3 < \ell_C$. □

Proposition A.9 (Lean)**.** *At the icosahedral design of §B.6 every triple C satisfies $\gamma(C) < 0$, while the triple $\{0, 2, 4\}$ dominates strictly.*

Proof. By (101) every leverage is 3 and every distinct pairing has square $\frac{9}{5}$. For a triple with pairings p_{ab}, p_{ac}, p_{bc} and product $\tau := p_{ab}p_{ac}p_{bc}$,

$$\mathrm{tr}\, \mathrm{Gram}_C = 9, \qquad \det \mathrm{Gram}_C = 27 + 2\tau - 3(p_{ab}^2 + p_{ac}^2 + p_{bc}^2) = 27 + 2\tau - \tfrac{81}{5} = \tfrac{54}{5} + 2\tau,$$

so by (93)

$$\gamma(C) = 4\Big(\tfrac{54}{5} + 2\tau\Big) - 64 = 8\tau - \tfrac{104}{5}.$$

Now $\tau^2 = (\frac{9}{5})^3 = \frac{729}{125}$, while $(\frac{13}{5})^2 = \frac{169}{25} = \frac{845}{125}$; hence $|\tau| < \frac{13}{5}$ and $8\tau < \frac{104}{5}$, so $\gamma(C) < 0$ at every one of the twenty triples. The domination of $\{0, 2, 4\}$, with $\lambda_{\min} = 3 - 3/\sqrt{5} > 1$, is computed in §B.6. □

The two-moment criterion is therefore silent at $(6, 3)$, and by Proposition A.8 so is every criterion in those two moments. The criterion discards exactly the middle elementary symmetric function of Gram_C: at the icosahedron the moduli of the pairings are all equal, and the verdict depends only on the sign of their product. Any certificate for GTZ$(6, 3)$ must consume $e_2(\mathrm{Gram}_C)$, or equivalently must be sensitive to that sign.

A.7. **Structure inherited from smaller sizes.**

Remark A.10. The signature of W carries no information: by Lemma 3.10 the matrix $W = P - D$ has inertia $(k, m - k, 0)$, and by Remark 3.11 there is a Hermitian matrix of inertia $(1, 1, 0)$ with no positive semidefinite 1×1 principal block. So no statement of the form "a Hermitian matrix of inertia $(k, m - k, 0)$ has a positive semidefinite $k \times k$ principal block" is available; a proof must use more of the projection than its inertia.

By Corollary 6.10 the statement "every extremal design contains two parallel atoms" is false at $(5, 3)$: the diamond of §B.4 is extremal and its five atoms are pairwise non-parallel. Every extremal design obtained by splitting does contain a parallel pair, so the statement is true on the family one constructs first and false in general. An induction branching on the presence of a parallel pair therefore owes a separate treatment of the diamond, and of every extremal family not obtained by splitting, a list not known to be finite. This is part of Problem 5.

A.8. **Conditioning.**

Remark A.11. The selection objective is ill-conditioned near the boundary of the weight simplex. As $t_c \to 0$ the Parseval condition forces $\ell_c \sim 1/t_c$, the entries of S_C grow like $1/\min_c t_c$, and the difference $\lambda_{\min}(S_C) - 1$ is the difference of two quantities of that size. A floating-point evaluation of $\lambda_{\min}$ on a formed Gram matrix therefore carries no information about the sign of the margin; the stable quantity is $\sigma_{\min}(V_C) - 1$, computed from the isometry V_C of (13) directly. Every constant asserted in this paper is exact: the witnesses above are rational, and the algebraic data of Appendix B carries its defining relations.

Appendix B. Exact data

Every design recorded here is given by data from which (1) can be checked in finite arithmetic, and every constant quoted is exact. Radicals occur in three places: the icosahedral design of §B.6, the whitener of a graphic design in §B.4, and the normalization of the tetrahedron. Each time, the algebraic relation that eliminates them is displayed. No entry rests on a floating-point evaluation.

B.1. **The tetrahedron at** $(4, 3)$**.** Atoms and weights:

$$g_0 = (1, 1, 1), \quad g_1 = (1, -1, -1), \quad g_2 = (-1, 1, -1), \quad g_3 = (-1, -1, 1), \qquad t_c = \tfrac{1}{4}.$$

These are the four sign vectors with an even number of minus signs. For $i \neq j$ the coordinatewise products of g_i and g_j sum to -1, and for each pair of distinct coordinates the four products $(g_c)_a(g_c)_b$ are $1, -1, -1, 1$; hence

$$\ell_c = 3, \qquad \langle g_i, g_j \rangle = -1 \ (i \neq j), \qquad \sum_c g_c g_c^{\mathsf{T}} = 4I_3,$$

so $\sum_c t_c g_c g_c^{\mathsf{T}} = I_3$ and this is a design. It is the design of Example 2.11 after the rescaling $g_c = \sqrt{3}\, u_c$ made there. Lemma 2.3 holds termwise: $t_c \ell_c = \frac{3}{4}$ and $\sum_c t_c \ell_c = 3 = k$.

For $C = \{0, 1, 2, 3\} \setminus \{j\}$,

$$S_C = 4I_3 - g_j g_j^{\mathsf{T}}, \qquad \operatorname{spec} S_C = \{1, 4, 4\}, \qquad \operatorname{spec}(S_C - I) = \{0, 3, 3\},$$

the eigenvalue 1 occurring along g_j. For $j = 3$ the gap is the singular positive form

$$x^{\mathsf{T}}(S_{\{0,1,2\}} - I)x = (x_0 - x_1)^2 + (x_0 + x_2)^2 + (x_1 + x_2)^2,$$

with null line $\mathbb{R}\,(1, 1, -1)$. Every triple dominates and none dominates strictly: the tetrahedron is extremal in the sense of Definition 6.1.

Chart and volume sampling. Here $v_c = \frac{1}{2} g_c$, so $P_{cd} = \frac{1}{4}\langle g_c, g_d \rangle$, that is $P = \frac{1}{4}(4I_4 - J_4)$ with J_n the all-ones $n \times n$ matrix. For a triple C, $P[C] = \frac{1}{4}(4I_3 - J_3)$, whose eigenvalues are $\frac{1}{4}\{1, 4, 4\}$, so by (55)

$$\pi_C = \det P[C] = \tfrac{16}{64} = \tfrac{1}{4} \qquad \text{for each of the four triples,}$$

confirming $\sum_C \pi_C = 1$. All four characteristic polynomials coincide, whence

$$q(x) = \mathbb{E}_\pi[\chi_{S_C}(x)] = (x - 1)(x - 4)^2 = x^3 - 9x^2 + 24x - 16,$$
$$q(1) = 0, \qquad q'(1) = 9, \qquad q''(1) = -12.$$

The subleading coefficient is $9 = \sum_c t_c \ell_c^2 = 4 \cdot \frac{1}{4} \cdot 9$, attaining the bound k^2 of Proposition 6.17.

Stationarity data. Take $\lambda_C = \frac{1}{4}$ on each of the four triples and $u_C = g_{c_0}/\sqrt{3}$ for C the complement of $\{c_0\}$. Then

$$\Lambda = \nu \sum_C \lambda_C\, u_C u_C^{\mathsf{T}} = 1 \cdot \sum_{c_0} \tfrac{1}{4} \cdot \tfrac{1}{3}\, g_{c_0} g_{c_0}^{\mathsf{T}} = \tfrac{1}{12} \cdot 4I_3 = \tfrac{1}{3} I_3, \qquad \operatorname{tr} \Lambda = 1 = \nu.$$

The quadric law (76) reads $g_c^{\mathsf{T}} \Lambda g_c = \ell_c/3 = 1 = \nu$, and the coverage law (85) reads, for each c, a sum over the three triples containing c of $\frac{1}{4}(-\frac{1}{\sqrt{3}})^2 = \frac{1}{12}$, that is $\frac{1}{4} = t_c \nu$.

Two moments. The Gram matrix of a triple is $4I_3 - J_3$, of trace 9 and determinant 16, so the quantity γ of (93) is $4 \cdot 16 - (9-1)^2 = 0$: the two-moment criterion of Proposition A.7 is exactly tight here.

B.2. **The corank-one extremal family.** Let $m = k+1$ and let t be any point of the open simplex. By Theorem 3.19 the extremal designs at $(k+1, k)$ with weights t are exactly those whose chart is $P = I - zz^{\mathsf{H}}$ with

$$|z_c|^2 = \frac{1-t_c}{k}, \qquad \text{equivalently} \qquad s_c = P_{cc} = \frac{k-1+t_c}{k}, \qquad \ell_c = \frac{k-1+t_c}{k\,t_c}.$$

The normalization is automatic: $\sum_c (1-t_c) = m-1 = k$. Over $\mathbb{R}$ the vector z is determined up to signs and an orthogonal change of the ambient frame; over $\mathbb{C}$, up to phases. At $k=3$, $t \equiv \frac{1}{4}$ one gets $|z_c|^2 = \frac{1}{4}$, $s_c = \frac{3}{4}$, $\ell_c = 3$ — the tetrahedron of §B.1. At $k=3$, $t = (\frac{2}{3}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9})$ one gets a member which is not a rotation of the tetrahedron:

$$g_0 = \tfrac{1}{3}(2,2,2), \quad g_1 = \tfrac{1}{3}(2,2,-7), \quad g_2 = \tfrac{1}{3}(-7,2,2), \quad g_3 = \tfrac{1}{3}(2,-7,2),$$

with $\ell = (\frac{4}{3}, \frac{19}{3}, \frac{19}{3}, \frac{19}{3})$, the single dependency $3g_0 + 2g_1 + 2g_2 + 2g_3 = 0$, and dominating triple $\{1,2,3\}$ whose gap is $\frac{8}{3}\big[(x_0-x_1)^2 + (x_0-x_2)^2 + (x_1-x_2)^2\big]$. The equality locus is therefore not a single orbit even at corank one.

B.3. **The chart point of §7.4.** Fix the symmetric matrix

$$P = \frac{1}{3}\begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \qquad M := 3P. \tag{95}$$

Then $M^2 = 3M$ entrywise, hence $P^2 = P$, and $\operatorname{tr} P = \frac{2+1+2+1}{3} = 2$. So P is a symmetric idempotent of rank two and (P,t) is a chart point of size four and rank two in the sense of Definition 7.2 for every t in the closed simplex. Take the escaping index $z = 1$ and the one-parameter family of weights

$$t^{(\tau)} = \Big(\tfrac{1-\tau}{3},\ \tau,\ \tfrac{1-\tau}{3},\ \tfrac{1-\tau}{3}\Big), \qquad 0 < \tau < 1, \tag{96}$$

whose member at $\tau = \frac{1}{4}$ is the uniform vector. With $W = P - \operatorname{diag}(t^{(\tau)})$,

$$W[\{0,1\}] = \begin{pmatrix} \dfrac{1+\tau}{3} & \dfrac{1}{3} \\ \dfrac{1}{3} & \dfrac{1}{3} - \tau \end{pmatrix}, \qquad \det W[\{0,1\}] = \frac{(1+\tau)(1-3\tau)-1}{9} = -\frac{\tau(2+3\tau)}{9}.$$

The determinant is strictly negative for every $\tau \in (0,1)$ and vanishes to first order as $\tau \to 0$. An explicit witness is $(-1,\, 1+\tau)$:

$$(-1, 1+\tau)\, W[\{0,1\}]\, (-1, 1+\tau)^{\mathsf{T}} = \frac{1+\tau}{3}\big[1 - 2 + (1+\tau)(1-3\tau)\big] = -\tfrac{1}{3}(1+\tau)\,\tau\,(2+3\tau).$$

At $\tau = \frac{1}{4}$ the block, its determinant and the probe $(1,-4)$ are

$$W[\{0,1\}] = \begin{pmatrix} 5/12 & 1/3 \\ 1/3 & 1/12 \end{pmatrix}, \qquad \det = -\tfrac{11}{144}, \qquad (1,-4)\, W[\{0,1\}]\, (1,-4)^{\mathsf{T}} = -\tfrac{11}{12}.$$

The leverages P_{cc}/t_c at $\tau = \frac{1}{4}$ are $\frac{8}{3}, \frac{4}{3}, \frac{8}{3}, \frac{4}{3}$, all exceeding 1, so the point is not disposed of by deletion of a light atom; and the deflation (65) at $z = 1$, where $P_{11} = \frac{1}{3} > 0$, is a symmetric idempotent of trace 1 whose gap at the surviving index 0 is exactly 0, at every τ. Theorem 7.21 exploits that frozen equality: the Schur term of (67) is matched in full at $t_z = 0$ and falls short by a quantity of first order in t_z thereafter, with no downstairs slack to close the gap.

> Exhibit a one-parameter deformation of (95) with compressed slack $\varepsilon > 0$, and display the resulting determinant $(3\varepsilon - 2\tau - 3\tau^2)/9$ of (68) changing sign at $\tau \asymp \varepsilon$.

B.4. **Graphic designs, and the diamond at** $(5, 3)$**.** The diamond is the second extremal design at rank three, and it is presented through a construction used again in §B.5. Let $\mathcal{G}$ be a connected multigraph on a vertex set V with a distinguished vertex, the *ground*; let $b_e \in \mathbb{R}^{V\setminus\{\text{ground}\}}$ be the reduced incidence vector of the edge e ($+1$ at its tail, -1 at its head, the ground coordinate deleted); let $c_e > 0$ be conductances and $t_e > 0$ weights summing to one. Put

$$L := \sum_e c_e\, b_e b_e^{\mathsf{T}},$$

the weighted reduced Laplacian, positive definite because $\mathcal{G}$ is connected. Choose R with $R^{\mathsf{T}}LR = I$ and set

$$g_e := \sqrt{c_e/t_e}\; R^{\mathsf{T}} b_e. \tag{97}$$

Then $\sum_e t_e g_e g_e^{\mathsf{T}} = R^{\mathsf{T}}LR = I$, so this is a design of size $|E|$ and rank $|V| - 1$. From $R^{\mathsf{T}}LR = I$ one reads $RR^{\mathsf{T}} = L^{-1}$, whence

$$s_e = t_e \ell_e = c_e\, b_e^{\mathsf{T}} L^{-1} b_e = c_e\, R_{\text{eff}}(e) : \tag{98}$$

the leverage score of an edge is its conductance times its effective resistance, and Lemma 2.3 becomes $\sum_e c_e R_{\text{eff}}(e) = |V| - 1$. Congruence by the invertible R removes the whitener from (2): an edge set E' with $|E'| = |V| - 1$ dominates if and only if

$$\sum_{e\in E'} \frac{c_e}{t_e}\, b_e b_e^{\mathsf{T}} \;\succeq\; L. \tag{99}$$

The atoms themselves are irrational and determined only up to the orthogonal group; the graph, the conductances, L, the effective resistances and the leverages are rational.

Take $\mathcal{G} = K_4 - e$ on vertices a, b, c, d with d the ground and cd the missing edge. The five edges and their reduced incidence rows in the coordinates (y_a, y_b, y_c) are

$$\begin{array}{lll} e_0 = ab: \ (1,-1,0), & e_1 = ac: \ (1,0,-1), & e_2 = ad: \ (1,0,0), \\ e_3 = bc: \ (0,1,-1), & e_4 = bd: \ (0,1,0). & \end{array}$$

Conductances $c = (2, 3, 3, 3, 3)$ and uniform weights $t_e = \frac{1}{5}$. Then

$$L = \begin{pmatrix} 8 & -2 & -3 \\ -2 & 8 & -3 \\ -3 & -3 & 6 \end{pmatrix}, \qquad \det L = 180, \qquad L^{-1} = \frac{1}{180}\begin{pmatrix} 39 & 21 & 30 \\ 21 & 39 & 30 \\ 30 & 30 & 60 \end{pmatrix},$$

so $R_{\text{eff}}(ab) = \frac{1}{180}(39 - 21 - 21 + 39) = \frac{1}{5}$ and $R_{\text{eff}}(e) = \frac{39}{180} = \frac{13}{60}$ for each of the four rim edges ac, ad, bc, bd. By (98),

$$s_{ab} = \tfrac{2}{5}, \quad s_{\text{rim}} = \tfrac{13}{20}, \qquad \ell_{ab} = 2, \quad \ell_{\text{rim}} = \tfrac{13}{4}, \qquad \tfrac{2}{5} + 4\cdot\tfrac{13}{20} = 3 = k.$$

Every leverage exceeds 1. The star at a, that is $E' = \{ab, ac, ad\}$, dominates: the left side of (99) is $5\big(2b_0b_0^{\mathsf{T}} + 3b_1b_1^{\mathsf{T}} + 3b_2b_2^{\mathsf{T}}\big)$, and subtracting L gives the form Q with

$$2\,Q(y) \;=\; (-8y_a + 2y_b + 3y_c)^2 + 9y_c^2,$$

positive semidefinite and singular, with null line $\mathbb{R}\,(1, 4, 0)$. No three-subset dominates strictly. The eight spanning trees each carry a null vector of their gap; the two circuits do worse. At $E' = \{ab, ac, bc\}$ the potential $y = (1, 1, 1)$ annihilates all three selected drops while $y^{\mathsf{T}}Ly = 3 + 3 = 6$, so the gap form equals -6; likewise at $E' = \{ab, ad, bd\}$ with $y = (0, 0, 1)$. The certificates for all ten subsets are

$$\begin{array}{llll} \{0,1,2\} : (1,4,0) & \{0,1,3\} : (1,1,1) & \{0,1,4\} : (4,1,5) & \{0,2,3\} : (1,4,5) \\ \{0,2,4\} : (0,0,1) & \{0,3,4\} : (4,1,0) & \{1,2,3\} : (2,8,5) & \{1,2,4\} : (3,-3,5) \\ \{1,3,4\} : (8,2,5) & \{2,3,4\} : (-3,3,5). & & \end{array}$$

The diamond is therefore an extremal design at $(5,3)$. Its five incidence rows are pairwise non-proportional, and by (97) proportionality of two atoms forces proportionality of the corresponding rows; so no two atoms of the diamond are parallel. This is Example 6.9.

The conductances are forced: keep the rim conductance at $c_{\mathrm{rim}} = 3$, let the spine carry $c_{ab} = p > 0$, and ask that the star at a dominate. With L recomputed at those conductances, (99) for $E' = \{ab, ac, ad\}$ reads $Q \succeq 0$ for a symmetric 3×3 form whose determinant is $324p - 648$: negative for $p < 2$, zero at $p = 2$, positive for $p > 2$. So $p = 2$ is the unique conductance at which the star dominates without dominating strictly, which is extremality; and then the three a-to-b paths in $K_4 - e$ — the edge ab of conductance 2 and two two-edge paths of conductance $\frac{3}{2}$ each — give $R_{\mathrm{eff}}(ab) = 1/(2+3) = \frac{1}{5}$ and $s_{ab} = \frac{2}{5}$, as computed above.

B.5. **Split families at** $(6,3)$ **and** $(7,3)$**.** Splitting an atom into parallel copies preserves (1), because the copies carry the same rank-one moment; and it preserves the value of the objective, because a k-subset repeating a direction is singular while a k-subset meeting k distinct directions has the gap of the original. Applied to the tetrahedron of §B.1 this gives the two families used throughout §6 and Appendix A.

At $(6,3)$. Directions $(0,1,2,2,3,3)$ on the four tetrahedron vertices, weights

$$\left(\tfrac{1}{4},\ \tfrac{1}{4},\ a,\ \tfrac{1}{4} - a,\ b,\ \tfrac{1}{4} - b\right), \qquad 0 < a, b < \tfrac{1}{4},$$

so that every direction carries total weight $\frac{1}{4}$. Every member is extremal. The active family — the triples attaining the value — is the twelve triples carrying three distinct directions,

$$\begin{array}{cccccc} \{0,1,2\} & \{0,1,3\} & \{0,1,4\} & \{0,1,5\} & \{0,2,4\} & \{0,2,5\} \\ \{0,3,4\} & \{0,3,5\} & \{1,2,4\} & \{1,2,5\} & \{1,3,4\} & \{1,3,5\} \end{array}$$

missing the directions $3,3,2,2,1,1$ and $1,1,0,0,0,0$ respectively.

At $(7,3)$. Directions $(0,1,2,3,0,1,2)$, weights $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$; again each direction carries $\frac{1}{4}$, every leverage is exactly 3, and (1) is exact. Of the 35 triples, the 20 carrying three distinct directions dominate, each with gap $3I_3 - g_d g_d^{\mathsf{T}}$ of spectrum $\{0,3,3\}$ where d is the missed direction; the remaining 15 do not. None of the 20 has any slack.

More generally, fix a surjection cl from $\{1, \ldots, m\}$ onto $\{1, \ldots, k+1\}$ and a weight vector t; give the index c the corank-one extremal atom of §B.2 attached to the class $\mathrm{cl}(c)$, evaluated at the class totals $T_j := \sum_{\mathrm{cl}(c)=j} t_c$. Atoms in one class are equal, so the fibrewise sums collapse and (1) is inherited from the corank-one design at T. A k-subset repeating a class is singular; a k-subset meeting k classes has the gap of that corank-one design, which is positive semidefinite and singular. Hence *at every size* $m \geq k+1$*, over every weight vector, and for every partition of the indices into* $k+1$ *nonempty classes there is an extremal design.* At $k = 3$, $m = 6$ the class profiles $(2,2,1,1)$ and $(3,1,1,1)$ occur, and at $m = 7$ the profile $(2,2,2,1)$. Since a member of this family has at most $k+1$ distinct directions and the diamond has $k+2 = 5$, the diamond is not one of them.

A rational extremal design with an exact certificate. Take the $(6,3)$ split at $(a,b) = (\frac{1}{6}, \frac{1}{8})$, so $t = (\frac{1}{4}, \frac{1}{4}, \frac{1}{6}, \frac{1}{12}, \frac{1}{8}, \frac{1}{8})$ and the atoms are the rational vectors of §B.1. A triple carrying three distinct directions and missing the direction d has, by §B.1,

$$S_C - I_3 \;=\; 3I_3 - g_d g_d^{\mathsf{T}}, \qquad x^{\mathsf{T}}(S_C - I_3)x \;=\; \sum_{i<j} (x_i - x_j)^2$$

after the relabelling that puts $g_d = (1,1,1)$: a sum of squares, singular on the line $\mathbb{R}g_d$. Every entry of the certificate is an integer, and its kernel is one-dimensional at every one of the twelve such triples. The equality locus therefore does not destroy the existence of a certificate; it destroys its slack.

B.6. **The icosahedral design at** $(6,3)$**.** Put

$$\rho := \frac{3}{\sqrt{5}}, \qquad \sigma := \sqrt{\frac{3-\rho}{2}}, \qquad \mu := \sqrt{\frac{3+\rho}{2}}, \tag{100}$$

so that $\sigma^2+\mu^2 = 3$ and $\sigma^2\mu^2 = \frac{9-\rho^2}{4} = \frac{9-9/5}{4} = \frac{9}{5} = \rho^2$, whence $\sigma\mu = \rho$. The six atoms are the six diagonals of a regular icosahedron, in the scaling that makes $\ell_c = 3$:

$$(\sigma,\mu,0),\ (-\sigma,\mu,0),\ (0,\sigma,\mu),\ (0,-\sigma,\mu),\ (\mu,0,\sigma),\ (\mu,0,-\sigma), \qquad t_c = \tfrac{1}{6}.$$

Each coordinate receives $2(\sigma^2+\mu^2) = 6$ from the six outer products and each off-diagonal entry cancels in pairs, so $\sum_c t_c g_c g_c^{\mathsf{T}} = I_3$. Every distinct pairing is $\pm\sigma\mu$, hence

$$\ell_c = 3, \qquad \langle g_c, g_d\rangle^2 = \tfrac{9}{5} \quad (c \neq d). \tag{101}$$

For $C = \{0,2,4\}$ all three pairings equal $+\rho$, so $S_C = (3-\rho)I_3 + \rho J_3$, with spectrum $\{3-\rho, 3-\rho, 3+2\rho\}$ and

$$\lambda_{\min}(S_{\{0,2,4\}}) = 3 - \frac{3}{\sqrt{5}} = 1.6584\ldots > 1:$$

the icosahedral design dominates, and dominates strictly. Exactly ten of its twenty triples dominate. It is used in Appendix A as the design on which the two-moment criterion is blind.

B.7. **Active-family census at** $(6,3)$ **and** $(7,3)$**.** By (85) the family $\mathcal{A}^+$ of active k-subsets carrying a positive multiplier covers the index set. The following are the numbers of covering families of 3-subsets, counted up to the symmetric group of the index set and listed by cardinality. At $m = 6$ a covering family has at least two members and at most twenty:

$\lvert\mathcal{A}^+\rvert$	2	3	4	5	6	7	8	9	10	11
classes	1	3	15	37	88	157	247	311	351	312
$\lvert\mathcal{A}^+\rvert$	12	13	14	15	16	17	18	19	20	
classes	249	161	94	43	21	7	3	1	1	

The total is 2102. The single class of cardinality two is the partition branch that Corollary 8.14 leaves undecided and does not exclude; setting it aside leaves 2101. By Carathéodory's theorem [4] the support of the multiplier vector may be taken of size one more than the dimension of the design manifold modulo the orthogonal group. That dimension is $18+6-6-1-3 = 14$ at $(6,3)$, so only the columns up to 15 bear on Conjecture 8.21; they account for 2069 of the 2102 classes, and for 2068 of the 2101 with at least three members.

At $m = 7$ a covering family has at least three members, and the counts are

$\lvert\mathcal{A}^+\rvert$	3	4	5	6	7	8	9	10
classes	3	17	94	415	1599	5308	15397	39081
$\lvert\mathcal{A}^+\rvert$	11	12	13	14	15	16	17	18
classes	87347	172684	303116	473733	660907	824389	920439	920443
$\lvert\mathcal{A}^+\rvert$	19	20	21	22	23	24	25	26
classes	824409	660949	473827	303277	172933	87659	39433	15709
$\lvert\mathcal{A}^+\rvert$	27	28	29	30	31	32	33	34
classes	5557	1760	509	137	38	10	3	1

together with a single class of cardinality 35, with total 7 011 184. The dimension count at $(7,3)$ is $21+7-6-1-3 = 18$, so the Carathéodory bound is 19, and the classes of cardinality at most 19 already number 5 249 381. Enumeration at this size is not a route; Problem 2 asks instead for an argument uniform in the active family.

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